

FC7xxx ETH User Manual

Rev.A0

Revision History

Revision	Date	Changes
A0	2025/01/16	Add support for FC7300F8MDQ

Table of Contents

Chapter 1 ETH Introduction	5
1.1 Requirement Tracing	5
1.2 Hardware Summary	5
Chapter 2 Software Design	6
2.1 Rejected Requirements	6
2.2 File Structure	9
2.3 Macros	10
2.3.1 Macros in Eth_Cfg.h	10
2.3.2 Macros in Eth_Types.h	13
2.3.3 Macros in Eth_Hw.h	14
2.3.4 Macros in Eth_Hal.h	17
2.3.5 Macros in Eth_Version.h	22
2.3.6 Macros in Eth_Hal.c	22
2.3.7 Macros in Eth.c	24
2.4 Enums	24
2.4.1 Enumerations in Eth_Types.h	24
2.4.2 Enumerations in Eth_GeneralTypes.h	25
2.4.3 Enumerations in Eth_Hw_Types.h	29
2.5 Typedefs	34
2.5.1 Typedefs in Eth_GeneralTypes.h	34
2.5.2 Typedefs in Eth_Types.h	34
2.6 Structures	34
2.6.1 Eth_ConfigType	34
2.6.2 Eth_CtrlCfgType	35
2.6.3 Eth_TimeAwareShaperType	37
2.6.4 Eth_GateControlType	38
2.6.5 Eth_MTLQueueCfgType	38
2.6.6 Eth_FifoConfigType	39
2.6.7 Eth_DMACHConfigType	39
2.6.8 Eth_DMADescType	39
2.6.9 Eth_HandleType	40
2.6.10 Eth_FifoStateType	40
2.6.11 Eth_BufIdxMapType	41
2.6.12 Eth_RxInfoType	42
2.6.13 Eth_DrvBufType	42
2.6.14 EthDemEventStatsType	43
2.6.15 Eth_TimeStampType	43
2.6.16 Eth_TimeIntDiffType	43
2.6.17 Eth_RateRatioType	44
2.6.18 Eth_MacVlanType	44

2.6.19	Eth_CounterType.....	44
2.6.20	Eth_RxStatsType	45
2.6.21	Eth_TxStatsType	47
2.6.22	Eth_TxErrorCounterValuesType	47
2.6.23	EthSwt_MgmtInfoType	48
2.6.24	EthSwt_PortMirrorCfgType.....	48
2.6.25	EthSwt_MgmtObjectValidType	49
2.6.26	EthSwt_MgmtObjectType	49
2.7	API Functions.....	49
2.7.1	Functions in Eth.h	49
2.7.2	Functions in SchM_Eth.h	65
2.7.3	Functions in Eth_Irq.c	66
2.7.4	Local Functions in Eth.c	68
2.8	Hardware Functions	75
2.8.1	Functions in Eth_Hal.h.....	75
2.8.2	Functions in Eth_Hw.h.....	84
2.9	API Sequence Diagram.....	129
2.9.1	Initialization	129
2.9.2	Communication Initialization	129
2.9.3	Data Transmission	130
2.9.4	Data Reception.....	131
2.9.5	Link State Change.....	132
Chapter 3	Tresos Configuration Items	134
3.1	Container Inclusion Relation	134
3.2	Containers and Variables.....	136
3.2.1	IMPLEMENTATION_CONFIG_VARIANT.....	136
3.2.2	EthGeneral.....	136
3.2.3	EthCtrlConfig	142
3.2.4	CommonPublishedInformation	166
Chapter 4	Configuration Guides.....	169
4.1	Configuration Item Constraint.....	169
	Eth Usage Common Steps	169

Chapter 1 ETH Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the ETH driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_EthernetDriver*.

1.2 Hardware Summary

The Ethernet (ETH) module is available for FC7300 series.

The Ethernet (ETH) module supports the following major features:

- 1) A standard ethernet controller compliant with the IEEE 802.3-2015 specification.
- 2) 10/100 Mbps data transfer rate with MII RMII interfaces.
- 3) 1000 Mbps data transfer rate with RGMII interface (FC7300F8MDT/ FC7300F8MDQ).
- 4) Both half-duplex and full-duplex are supported.
- 5) Optional MDIO master interface for PHY device configuration.
- 6) Two queues on the Receive path and Transmit path.
- 7) Automatic CRC generation and checking.
- 8) Checksum offloading.
- 9) Hardware packet filtering.
- 10) Global time support (IEEE 1588 Precision Time Protocol).
- 11) MAC Management Counters.
- 12) Support both interrupt and polling mode data transmission.
- 13) Support time sensitive networking.
- 14) Support automotive safety features.

The ETH driver provides several services including initializing, transmission, global time APIs, frame statistics and others.

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_Eth_00005**

Description	The Ethernet Driver module shall support link time configuration.
Rejection Reason	Pre-compile time, and post-build time configurations are implemented, link time configuration is not supported.

Rejected Requirement 2 **SWS_Eth_00120**

Description	API function Description Det_ReportError Service to report development errors. EthSwt_EthRxFinishedIndication Indication for a finished receive process for a specific Ethernet frame, which results in providing the management information retrieved during EthSwt_EthRxProcessFrame(). EthSwt_EthRxProcessFrame Function inspects the Ethernet frame passed by the data pointer for management information and stores it for later use in EthSwt_EthRxFinishedIndication(). EthSwt_EthTxAdaptBufferLength Modifies the buffer length to be able to insert management information. EthSwt_EthTxFinishedIndication Indication for a finished transmit process for a specific Ethernet frame. EthSwt_EthTxPrepareFrame Prepares the Ethernet frame for common Ethernet communication (frame shall be handled by switch according to the common address resolution behavior) and stores the information for processing of EthSwt_EthTxFinishedIndication().
Rejection Reason	These optional switch Interfaces except for Det_ReportError are not supported in the current version.

Rejected Requirement 3 **SWS_Eth_00247**

Description	The Switch Driver management API' s: EthSwt_EthRxProcessFrame(), EthSwt_EthRxFinishedIndication(), EthSwt_EthTxPrepareFrame(), EthSwt_EthTxAdaptBufferLength(), EthSwt_EthTxProcessFrame() and EthSwt_EthTxFinishedIndication() shall be used to to inform the Switch Driver about a required special treatment for Switch management purpose (see document AUTOSAR_SWS_EthernetInterface).
Rejection Reason	These optional switch Interfaces are not supported in the current version.

Rejected Requirement 4 **SWS_Eth_00263**

Description	If the configuration parameter EthCtrlConfigSwBufferHandling is set to TRUE, then the optional SW buffer handling shall be enabled. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 5 **SWS_Eth_00264**

Description	If the config parameter EthCtrlConfigSwBufferHandling is set to TRUE, each SW FIFO size shall
--------------------	---

	be configured using a multiple of the parameter EthCtrlConfigEgressFifoBufLenByte. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 6 SWS_Eth_00265

Description	All SW FIFOs shall follow the criteria listed here: a) Each SW FIFO shall be reserved for a specific priority. b) The SW FIFOs shall be filled and read out according to FIFO principles. c) Each SW FIFO shall have identical settings except for EthCtrlConfigShaperIdleSlope and EthCtrlConfigEgressFifoBufLenByte. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 7 SWS_Eth_00266

Description	SW FIFOs shall be iterated and their credits account be updated in the following way and order: a) Iterate through all SW FIFOs, starting at the highest priority SW FIFO and descending, and add the amount of credits accumulated since the last Eth_MainFunction() call. The amount of credits accumulated is given by EthCtrlConfigShaperIdleSlope. b) Credits are only accumulated for SW FIFOs which have at least one message queued inside them. Empty SW FIFOs do not accumulate credits. c) If a SW FIFO is empty but has still credits left from previous iterations, these credits shall neither be deleted nor increased. d) If a SW FIFO reaches EthCtrlConfigShaperMaxCredit then the credit accumulation shall stop at that point and the next SW FIFO in the row is handled. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 8 SWS_Eth_00267

Description	If Eth_ProvideTxBuffer() is called and EthCtrlConfigSwBufferHandling is set to TRUE, a tuple of BuffIdx pointer to the SW buffer pool (which is returned) and priority (provided by argument of the current function call) shall be stored. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 9 SWS_Eth_00268

Description	When Eth_Transmit() is called, the given BuffIdx pointer shall be assigned to the SW FIFO with the EthCtrlConfigEgressFifoPriorityAssignment matches the priority given previously by the previous Eth_ProvideTxBuffer() call (see SWS_Eth_00267). (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 10 SWS_Eth_00269

Description	Upon calling Eth_Transmit(), messages from the SW FIFOs shall be moved to the HW FIFO as described in SWS_Eth_00271. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 11 SWS_Eth_00270

Description	In the context of Eth_MainFunction(), the following actions shall be executed in the given order: a) All SW FIFOs shall be iterated and their credits account updated as specified in SWS_Eth_00266. b) All SW FIFOs shall be iterated and checked for messages which are ready for transmission. c) For each SW FIFO iterated, transmission shall be attempted as specified in SWS_Eth_00271. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 12 SWS_Eth_00271

Description	Messages queued inside SW FIFOs shall be moved to the HW FIFO in the following way and order: a) Loop through each SW FIFO, starting at the highest priority in descending order. b) Move the first message inside a SW FIFO whose credit account is at least EthCtrlConfigShaperMinCredit to the HW FIFO. c) If EthTrcvPhysLayerPLCAMaxBurstCount is set to 0 then only one message is moved to the HW FIFO and the iteration to the next SW FIFOs is stopped. d) Deduct the size of the message moved in bytes from the credits account of that SW FIFO. e) If EthTrcvPhysLayerPLCAMaxBurstCount is higher than 0 then proceed on top as specified in SWS_Eth_00272. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 13 SWS_Eth_00272

Description	If frame transmission is triggered (see SWS_Eth_00269 and SWS_Eth_00270) and EthTrcvPhysLayerPLCAMaxBurstCount is higher than 0 then as many messages as EthTrcvPhysLayerPLCAMaxBurstCount indicate shall be moved additionally to the HW FIFO in the following way: a) Move messages from the SW FIFO, deducting right after each message from the credits account, until the SW FIFO has reached EthCtrlConfigShaperMinCredit. b) Check the amount of available credits and only continue with the next message if the credits account is at least EthCtrlConfigShaperMinCredit. c) If the credits account for this SW FIFO has dropped below EthCtrlConfigShaperMaxCredit then move to the next SW FIFO based on its priority and repeat the previous step until having reached a total of EthTrcvPhysLayerPLCAMaxBurstCount messages moved from the SW FIFO(s) to the HW FIFO. (SRS_Eth_00146)
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 14 SWS_Eth_00066

Description	Enables the transmission with PLCA (Physical Layer Collision Avoidance) TRUE: PLCA enabled FALSE: PLCA disabled
Rejection Reason	The status of this parameter is draft.

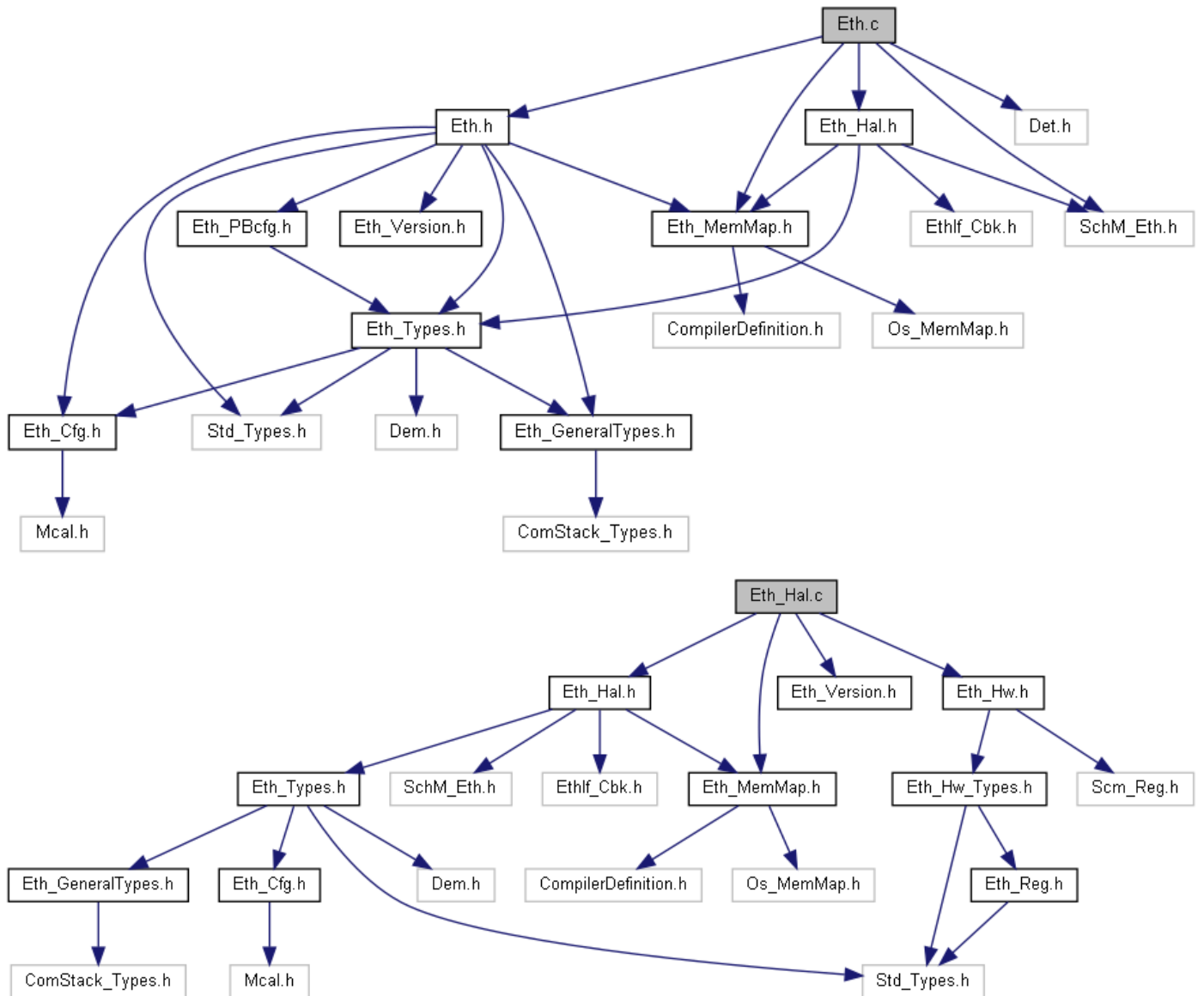
Rejected Requirement 15 SWS_Eth_00071

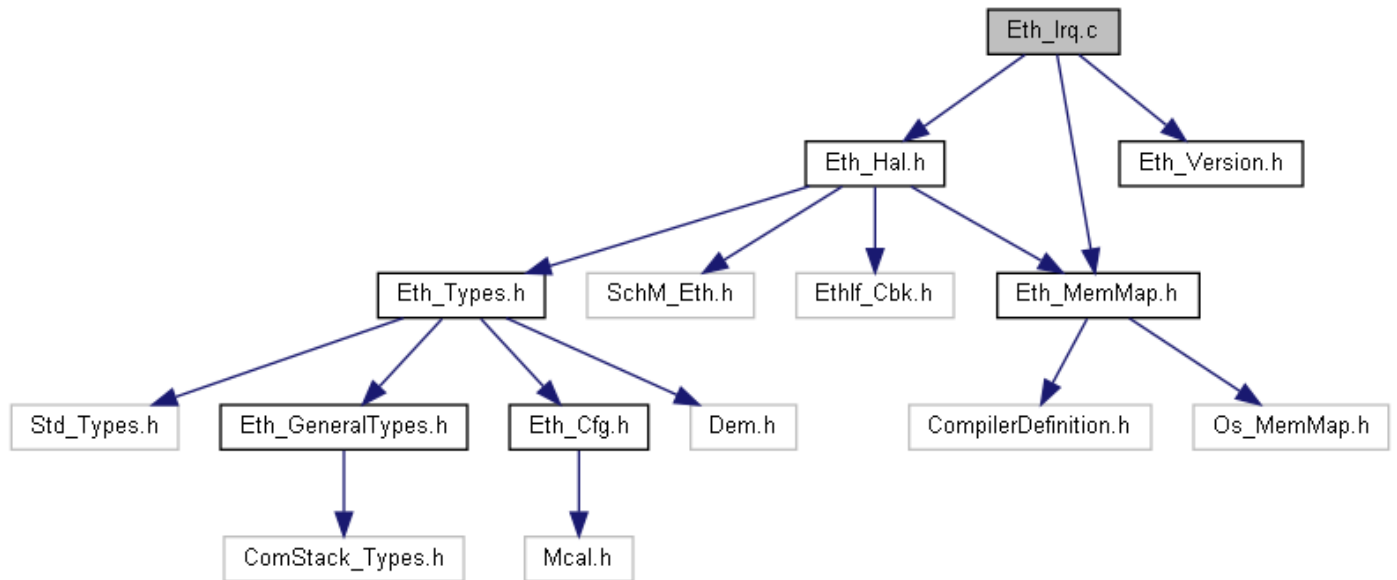
Description	Enables / Disables software buffer management.
Rejection Reason	The buffer handling is supported by hardware such as AVB and Time Aware Shaper.

Rejected Requirement 16 SWS_Eth_00261

Description	EthCtrlConfig, EthTrcvConfig and EthSwtConfig (if existent in configuration) of one communication channel shall all reference the same ECUC partition.
Rejection Reason	The Eth controller is referenced by reansceiver or switch, this requirement is best implemented in the transceiver or switch module.

2.2 File Structure





2.3 Macros

2.3.1 Macros in Eth_Cfg.h

- #define **ETH_CFG_AR_RELEASE_MAJOR_VERSION** 4
- #define **ETH_CFG_AR_RELEASE_MINOR_VERSION** 6
- #define **ETH_CFG_AR_RELEASE_REVISION_VERSION** 0
- #define **ETH_CFG_SW_MAJOR_VERSION** 1
- #define **ETH_CFG_SW_MINOR_VERSION** 1
- #define **ETH_CFG_SW_PATCH_VERSION** 0
- #define **ETH_CFG_VENDOR_ID** 174
- #define **ETH_CFG_MODULE_ID** 88
- #define **ETH_PRECOMPILE_SUPPORT** (STD_OFF)
- #define **ETH_CTRLIDX_0** (0U)
- #define **ETH_CHANNELIDX_0** (0U)
- #define **ETH_CHANNELIDX_1** (1U)
- #define **ETH_DEV_ERROR_DETECT** (STD_ON)
Eth Dev error detect switch.

- **#define ETH_INDEX** (0U)
Instance Id of this Eth driver.
 - **#define ETH_MAX_CTRL_S_SUPPORTED** (1U)
Limits the total number of supported controllers.
 - **#define ETH_ENABLE_PLCA** (1U)
Enables the transmission with PLCA (Physical Layer Collision Avoidance)
 - **#define ETH_GET_COUNTER_VALUES_API** (STD_ON)
Enables / Disables Eth_GetCounterValues API.
 - **#define ETH_GET_TX_ERROR_COUNTER_VALUES_API** (STD_ON)
Enables / Disables Eth_GetTxErrorCounterValues API.
 - **#define ETH_GET_TX_STATS_API** (STD_ON)
Enables / Disables Eth_GetTxStats API.
 - **#define ETH_GET_RX_STATS_API** (STD_ON)
Enables / Disables Eth_GetRxStats API.
 - **#define ETH_GLOBAL_TIME_SUPPORT** (STD_ON)
Enables / Disables the GlobalTime APIs.
 - **#define ETH_UPDATE_PHYS_ADDR_FILTER** (STD_ON)
Enables/Disables optional API Eth_UpdatePhysAddrFilter.
 - **#define ETH_VERSION_INFO_API** (STD_ON)
Enables / Disables version info API.
 - **#define ETH_DEM_EVENT_DETECT** (STD_ON)
Diagnostic Event switch.
 - **#define ETH_PHYS_ADDR_LEN_BYTE** (6U)
Eth physical address length in byte.
 - **#define ETH_HEADER_LEN_BYTE** (14U)
Eth MAC layer header length in byte.
 - **#define ETH_TIMEOUT_LOOP_COUNT** (1500U)
Eth hardware wait timeout loop cycles.
- #define ETH_CTRL_ENABLE_OFFLOAD_CHECKSUM_ICMP** (STD_OFF)
Enables / Disables hardware offloading for ICMP checksums.

#define **ETH_CTRL_ENABLE_OFFLOAD_CHECKSUM_IPV4** (STD_OFF)

Enables / Disables hardware offloading for IPV4 checksums.

- #define **ETH_CTRL_ENABLE_OFFLOAD_CHECKSUM_TCP** (STD_OFF)
Enables / Disables hardware offloading for TCP checksums.
- #define **ETH_CTRL_ENABLE_OFFLOAD_CHECKSUM_UDP** (STD_OFF)
Enables / Disables hardware offloading for UDP checksums.
- #define **ETH_CTRL_ENABLE_OFFLOAD_CHECKSUM** (STD_OFF)
Enables / Disables hardware offloading for checksums.
- #define **ETH_CTRL_ENABLE_MII** (STD_ON)
Enables / Disables Media Independent Interface (MII) for transceiver.
- #define **ETH_CTRL_ENABLE_MMD** (STD_ON)
Enables / Disables MDIO clause 45 read write.
- #define **ETH_LIMIT_MAX_FIFO_BUF_LENBYTES** (1536U)
Limits the max length bytes of FIFO buffers.
- #define **ETH_LIMIT_MIN_FIFO_BUF_LENBYTES** (512U)
Limits the min length bytes of FIFO buffers.
- #define **ETH_LIMIT_MAX_FIFO_BUF_COUNTS** (64U)
Limits the max count of FIFO buffers.
- #define **ETH_LIMIT_MIN_FIFO_BUF_COUNTS** (4U)
Limits the min count of FIFO buffers.
- #define **ETH_FEATURE_MAX_CHANNEL_COUNT** (2U)
Hardware feature max channel count of Eth controllers.
- #define **ETH_FEATURE_SAFETY_SUPPORT** STD_ON
Hardware feature safety function support.
- #define **ETH_FEATURE_TIME_AWARE_SHAPER_SUPPORT** STD_ON
Hardware feature time aware shaper function support.
- #define **ETH_FEATURE_TOTAL_TX_FIFO_SIZE** (8192U)
Hardware feature total transmit fifo memory size.
- #define **ETH_FEATURE_TOTAL_RX_FIFO_SIZE** (8192U)
Hardware feature total receive fifo memory size.
- #define **ETH_NUM_ECU_PARTITIONS** (3U)
The number of ECU partitions supported by Eth driver.

- #define **ETH_CHECK_ECU_PARTITION** (STD_ON)
ECU partitions check switch.

2.3.2 Macros in Eth_Types.h

- #define **ETH_DRV_SID_INIT** 0x01
- #define **ETH_DRV_SID_SETCONTROLLERMODE** 0x03
- #define **ETH_DRV_SID_GETCONTROLLERMODE** 0x04
- #define **ETH_DRV_SID_WRITEMII** 0x05
- #define **ETH_DRV_SID_READMII** 0x06
- #define **ETH_DRV_SID_GETPHYSADDR** 0x08
- #define **ETH_DRV_SID_PROVIDETXBUFFER** 0x09
- #define **ETH_DRV_SID_TRANSMIT** 0x0A
- #define **ETH_DRV_SID_RECEIVE** 0x0B
- #define **ETH_DRV_SID_TXCONFIRMATION** 0x0C
- #define **ETH_DRV_SID_GETVERSIONINFO** 0x0D
- #define **ETH_DRV_SID_IRQHANDLER** 0x0F
- #define **ETH_DRV_SID_TXIRQHANDLER** 0x10
- #define **ETH_DRV_SID_RXIRQHANDLER** 0x11
- #define **ETH_DRV_SID_UPDATEADDRFILTER** 0x12
- #define **ETH_DRV_SID_SETPHYSADDR** 0x13
- #define **ETH_DRV_SID_GETCOUNTERVALUES** 0x14
- #define **ETH_DRV_SID_GETRXSTATS** 0x15
- #define **ETH_DRV_SID_GETCURRENTTIME** 0x16
- #define **ETH_DRV_SID_ENABLEEGRESSTIMESTAMP** 0x17
- #define **ETH_DRV_SID_GETEGRESSTIMESTAMP** 0x18
- #define **ETH_DRV_SID_GETINGRESSTIMESTAMP** 0x19

- `#define ETH_DRV_SID_SETCORRECTIONTIME 0x1A`
- `#define ETH_DRV_SID_SETGLOBALTIME 0x1B`
- `#define ETH_DRV_SID_GETTXSTATS 0x1C`
- `#define ETH_DRV_SID_GETTXERRORCOUNTERVALUES 0x1D`
- `#define ETH_DRV_SID_MAINFUNCTION 0x20`
- `#define ETH_DRV_SID_WRITEMMD 0x21`
- `#define ETH_DRV_SID_READMMD 0x22`
- `#define ETH_DRV_SID_TIMEAWARESHAPER 0x23`
- `#define ETH_E_INV_CTRL_IDX ((uint8)0x01)`
- `#define ETH_E_UNINIT ((uint8)0x02)`
- `#define ETH_E_PARAM_POINTER ((uint8)0x03)`
- `#define ETH_E_INV_PARAM ((uint8)0x04)`
- `#define ETH_E_INV_MODE ((uint8)0x05)`
- `#define ETH_E_COMMUNICATION ((uint8)0x06)`
- `#define ETH_E_INV_PARTITION ((uint8)0x10)`
- `#define ETH_E_NOT_SUPPORT ((uint8)0x11)`
- `#define ETH_E_UNEXPECTED_ISR ((uint8)0x12)`
- `#define ETH_E_ALREADY_INITIALIZED ((uint8)0x13)`

2.3.3 Macros in Eth_Hw.h

- `#define ENET_TDES2_IOC_MASK (0x80000000U)`
DMA TX descriptor Transmit Complete Interrupt Enable.
- `#define ENET_TDES2_TTSE_MASK (0x40000000U)`
DMA TX descriptor TX Timestamp Enable.
- `#define ENET_TDES2_VTIR(x) (((uint32)((uint32)(x) < 14U)) & 0x0000C000U)`
DMA TX descriptor TX VLAN Tag Control.

- **#define ENET_TDES2_BUFFER1_LEN(x)** (((uint32)((uint32)(x) < 0U)) & 0x00003FFFU)
DMA TX descriptor TX Buffer1 Length.
- **#define ENET_TDES2_BUFFER2_LEN(x)** (((uint32)((uint32)(x) < 16U)) & 0x3FFF0000U)
DMA TX descriptor TX Buffer2 Length.
- **#define ENET_TDES3_CPC(x)** (((uint32)((uint32)(x) < 26U)) & 0x0C000000U)
DMA TX descriptor CRC Pad Control.
- **#define ENET_TDES3_CIC(x)** (((uint32)((uint32)(x) < 16U)) & 0x00030000U)
DMA TX descriptor Checksum Insertion Control.
- **#define ENET_TDES3_FD_MASK** (0x20000000U)
DMA TX descriptor First Descriptor.
- **#define ENET_TDES3_LD_MASK** (0x10000000U)
DMA TX descriptor Last Descriptor.
- **#define ENET_TDES3_THL(x)** (((uint32)((uint32)(x) < 19U)) & 0x00780000U)
DMA TX descriptor TCP/UDP Header Length.
- **#define ENET_TDES3_TSE_MASK** (0x00040000U)
DMA TX descriptor TCP Segmentation Enable.
- **#define ENET_TDES3_TPL_MASK** (0x0003FFFFU)
DMA TX descriptor TCP Payload Length Mask.
- **#define ENET_TDES3_TPL(x)** (((uint32)((uint32)(x) < 0U)) & 0x0003FFFFU)
DMA TX descriptor TCP Payload Length.
- **#define ENET_TDES3_FL_MASK** (0x00007FFFU)
DMA TX descriptor Frame Length Mask.
- **#define ENET_TDES3_FL(x)** (((uint32)((uint32)(x) < 0U)) & 0x00007FFFU)
DMA TX descriptor Frame Length.
- **#define ENET_TDES3_TTSS_MASK** (0x00020000U)
DMA TX descriptor Tx Timestamp Status.
- **#define ENET_TDES2_IVT(x)** (((uint32)((uint32)(x) < 16U)) & 0xFFFF0000U)
DMA TX descriptor Inner VLAN Tag.
- **#define ENET_TDES2_MSS(x)** (((uint32)((uint32)(x) < 0U)) & 0x00003FFFU)
DMA TX descriptor Maximum Segment Size.
- **#define ENET_TDES3_TCMSSV_MASK** (0x04000000U)
DMA TX descriptor One-Step Timestamp Correction Input or MSS Valid.

- #define **ENET_TDES3_IVTIR(x)** (((uint32)((uint32)(x) < 18U)) & 0x000C0000U)
DMA TX descriptor Inner VLAN Tag Control.
- #define **ENET_TDES3_IVLTV_MASK** (0x00020000U)
DMA TX descriptor Inner VLAN Tag Valid.
- #define **ENET_TDES3_VLTV_MASK** (0x00010000U)
DMA TX descriptor VLAN Tag Valid.
- #define **ENET_TDES3_VT(x)** (((uint32)((uint32)(x) < 0U)) & 0x0000FFFFU)
DMA TX descriptor VLAN Tag.
- #define **ENET_RDES3_IOC_MASK** (0x40000000U)
DMA RX descriptor Interrupt Enabled on Completion.
- #define **ENET_RDES3_BUFFER1_VALID_MASK** (0x01000000U)
DMA RX descriptor Buffer 1 Address Valid.
- #define **ENET_RDES3_BUFFER2_VALID_MASK** (0x02000000U)
DMA RX descriptor Buffer 2 Address Valid.
- #define **ENET_RDES0_INNER_VLAN_TAG_MASK** (0xFFFF0000U)
DMA RX descriptor Inner VLAN Tag Mask.
- #define **ENET_RDES0_INNER_VLAN_TAG_SHIFT** (16U)
DMA RX descriptor Inner VLAN Tag Shift.
- #define **ENET_RDES0_OUTER_VLAN_TAG_MASK** (0x0000FFFFU)
DMA RX descriptor Outer VLAN Tag.
- #define **ENET_RDES1_TIMESTAMP_AVAILABLE_MASK** (0x00004000U)
DMA RX descriptor Timestamp Available.
- #define **ENET_RDES1_IP_PAYLOAD_ERROR_MASK** (0x00000080U)
DMA RX descriptor IP Payload Error.
- #define **ENET_RDES1_IPV6_HEADER_MASK** (0x00000020U)
DMA RX descriptor IPv6 header Present.
- #define **ENET_RDES1_IPV4_HEADER_MASK** (0x00000010U)
DMA RX descriptor IPv4 Header Present.
- #define **ENET_RDES1_IP_HEADER_ERROR_MASK** (0x00000008U)
DMA RX descriptor IP Header Error.
- #define **ENET_RDES1_IP_PAYLOAD_TYPE_MASK** (0x00000007U)
DMA RX descriptor Payload Type.

- **#define ENET_RDES3_FD_MASK** (0x20000000U)
DMA RX descriptor First Descriptor.
- **#define ENET_RDES3_LD_MASK** (0x10000000U)
DMA RX descriptor Last Descriptor.
- **#define ENET_RDES3_PACKET_LENGTH_MASK** (0x00007FFFU)
DMA RX descriptor Packet Length.
- **#define ENET_RDES3_RDES0_VALID_MASK** (0x02000000U)
DMA RX descriptor Receive Status RDES0 Valid.
- **#define ENET_RDES3_RDES1_VALID_MASK** (0x04000000U)
DMA RX descriptor Receive Status RDES1 Valid.
- **#define ENET_RDES3_RDES2_VALID_MASK** (0x08000000U)
DMA RX descriptor Receive Status RDES2 Valid.
- **#define ENET_DES3_OWN_MASK** (0x80000000U)
DMA descriptor owner bit for all format.
- **#define ENET_DES3_CTXT_MASK** (0x40000000U)
DMA descriptor Context Descriptor bit.
- **#define ENET_EST_INTERVAL_WIDTH** (24U)
ENET GCL interval time width.
- **#define ENET_EST_INTERVAL_MASK** (0xFFFFFFFU)
ENET GCL interval time mask.
- **#define ENET_EST_GATE_MASK** (0x3U)
ENET GCL gate mask.

2.3.4 Macros in Eth_Hal.h

- **#define Eth_GetPartitionID()** GET_CPU_ID ()
- **#define ETH_DATA_BARRIER()** MCAL_DATA_MEMORY_BARRIER()
- **#define ETH_ENTER_EXCLUSIVE_AREA_TX_BUFFER()** SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_00()
- **#define ETH_EXIT_EXCLUSIVE_AREA_TX_BUFFER()** SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_00()
- **#define ETH_ENTER_EXCLUSIVE_AREA_TRANSMIT()** SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_01()
- **#define ETH_EXIT_EXCLUSIVE_AREA_TRANSMIT()** SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_01()
- **#define ETH_ENTER_EXCLUSIVE_AREA_TX_REPORT()** SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_02()

- `#define ETH_EXIT_EXCLUSIVE_AREA_TX_REPORT() SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_02()`
- `#define ETH_ENTER_EXCLUSIVE_AREA_CTRLMODE() SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_03()`
- `#define ETH_EXIT_EXCLUSIVE_AREA_CTRLMODE() SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_03()`
- `#define ETH_ENTER_EXCLUSIVE_AREA_RECEIVE() SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_04()`
- `#define ETH_EXIT_EXCLUSIVE_AREA_RECEIVE() SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_04()`
- `#define ETH_ENTER_EXCLUSIVE_AREA_GETTIME() SchM_Enter_Eth_ETH_EXCLUSIVE_AREA_05()`
- `#define ETH_EXIT_EXCLUSIVE_AREA_GETTIME() SchM_Exit_Eth_ETH_EXCLUSIVE_AREA_05()`
- `#define ETH_CONFIG_PTR (Eth_Config_Ptrs[CoreId])`
- `#define ETH_CFG_PARTITION_ID(CtrlIdx,CoreId) ((Eth_Config_Ptrs[CoreId])->CtrlPtr[CtrlIdx].EcuPartition)`
- `#define ETH_CFG_NUM_CONTROLS(CoreId) ((Eth_Config_Ptrs[CoreId])->NumController)`
- `#define ETH_CFG_CTRL_CFG_PTR(CtrlIdx,CoreId) (&((Eth_Config_Ptrs[CoreId])->CtrlPtr[CtrlIdx]))`
- `#define ETH_CFG_BUS_CLOCK_FREQ(CtrlCfgPtr) ((CtrlCfgPtr)->BusClkFreq)`
- `#define ETH_CFG_ETHIF_CTRLIDX(CtrlCfgPtr) ((CtrlCfgPtr)->EthIfIdx)`
- `#define ETH_CFG_CTRL_NUM_TX_CHANNEL(CtrlCfgPtr) ((CtrlCfgPtr)->EgressFifoCount)`
- `#define ETH_CFG_CTRL_NUM_RX_CHANNEL(CtrlCfgPtr) ((CtrlCfgPtr)->IngressFifoCount)`
- `#define ETH_CFG_NUM_TX_FIFO(CtrlCfgPtr) ((CtrlCfgPtr)->EgressFifoCount)`
- `#define ETH_CFG_NUM_RX_FIFO(CtrlCfgPtr) ((CtrlCfgPtr)->IngressFifoCount)`
- `#define ETH_CFG_CTRL_ENABLE_MII(CtrlCfgPtr) ((CtrlCfgPtr)->EnableMii)`
- `#define ETH_CFG_CTRL_ENABLE_MMD(CtrlCfgPtr) ((CtrlCfgPtr)->EnableMmd)`
- `#define ETH_CFG_CTRL_READ_MII_INDICATION(CtrlCfgPtr,CtrlIdx,TrcvIdx,RegIdx,RegVal) \`
`if(CtrlCfgPtr->ReadMiiCbK != NULL_PTR){ CtrlCfgPtr->ReadMiiCbK(CtrlIdx,TrcvIdx,RegIdx,RegVal); }`
- `#define ETH_CFG_CTRL_WRITE_MII_INDICATION(CtrlCfgPtr,CtrlIdx,TrcvIdx,RegIdx) \`
`if(CtrlCfgPtr->WriteMiiCbK!= NULL_PTR){ CtrlCfgPtr->WriteMiiCbK (CtrlIdx,TrcvIdx,RegIdx); }`

- **#define ETH_CFG_CTRL_ENABLE_TX_INTERRUPT(CtrlCfgPtr)** ((CtrlCfgPtr)->TxInterruptEn)
- **#define ETH_CFG_CTRL_ENABLE_RX_INTERRUPT(CtrlCfgPtr)** ((CtrlCfgPtr)->RxInterruptEn)
- **#define ETH_CFG_PHY_INTERFACE(CtrlCfgPtr)** ((CtrlCfgPtr)->PhyIf)
- **#define ETH_CFG_CTRL_MAC_ADDRESS(CtrlCfgPtr)** ((CtrlCfgPtr)->MacAddress)
- **#define ETH_CFG_CTRL_MAC_CONFIGURATION(CtrlCfgPtr)** ((CtrlCfgPtr)->MacConfiguration)
- **#define ETH_CFG_CTRL_MAC_EXT_CONFIGURATION(CtrlCfgPtr)** ((CtrlCfgPtr)->MacExtConfiguration)
- **#define ETH_CFG_CTRL_MAC_PACKET_FILTER(CtrlCfgPtr)** ((CtrlCfgPtr)->MacFilterRegValue)
- **#define ETH_CFG_CTRL_TX_FIFO_BUF_LENBYTE(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->EgressFifoCfgPtr[Fifold].FifoBufLenByte)
- **#define ETH_CFG_CTRL_RX_FIFO_BUF_LENBYTE(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->IngressFifoCfgPtr[Fifold].FifoBufLenByte)
- **#define ETH_CFG_CTRL_TX_FIFO_BUF_TOTAL(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->EgressFifoCfgPtr[Fifold].FifoBufTotal)
- **#define ETH_CFG_CTRL_RX_FIFO_BUF_TOTAL(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->IngressFifoCfgPtr[Fifold].FifoBufTotal)
- **#define ETH_CFG_CTRL_TX_DESC_START_ADDRESS(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->EgressFifoCfgPtr[Fifold].DescListAddress)
- **#define ETH_CFG_CTRL_RX_DESC_START_ADDRESS(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->IngressFifoCfgPtr[Fifold].DescListAddress)
- **#define ETH_CFG_CTRL_TX_BUF_START_ADDRESS(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->EgressFifoCfgPtr[Fifold].FifoBufStartAddress)
- **#define ETH_CFG_CTRL_RX_BUF_START_ADDRESS(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->IngressFifoCfgPtr[Fifold].FifoBufStartAddress)
- **#define ETH_CFG_CTRL_TX_FIFO_PRIORITY_MASK(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->EgressFifoCfgPtr[Fifold].PriorityMask)
- **#define ETH_CFG_CTRL_RX_FIFO_PRIORITY_MASK(CtrlCfgPtr,Fifold)** ((CtrlCfgPtr)->IngressFifoCfgPtr[Fifold].PriorityMask)
- **#define ETH_CFG_CTRL_DMA_MODE(CtrlCfgPtr)** ((CtrlCfgPtr)->DmaMode)
- **#define ETH_CFG_CTRL_DMA_SYSBUS_MODE(CtrlCfgPtr)** ((CtrlCfgPtr)->DmaSysBusMode)

- **#define ETH_CFG_CTRL_DMA_CHANNEL_CONTROL(CtrlCfgPtr,Channel)** ((CtrlCfgPtr)->DMAChannelCfgPtr[Channel].Control)
- **#define ETH_CFG_CTRL_DMA_TX_CHANNEL_CONTROL(CtrlCfgPtr,Channel)** ((CtrlCfgPtr)->DMAChannelCfgPtr[Channel].TxControl)
- **#define ETH_CFG_CTRL_DMA_RX_CHANNEL_CONTROL(CtrlCfgPtr,Channel)** ((CtrlCfgPtr)->DMAChannelCfgPtr[Channel].RxControl)
- **#define ETH_CFG_CTRL_MTL_OPERATION_MODE(CtrlCfgPtr)** ((CtrlCfgPtr)->MtlOperationMode)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_MODE(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].TxQueueMode)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_WEIGHTS(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].TxQueueWeights)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_IDLE_SLOPE_CREDIT(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].IdleSlopeCredit)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_SEND_SLOPE_CREDIT(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].SendSlopeCredit)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_HIGH_CREDIT(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].HiCredit)
- **#define ETH_CFG_CTRL_MTL_TX_QUEUE_LOW_CREDIT(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].LoCredit)
- **#define ETH_CFG_CTRL_MTL_RX_QUEUE_MODE(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].RxQueueMode)
- **#define ETH_CFG_CTRL_MTL_FORWARD_TCPIP_CHECKSUM_ERROR(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].ForwardTCPIPChecksumErr)
- **#define ETH_CFG_CTRL_MTL_FORWARD_ERROR_PACKETS(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].ForwardErrorPackets)
- **#define ETH_CFG_CTRL_MTL_FORWARD_UNDERSIZE_GOOD_PACKETS(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].ForwardUndersizedGood)
- **#define ETH_CFG_CTRL_MTL_RX_QUEUE_WEIGHT(CtrlCfgPtr,Queue)** ((CtrlCfgPtr)->MTLQueueCfgPtr[Queue].RxQueueWeight)
- **#define ETH_CFG_CTRL_MMC_RESET_ON_READ(CtrlCfgPtr)** ((CtrlCfgPtr)->MMCResetOnRead)
- **#define ETH_CFG_CTRL_MMC_STOP_ROLLOVER(CtrlCfgPtr)** ((CtrlCfgPtr)->MMCStopRollover)

- **#define ETH_CFG_CTRL_PTP_CONTROL(CtrlCfgPtr)** ((CtrlCfgPtr)->PtpCtrl)
- **#define ETH_CFG_CTRL_PTP_INIT_TIME_NANOSECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->InitTime.nanoseconds)
- **#define ETH_CFG_CTRL_PTP_INIT_TIME_SECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->InitTime.seconds)
- **#define ETH_CFG_CTRL_PTP_INIT_TIME_SECONDSHI(CtrlCfgPtr)** ((CtrlCfgPtr)->InitTime.secondsHi)
- **#define ETH_CFG_CTRL_PTP_SUBSECONDS_INCREMENT(CtrlCfgPtr)** ((CtrlCfgPtr)->SubSecondInc)
- **#define ETH_CFG_CTRL_PTP_TIMESTAMP_ADDEND(CtrlCfgPtr)** ((CtrlCfgPtr)->TimestampAddend)
- **#define ETH_CFG_CTRL_PTP_EGRESS_CORR_NANOSECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->EgressCorrNanoseconds)
- **#define ETH_CFG_CTRL_PTP_INGRESS_CORR_NANOSECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->IngressCorrNanoseconds)
- **#define ETH_CFG_CTRL_TIME_AWARE_SHAPER_PRESENT(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr != NULL_PTR)
- **#define ETH_CFG_CTRL_GCL_BASE_TIME_SECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->BaseTimeSeconds)
- **#define ETH_CFG_CTRL_GCL_BASE_TIME_NANOSECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->BaseTimeNanoseconds)
- **#define ETH_CFG_CTRL_GCL_LIST_LENGTH(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->GateControlListLength)
- **#define ETH_CFG_CTRL_GCL_CYCLE_TIME_SECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->CycleTimeSeconds)
- **#define ETH_CFG_CTRL_GCL_CYCLE_TIME_NANOSECONDS(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->CycleTimeNanoseconds)
- **#define ETH_CFG_CTRL_GCL_TIME_EXTENSION(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->TimeExtension)
- **#define ETH_CFG_CTRL_GCL_GATE_CONTROL(CtrlCfgPtr,GclIdx)** ((CtrlCfgPtr)->TimeAwareShaperPtr->GateControlList[GclIdx].GateControl)
- **#define ETH_CFG_CTRL_GCL_TIME_INTERVAL(CtrlCfgPtr,GclIdx)** ((CtrlCfgPtr)->TimeAwareShaperPtr->GateControlList[GclIdx].TimeInterval)
- **#define ETH_CFG_CTRL_ENABLE_PREEMPTION(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->EnablePreemption)

- **#define ETH_CFG_CTRL_PREEMPTION_CLASSIFICATION(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->PreemptionClassification)
- **#define ETH_CFG_CTRL_PREEMPTION_RELEASE_ADVANCE_TIME(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->ReleaseAdvanceTime)
- **#define ETH_CFG_CTRL_PREEMPTION_HOLD_ADVANCE_TIME(CtrlCfgPtr)** ((CtrlCfgPtr)->TimeAwareShaperPtr->HoldAdvanceTime)
- **#define ETH_CFG_CTRL_SAFETY_ECC_ENABLE(CtrlCfgPtr)** ((CtrlCfgPtr)->SafetyEccEn)
- **#define ETH_CFG_CTRL_SAFETY_FSM_PARITY_ENABLE(CtrlCfgPtr)** ((CtrlCfgPtr)->SafetyFsmParityEn)
- **#define ETH_CFG_CTRL_SAFETY_FSM_TIMEOUT_ENABLE(CtrlCfgPtr)** ((CtrlCfgPtr)->SafetyFsmTimeoutEn)
- **#define ETH_CFG_CTRL_SAFETY_DPP_MODE(CtrlCfgPtr)** ((CtrlCfgPtr)->SafetyDppMode)
- **#define Eth_ReportDemEvent(CtrlCfgPtr, DemId, Status)** if((uint32)STD_ON == CtrlCfgPtr->Cfg_##DemId.state){ (void)Dem_SetEventStatus((Dem_EventIdType)CtrlCfgPtr->Cfg_##DemId.id, Status); }
- **#define ETH_INVALID_FIFO_IDX** (0xFFU)

2.3.5 Macros in Eth_Version.h

- **#define ETH_VENDOR_ID** 174
- **#define ETH_MODULE_ID** 88
- **#define ETH_AR_RELEASE_MAJOR_VERSION** 4
- **#define ETH_AR_RELEASE_MINOR_VERSION** 6
- **#define ETH_AR_RELEASE_REVISION_VERSION** 0
- **#define ETH_SW_MAJOR_VERSION** 1
- **#define ETH_SW_MINOR_VERSION** 1
- **#define ETH_SW_PATCH_VERSION** 0

2.3.6 Macros in Eth_Hal.c

- **#define ETH_MAC_HASH_TABLE_INDEX_SHIFT** (31U)
MAC hash table index shift.
- **#define ETH_MAC_HASH_TABLE_INDEX_MASK** (0x1U)
MAC hash table index mask.

- **#define ETH_MAC_HASH_TABLE_BIT_SHIFT** (26U)
MAC hash table bit shift.
- **#define ETH_MAC_HASH_TABLE_BIT_MASK** (0x1FU)
MAC hash table bit mask.
- **#define ETH_MDC_CSR_VALUE_CNT** (6U)
The number of max MDC csr value.
- **#define ETH_MII_OPT_WRITE** (ENET_MAC_MDIO_ADDRESS_GOC_0_MASK)
MII write operation.
- **#define ETH_MII_OPT_READ** (ENET_MAC_MDIO_ADDRESS_GOC_1_MASK |
ENET_MAC_MDIO_ADDRESS_GOC_0_MASK)
MII read operation.
- **#define ETH_ENABLE** ((uint8)1U)
Eth enable state.
- **#define ETH_DISABLE** ((uint8)0U)
Eth disable state.
- **#define ETH_DMA_DESC_LEN_BYTES** (16U)
Eth DMA descriptor words length in bytes.
- **#define ETH_MAX_BUF_IDX_SIZE** (ETH_FEATURE_MAX_CHANNEL_COUNT * ETH_LIMIT_MAX_FIFO_BUF_COUNTS)
Max number of TX buffers for all channels.
- **#define ETH_FRAME_IS_BOARDCAST(dstmac)** ((dstmac[0U] == 0xFFU) && (dstmac[1U] == 0xFFU) &&
(dstmac[2U] == 0xFFU) && (dstmac[3U] == 0xFFU) && (dstmac[4U] == 0xFFU) && (dstmac[5U] == 0xFFU))
Check destination MAC is broadcast address.
- **#define ETH_TXCONFIRM_TIMEOUT** (2U)
TX confirmation timeout value.
- **#define ETH_FRAME_MAC_LENGTH** (6U)
Eth frame MAC address length in bytes.
- **#define ETH_FRAME_DSTMAC_OFFSET** (0U)
Eth frame destination MAC address offset.
- **#define ETH_FRAME_SRCMAC_OFFSET** (6U)
Eth frame source MAC address offset.
- **#define ETH_FRAME_TYPE_OFFSET** (12U)
Eth frame type field offset.

- #define **ETH_FRAME_HEADER_LENGTH** ETH_HEADER_LEN_BYTE
Eth frame header length in bytes.
- #define **ETH_FRAME_PAYLOAD_OFFSET** ETH_HEADER_LEN_BYTE
Eth frame payload data offset.

2.3.7 Macros in Eth.c

- #define **ETH_C_VENDOR_ID** 174
- #define **ETH_C_MODULE_ID** 88
- #define **ETH_C_AR_RELEASE_MAJOR_VERSION** 4
- #define **ETH_C_AR_RELEASE_MINOR_VERSION** 6
- #define **ETH_C_AR_RELEASE_REVISION_VERSION** 0
- #define **ETH_C_SW_MAJOR_VERSION** 1
- #define **ETH_C_SW_MINOR_VERSION** 1
- #define **ETH_C_SW_PATCH_VERSION** 0

2.4 Enums

2.4.1 Enumerations in Eth_Types.h

2.4.1.1 Eth_PhylfType

Enumeration	Eth_PhylfType	
Description	This type specifies PHY interface of the controller.	
Values	Value	Description
	ETH_PHYIF_MII = 0x0U	MII interface.
	ETH_PHYIF_RMII = 0x1U	RMII interface.
	ETH_PHYIF_RGMII = 0x2U	RGMII interface.

2.4.1.2 Eth_MTLQueueEnableType

Enumeration	Eth_MTLQueueEnableType	
Description	ETH MTL queue enable mode.	
Values	Value	Description
	ETH_MTL_QUEUE_DISABLE = 0U	MTL Queue not enabled.
	ETH_MTL_QUEUE_ENABLE_AVB = 1U	MTL Queue enabled for AV.
	ETH_MTL_QUEUE_ENABLE_GENERIC = 2U	MTL Queue enabled for DCB/Generic.

2.4.1.3 Eth_MTLDppModeType

Enumeration	Eth_MTLDppModeType	
Description	ETH MTL Data Path Parity Protection enable mode.	

Values	Value	Description
	ETH_MTL_DPP_DISABLE_PARITY = 0U	Disable Data Path Parity Protection.
	ETH_MTL_DPP_EVEN_PARITY = 1U	Even Data Path Parity Protection.
	ETH_MTL_DPP_ODD_PARITY = 2U	Odd Data Path Parity Protection.

2.4.2 Enumerations in Eth_GeneralTypes.h

2.4.2.1 Eth_StateType

Enumeration	Eth_StateType	
Description	A variable of this type holds the state of the Ethernet driver module.	
Values	Value	Description
	ETH_STATE_UNINIT = 0U	The driver is not yet configured.
	ETH_STATE_INIT = 1U	The driver has been initialized.

2.4.2.2 Eth_ModeType

Enumeration	Eth_ModeType	
Description	This type indicates whether the Ethernet controller is stopped or running.	
Values	Value	Description
	ETH_MODE_DOWN = 0U	Disable the Ethernet Rx/Tx communication and set its corresponding hardware to a lowpower sleep mode and initiate a sleep process, if the Ethernet hardware provide such a feature. E.g. request a sleep on data line for OATC10 compatible Ethernet hardware.
	ETH_MODE_ACTIVE	Enable the Ethernet Rx/Tx communication and set its corresponding hardware to an power on mode.
	ETH_MODE_ACTIVE_WITH_WAKEUP_REQUEST	Enable the Ethernet Rx/Tx communication and set its corresponding hardware to an power on mode and request an wake-up on the network, if the Ethernet hardware provide a wake-up feature. E.g. wake-up on data line for OA TC10 compatible Ethernet hardware.

2.4.2.3 Eth_RxStatusType

Enumeration	Eth_RxStatusType	
Description	The Ethernet reception status.	
Values	Value	Description
	ETH_RECEIVED = 0U	Ethernet frame has been received, no further frames available.
	ETH_NOT_RECEIVED	Ethernet frame has not been received, no further frames available.
	ETH_RECEIVED_MORE_DATA_AVAILABLE	Ethernet frame has been received, more frames are available.

2.4.2.4 Eth_FilterActionType

Enumeration	Eth_FilterActionType	
Description	Action type for MAC address filtering.	

Values	Value	Description
	ETH_ADD_TO_FILTER = 0U	Add MAC address to the filter.
	ETH_REMOVE_FROM_FILTER	Remove MAC address from the filter.

2.4.2.5 Eth_TimeStampQualType

Enumeration	Eth_TimeStampQualType	
Description	The Ethernet quality of timestamp type.	
Values	Value	Description
	ETH_VALID = 0U	Timestamp valid.
	ETH_INVALID	Timestamp invalid.
	ETH_UNCERTAIN	Timestamp uncertain.

2.4.2.6 EthTrcv_LinkStateType

Enumeration	EthTrcv_LinkStateType	
Description	This type defines the Ethernet link state.	
Values	Value	Description
	ETHTRCV_LINK_STATE_DOWN = 0U	No physical Ethernet connection established.
	ETHTRCV_LINK_STATE_ACTIVE	Physical Ethernet connection established.

2.4.2.7 EthTrcv_StateType

Enumeration	EthTrcv_StateType	
Description	This type defines the transceiver state.	
Values	Value	Description
	ETHTRCV_STATE_UNINIT = 0U	Driver is not yet configured.
	ETHTRCV_STATE_INIT	Driver is configured.

2.4.2.8 EthTrcv_BaudRateType

Enumeration	EthTrcv_BaudRateType	
Description	This type defines the Ethernet baud rate.	
Values	Value	Description
	ETHTRCV_BAUD_RATE_10MBIT = 0U	10MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_100MBIT	100MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_1000MBIT	1000MBIT Ethernet connection.
	ETHTRCV_BAUD_RATE_2500MBIT	2500MBIT Ethernet connection

2.4.2.9 EthTrcv_DuplexModeType

Enumeration	EthTrcv_DuplexModeType	
Description	This type defines the Ethernet duplex modes.	
Values	Value	Description
	ETHTRCV_DUPLEX_MODE_HALF = 0U	Half duplex Ethernet connection.
	ETHTRCV_DUPLEX_MODE_FULL	Full duplex Ethernet connection.

2.4.2.10 EthTrcv_WakeupModeType

Enumeration	EthTrcv_WakeupModeType	
-------------	------------------------	--

Description	This type controls the transceiver wake up modes and/or clears the wake-up reason.	
Values	Value	Description
	ETHTRCV_WUM_DISABLE = 0U	Transceiver wake up disabled.
	ETHTRCV_WUM_ENABLE	Transceiver wake up enabled.
	ETHTRCV_WUM_CLEAR	Transceiver wake up reason cleared.

2.4.2.11 EthTrcv_WakeupReasonType

Enumeration	EthTrcv_WakeupReasonType	
Description	This type defines the transceiver wake up reasons.	
Values	Value	Description
	ETHTRCV_WUR_NONE = 0U	No wake up reason detected.
	ETHTRCV_WUR_GENERAL	General wake up detected, no distinct reason supported by hardware.
	ETHTRCV_WUR_BUS	Bus wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_INTERNAL	Internal wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_RESET	Reset wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_POWER_ON	Power on wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_PIN	Pin wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_SYSERR	System error wake up detected. Available if supported by hardware.
	ETHTRCV_WUR_WODL_WUP	Wake-up on data line (WUP = wake up pulse) detected while link is down of the corresponding Ethernet hardware e.g. PHY). Only available if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
	ETHTRCV_WUR_WODL_WUR	Wake-up on data line (WUR = wake up request) detected while link is active of the used Ethernet hardware (e.g. PHY). Only available if EthTrcvWakeupSleepOnDatalineEnabled is set to TRUE.
	ETHTRCV_WUR_TRANSFER	Device internal transfer of wake-up on data line from a neighboring PHY. Only possible if Ethernet hardware is compliant to OA TC10.

2.4.2.12 EthTrcv_PhyTestModeType

Enumeration	EthTrcv_PhyTestModeType	
Description	Describes the possible PHY test modes.	
Values	Value	Description
	ETHTRCV_PHYTESTMODE_NONE = 0U	Normal operation.
	ETHTRCV_PHYTESTMODE_1	Test transmitter droop.
	ETHTRCV_PHYTESTMODE_2	Test master timing jitter.
	ETHTRCV_PHYTESTMODE_3	Test slave timing jitter.
	ETHTRCV_PHYTESTMODE_4	Test transmitter distortion.
	ETHTRCV_PHYTESTMODE_5	Test power spectral density (PSD) mask.

2.4.2.13 EthTrcv_PhyLoopbackModeType

Enumeration	EthTrcv_PhyLoopbackModeType	
Description	Describes the possible PHY loopback modes.	
Values	Value	Description
	ETHTRCV_PHYLOOPBACK_NONE = 0U	Normal operation.
	ETHTRCV_PHYLOOPBACK_INTERNAL	Internal loopback.
	ETHTRCV_PHYLOOPBACK_EXTERNAL	External loopback.
	ETHTRCV_PHYLOOPBACK_REMOTE	Remote loopback.

2.4.2.14 EthTrcv_PhyTxModeType

Enumeration	EthTrcv_PhyTxModeType	
Description	Describes the possible PHY transmit modes.	
Values	Value	Description
	ETHTRCV_PHYTXMODE_NORMAL = 0U	Normal operation.
	ETHTRCV_PHYTXMODE_TX_OFF	Transmitter disabled.
	ETHTRCV_PHYTXMODE_SCRAMBLER_OFF	Scrambler disabled.

2.4.2.15 EthTrcv_CableDiagResultType

Enumeration	EthTrcv_CableDiagResultType	
Description	Describes the results of the cable diagnostics.	
Values	Value	Description
	ETHTRCV_CABLEDIAG_OK = 0U	Cable diagnostic ok.
	ETHTRCV_CABLEDIAG_ERROR	Cable diagnostic failed.
	ETHTRCV_CABLEDIAG_SHORT	Short circuit detected.
	ETHTRCV_CABLEDIAG_OPEN	Open circuit detected.
	ETHTRCV_CABLEDIAG_PENDING	Cable diagnostic is still running.
	ETHTRCV_CABLEDIAG_WRONG_POLARITY	Cable diagnostics has detected wrong polarity of the "Ethernet physical+" or "Ethernet physical-" lines.

2.4.2.16 EthTrcv_MacMethodType

Enumeration	EthTrcv_MacMethodType	
Description	This type defines the media access control type in half duplex.	
Values	Value	Description
	ETHTRCV_MAC_TYPE_CSMA_CD = 0U	Carrier sense multiple access with collision detection.
	ETHTRCV_MAC_TYPE_PLCA	Physical layer collision avoidance.

2.4.2.17 EthSwt_StateType

Enumeration	EthSwt_StateType	
Description	Status supervision used for Development Error Detection.	
Values	Value	Description
	ETHSWT_STATE_UNINIT = 0U	The Switch is not yet configured.
	ETHSWT_STATE_INIT	The Switch is configured.
	ETHSWT_STATE_PORTINIT_COMPLETED	Port initialization is completed.
	ETHSWT_STATE_ACTIVE	The Switch is active.

2.4.2.18 EthSwt_MacLearningType

Enumeration	EthSwt_MacLearningType	
Description	This type defines the MAC addresses learning modes.	
Values	Value	Description
	ETHSWT_MACLEARNING_HWDISABLED = 0U	If hardware learning disabled, the switch must not learn new MAC addresses.
	ETHSWT_MACLEARNING_HWENABLED	If hardware learning enabled, the switch learns new MAC addresses.
	ETHSWT_MACLEARNING_SWENABLED	If software learning enabled, the hardware learning is disabled and the switch forwards packets with an unknown source address to a host CPU.

2.4.2.19 EthSwt_PortMirrorStateType

Enumeration	EthSwt_PortMirrorStateType	
Description	It shall be possible to set and get the mirror state (enable / disable) of the ethernet switch port.	
Values	Value	Description
	PORT_MIRROR_DISABLED = 0U	Port mirroring disabled.
	PORT_MIRROR_ENABLED	Port mirroring enabled.
	PORT_MIRRORING_CONFIGURATION_NOT_SUPPORTED	Port mirroring configuration is not supported by Ethernet switch driver or by the Ethernet switch hardware.

2.4.2.20 EthSwt_MgmtOwner

Enumeration	EthSwt_MgmtOwner	
Description	Holds information if upper layer or EthSwt is owner of mgmt_obj.	
Values	Value	Description
	ETHSWT_MGMT_OBJ_UNUSED = 0U	Object unused.
	ETHSWT_MGMT_OBJ_OWNED_BY_ETHSWT	Object used and EthSwt collects needed data.
	ETHSWT_MGMT_OBJ_OWNED_BY_UPPER_LAYER	Object used and the upper layer does calculations.

2.4.3 Enumerations in Eth_Hw_Types.h

2.4.3.1 ENET_MAC_StatusType

Enumeration	ENET_MAC_StatusType	
Description	ENE MAC status type definition.	
Values	Value	Description
	ENET_MAC_STATUS_IDLE = 0U	MAC is idle.
	ENET_MAC_STATUS_BUSY	MAC is busy.

2.4.3.2 ENET_MAC_HashOrPerfectModeType

Enumeration	ENET_MAC_HashOrPerfectModeType	
Description	MAC perfect or hash filter mode type definition.	
Values	Value	Description
	ENET_MAC_USE_PERFECT_FILTER = 0U	MAC use address perfect match filter.
	ENET_MAC_USE_HASH_FILTER	MAC use address hash match filter.

2.4.3.3 ENET_MTL_QueueEnableType

Enumeration	ENET_MTL_QueueEnableType	
Description	MTL queue enable mode type.	
Values	Value	Description
	ENET_MTL_QUEUE_DISABLE = 0U	MTL queue is disabled.
	ENET_MTL_QUEUE_ENABLE_AVB	MTL queue is enabled as AVB mode.
	ENET_MTL_QUEUE_ENABLE_GENERIC	MTL queue is enabled as generic mode.

2.4.3.4 ENET_MAC_MiiModeType

Enumeration	ENET_MAC_MiiModeType	
Description	MAC PHY interface mode type.	
Values	Value	Description
	ENET_MAC_PHY_MII = 0U	MII mode.
	ENET_MAC_PHY_RGMII = 1U	RGMII mode.
	ENET_MAC_PHY_RMII = 4U	RMII mode.

2.4.3.5 ENET_MAC_CSRClockRangeType

Enumeration	ENET_MAC_CSRClockRangeType	
Description	MAC MDIO access clock source range type.	
Values	Value	Description
	ENET_MAC_CSR_CLOCK_RANGE_60_100MHZ = 0U	CSR clock range 60-100MHz.
	ENET_MAC_CSR_CLOCK_RANGE_100_150MHZ	CSR clock range 100-150MHz.
	ENET_MAC_CSR_CLOCK_RANGE_20_35MHZ	CSR clock range 20-35MHz.
	ENET_MAC_CSR_CLOCK_RANGE_35_60MHZ	CSR clock range 35-60MHz.
	ENET_MAC_CSR_CLOCK_RANGE_150_250MHZ	CSR clock range 150-250MHz.
	ENET_MAC_CSR_CLOCK_RANGE_250_300MHZ	CSR clock range 250-300MHz.

2.4.3.6 ENET_MAC_TimestampRolloverType

Enumeration	ENET_MAC_TimestampRolloverType	
Description	PTP counter rollover mode type.	
Values	Value	Description
	ENET_MAC_TIMESTAMP_ROLLOVER_BINARY = 0U	PTP time nanoseconds binary rollover.
	ENET_MAC_TIMESTAMP_ROLLOVER_DIGITAL = 1U	PTP time nanoseconds digital rollover.

2.4.3.7 ENET_MAC_TimestampUpdateMethodType

Enumeration	ENET_MAC_TimestampUpdateMethodType	
Description	PTP timestamp update mode type.	
Values	Value	Description
	ENET_MAC_TIMESTAMP_UPDATE_COARSE = 0U	PTP coarse update mode.
	ENET_MAC_TIMESTAMP_UPDATE_FINE	PTP fine update mode.

2.4.3.8 ENET_MAC_SystemTimeOperationType

Enumeration	ENET_MAC_SystemTimeOperationType	
Description	PTP system time offset update operation type.	
Values	Value	Description
	ENET_MAC_SYSTEM_TIME_ADD_TIME = 0U	Add to current system time.
	ENET_MAC_SYSTEM_TIME_SUBTRACT_TIME	Subtrace from current system time.

2.4.3.9 ENET_MTL_QueueType

Enumeration	ENET_MTL_QueueType	
Description	ENET MTL queue type.	
Values	Value	Description
	ENET_MTL_QUEUE_0 = 0U	ENET MTL queue 0.
	ENET_MTL_QUEUE_1	ENET MTL queue 1.

2.4.3.10 ENET_MTL_TxThresholdType

Enumeration	ENET_MTL_TxThresholdType	
Description	MTL TX threshold type.	
Values	Value	Description
	ENET_MTL_TX_THRESHOLD_32B = 0U	N/A
	ENET_MTL_TX_THRESHOLD_64B = 1U	N/A
	ENET_MTL_TX_THRESHOLD_96B = 2U	N/A
	ENET_MTL_TX_THRESHOLD_128B = 3U	N/A
	ENET_MTL_TX_THRESHOLD_192B = 4U	N/A
	ENET_MTL_TX_THRESHOLD_256B = 5U	N/A
	ENET_MTL_TX_STORE_FORWARD = (1U << ENET_MTL_TXQ0_OPERATION_MODE_TTC_WIDTH)	N/A

2.4.3.11 ENET_MTL_RxFlowCtrlThresholdType

Enumeration	ENET_MTL_RxFlowCtrlThresholdType	
Description	MTL RX flow control threshold type.	
Values	Value	Description
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_1_KB = 0U	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_1_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_2_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_2_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_3_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_3_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_4_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_4_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_5_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_6_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_6_5_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_7_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_7_5_KB	N/A

	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_8_KB	N/A
	ENET_MTL_RX_FLOWCTRL_THRESHOLD_FULL_8_5_KB	N/A

2.4.3.12 ENET_MTL_RxThresholdType

Enumeration	ENET_MTL_RxThresholdType	
Description	MTL RX threshold type.	
Values	Value	Description
	ENET_MTL_RX_THRESHOLD_64B = 0U	N/A
	ENET_MTL_RX_THRESHOLD_32B = 1U	N/A
	ENET_MTL_RX_THRESHOLD_96B = 2U	N/A
	ENET_MTL_RX_THRESHOLD_128B = 3U	N/A
	ENET_MTL_RX_STORE_FORWARD = (1U << ENET_MTL_RXQ0_OPERATION_MODE_RTC_WIDTH)	N/A

2.4.3.13 ENET_MTL_GclRegistersType

Enumeration	ENET_MTL_GclRegistersType	
Description	Gcl related registers type.	
Values	Value	Description
	ENET_MTL_GCL_BTR_LOW = 0x0U	GCL Base Time Register Low 32 bits.
	ENET_MTL_GCL_BTR_HIGH = 0x1U	GCL Base Time Register High 32 bits.
	ENET_MTL_GCL_CTR_LOW = 0x2U	GCL Cycle Time Low 32 bits.
	ENET_MTL_GCL_CTR_HIGH = 0x3U	GCL Cycle Time High 8 bits.
	ENET_MTL_GCL_TER = 0x4U	GCL Time Extension.
	ENET_MTL_GCL_LLRL = 0x5U	Gate Control List Length.

2.4.3.14 ENET_MTL_GclOperationType

Enumeration	ENET_MTL_GclOperationType	
Description	Gate Control List operation type.	
Values	Value	Description
	ENET_MTL_GCL_WRITE = 0x0U	GCL write operation.
	ENET_MTL_GCL_READ = 0x1U	GCL read operation.

2.4.3.15 ENET_DMA_ChannelType

Enumeration	ENET_DMA_ChannelType	
Description	ENET DMA channel index type.	
Values	Value	Description
	ENET_DMA_CHANNEL_0 = 0U	DMA channel 0.
	ENET_DMA_CHANNEL_1	DMA channel 1.

2.4.3.16 ENET_DMA_StatusType

Enumeration	ENET_DMA_StatusType	
Description	ENET DMA running status.	
Values	Value	Description

	ENET_DMA_STATUS_IDLE = 0U	The ENET DMA engine is stopped..
	ENET_DMA_STATUS_ACTIVE	The ENET DMA engine is running.

2.4.3.17 ENET_DMA_IntModeType

Enumeration	ENET_DMA_IntModeType	
Description	The interrupt mode of ENET DMA.	
Values	Value	Description
	ENET_DMA_INTM_SBD_PERCH_PULSE = 0U	sbd_perch_* are pulse signals and sbd_inter_o will assert.
	ENET_DMA_INTM_SBD_PERCH_LEVEL	sbd_perch_* are level signals and sbd_inter_o will not assert.
	ENET_DMA_INTM_SBD_PERCH_LEVEL_REENFRANT	sbd_perch_* are level signals and will regenerate if the same event occur again before the signals cleared. sbd_inter_o will not assert.

2.4.3.18 ENET_DMA_InterruptFlagType

Enumeration	ENET_DMA_InterruptFlagType	
Description	ENET DMA interrupt flag type.	
Values	Value	Description
	ENET_DMA_TX_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_TIE_MASK	DMA TX interrupt.
	ENET_DMA_TX_STOPPED_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_TXSE_MASK	DMA TX stopped interrupt.
	ENET_DMA_TX_BUFFER_UNAVAILABLE_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_TBUE_MASK	DMA TX buffer unavailable interrupt.
	ENET_DMA_RX_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_RIE_MASK	DMA RX interrupt.
	ENET_DMA_RX_BUFFER_UNAVAILABLE_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_RBUE_MASK	DMA RX buffer unavailable interrupt.
	ENET_DMA_RX_STOPPED_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_RSE_MASK	DMA RX stopped interrupt.
	ENET_DMA_RX_WATCHDOG_TIMEOUT_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_RWTE_MASK	DMA RX watchdog timeout interrupt.
	ENET_DMA_EARLY_TX_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_ETIE_MASK	DMA early TX interrupt.
	ENET_DMA_EARLY_RX_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_ERIE_MASK	DMA early RX interrupt.

	K	
	ENET_DMA_FATAL_BUS_ERROR_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_FBEE_MASK K	DMA fatal bus error interrupt.
	ENET_DMA_CONTEXT_DESCRIPTOR_ERROR_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_CDEE_MASK	DMA context descriptor error interrupt.
	ENET_DMA_ABNORMAL_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_AIE_MASK	DMA abnormal interrupt.
	ENET_DMA_NORMAL_INTERRUPT = ENET_DMA_CH0_INTERRUPT_ENABLE_NIE_MASK K	DMA normal interrupt.

2.5 Typedefs

2.5.1 Typedefs in Eth_GeneralTypes.h

- typedef uint16 **Eth_FrameType**
This type is used to pass the uint16 value of type or length field in the Ethernet frame header.
- typedef uint8 **Eth_DataType**
Type used to pass transmit or receive data to or from the driver.
- typedef uint32 **Eth_BufIdxType**
Indicate the index of buffer used in transmitted and received data.

2.5.2 Typedefs in Eth_Types.h

- typedef void (***Eth_ReadMiiCbkType**)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal)
This type is used to save the pointer to the EthTrcv_ReadMiiIndication function.
- typedef void (***Eth_WriteMiiCbkType**)(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx)
This type is used to save the pointer to the EthTrcv_WriteMiiIndication function.

2.6 Structures

2.6.1 Eth_ConfigType

Structure	Eth_ConfigType
Description	Eth configuration type. This is the type of the data structure including the configuration set required for initializing the Eth driver.

Diagram	
Data Fields	• uint8 NumController The number of enabled controller. • const Eth_CtrlCfgType *CtrlPtr The controllers configurations.

2.6.2 Eth_CtrlCfgType

Structure	Eth_CtrlCfgType
Description	Implementation specific structure of the controller configuration.
Diagram	
Data Fields	• uint32 BusClkFreq Eth interface bus clock. • uint8 EcucPartition The referenced ECUC partition ID. • uint8 EthIfIdx Index of the controller within the context of the Ethernet Interface. • boolean EnableMii Enable MII clause 22 access. • boolean EnableMmd Enable MII clause 45 access. • Eth_ReadMiiCbkJType ReadMiiCbkJ EthTrcv_ReadMiiIndication. • Eth_WriteMiiCbkJType WriteMiiCbkJ EthTrcv_WriteMiiIndication. • boolean TxInterruptEn Enable TX interrupt. • boolean RxInterruptEn Enable RX interrupt • uint8 EgressFifoCount

	TX FIFO rings count. • uint8 IngressFifoCount RX FIFO rings count. • const Eth_FifoConfigType *EgressFifoCfgPtr Egress FIFO configuration. • const Eth_FifoConfigType *IngressFifoCfgPtr Ingress FIFO configuration. • Eth_PhylfType Phylf PHY interface. • uint8 MacAddress[ETH_PHYS_ADDR_LEN_BYTE] MAC address. • uint32 MacConfiguration; MAC configuration register value. • uint32 MacExtConfiguration MAC extended config register. • uint32 MacFilterRegValue MAC packet filter register value. • uint32 DmaMode DMA mode register value. • uint32 DmaSysBusMode DMA sysbus mode register value. • const Eth_DMACHConfigType *DMAChannelCfgPtr DMA channel configurations. • uint32 MtlOperationMode MTL operation mode register. • const Eth_MTLQueueCfgType *MTLQueueCfgPtr MTL queues configurations. • boolean MMCResetOnRead Counters reset to 0 after read. • boolean MMCStopRollover Counters does not roll over to zero after reaching the maximum value. • uint32 PtpCtrl PTP control register. • Eth_TimeStampType InitTime PTP initialize time. • uint32 SubSecondInc MAC_Sub_Second_Increment value. • uint32 TimestampAddend MAC_Timestamp_Addend value. • uint32 EgressCorrNanoseconds Egress_Corr_Nanosecond value. • uint32 IngressCorrNanoseconds Ingress_Corr_Nanosecond value. • const Eth_TimeAwareShaperType *TimeAwareShaperPtr Time aware shaper. • boolean SafetyEccEn ECC Error Protection. • boolean SafetyFsmParityEn
--	---

	FSM parity protection. - boolean SafetyFsmTimeoutEn FSM time out protection. - Eth_MTLDPpModeType SafetyDppMode Data Path Parity Protection. - Mcal_DemErrorType Cfg_Dem_ESafety DEM safety error. - Mcal_DemErrorType Cfg_Dem_EAccess DEM access error. - Mcal_DemErrorType Cfg_Dem_EAlignment DEM alignment error. - Mcal_DemErrorType Cfg_Dem_ECrc DEM CRC error. - Mcal_DemErrorType Cfg_Dem_ELateCollision DEM late collision. - Mcal_DemErrorType Cfg_Dem_EMultipleCollision DEM multiple collision. - Mcal_DemErrorType Cfg_Dem_EOverSizeFrame DEM over size frame. - Mcal_DemErrorType Cfg_Dem_ERxFramesLost DEM RX frames lost. - Mcal_DemErrorType Cfg_Dem_ESingleCollision DEM single collision. - Mcal_DemErrorType Cfg_Dem_EUnderSizeFrame DEM under size frame.
--	---

2.6.3 Eth_TimeAwareShaperType

Structure	Eth_TimeAwareShaperType
Description	ETH Time Aware Shaper configuration type.
Diagram	<pre> classDiagram class Eth_TimeAwareShaperType class Eth_TimeAwareShaperType_struct class Eth_GateControlType class Eth_GateControlType_struct Eth_TimeAwareShaperType --> Eth_TimeAwareShaperType_struct Eth_GateControlType --> Eth_GateControlType_struct Eth_TimeAwareShaperType ..> Eth_GateControlType : GateControlList </pre>
Data Fields	- uint32 BaseTimeSeconds The upper 32 bits of the start time to execute Gate Control. - uint32 BaseTimeNanoseconds The lower 32 bits of the start time to execute Gate Control. - uint16 GateControlListLength The number of items in the Gate Control list. - uint16 CycleTimeSeconds

	The upper 8 bits of the cycle time to repeat Gate Control list. - uint32 CycleTimeNanoseconds The lower 32 bits of the cycle time to repeat Gate Control list. - uint32 TimeExtension The time in nanoseconds the current Gate Control List can be extended before switching to the new List. - boolean EnablePreemption Enable frame preemption. - uint8 PreemptionClassification Indicates the Queues be classified as preemptable or express. - uint16 ReleaseAdvanceTime The maximum time between issuing a RELEASE and the MAC being ready to resume transmission of preemptable frames. - uint16 HoldAdvanceTime The maximum time between issuing a HOLD and the MAC ceasing to transmit any preemptable frame. - const Eth_GateControlType *GateControlList The Gate Control list items.
--	---

2.6.4 Eth_GateControlType

Structure	Eth_GateControlType
Description	ETH Gate Control item type of the Gate Control List.
Diagram	N/A
Data Fields	- uint8 GateControl The Gate Control state. - uint32 TimeInterval The Gate Control time interval.

2.6.5 Eth_MTLQueueCfgType

Structure	Eth_MTLQueueCfgType
Description	ETH MTL queue configuration type.
Diagram	N/A
Data Fields	- Eth_MTLQueueEnableType TxQueueMode MTL TX queue mode. - uint32 TxQueueWeights MTL TX queue weights,max 0x64. - uint32 IdleSlopeCredit Idle slope credit for AVB queues - uint32 SendSlopeCredit Send slope credit for AVB queues - uint32 HiCredit High credit limit for AVB queues - sint32 LoCredit Low credit limit for AVB queues - Eth_MTLQueueEnableType RxQueueMode MTL TX queue mode . - boolean ForwardTCPIPChecksumErr

	Forward TCP checksum error packets. • boolean ForwardErrorPackets Forward error packets. • boolean ForwardUndersizedGood Forward undersized good packets. • uint8 RxQueueWeight RX queue Weight.
--	---

2.6.6 Eth_FifoConfigType

Structure	Eth_FifoConfigType
Description	ETH FIFO configuration type.
Diagram	<pre> graph TD A[Eth_FifoConfigType_struct] --> B[Eth_FifoConfigType] B -.-> DescListAddress C[Eth_DMADescType] </pre>
Data Fields	• uint16 FifoBufLenByte FIFO buffer length. • uint16 FifoBufTotal FIFO total count. • Eth_DMADescType *DescListAddress Descriptor start address, length = sizeof(Eth_DMADescType) * FifoBufTotal. • uint8 *FifoBufStartAddress FIFO buf start address, length = FifoBufLenByte * FifoBufTotal. • uint8 PriorityMask Priority mask of the FIFO.

2.6.7 Eth_DMACHConfigType

Structure	Eth_DMACHConfigType
Description	ETH DMA channel configuration type.
Diagram	N/A
Data Fields	• uint32 Control DMA channel control register value. • uint32 TxControl DMA TX channel control register value. • uint32 RxControl DMA RX channel control register value.

2.6.8 Eth_DMADescType

Structure	Eth_DMADescType
Description	DMA descriptor type.
Diagram	N/A
Data Fields	• __IO uint32 Des0 DMA Descriptor word 0.

- __IO uint32 Des1
DMA Descriptor word 1.
- __IO uint32 Des2
DMA Descriptor word 2.
- __IO uint32 Des3
DMA Descriptor word 3.
- uint32 Backup0
DMA Descriptor backup data0.
- uint32 Backup1
DMA Descriptor backup data1.

2.6.9 Eth_HandleType

Structure	Eth_HandleType
Description	ETH Hal driver handle type.
Diagram	<pre> graph TD Eth_HandleType -- RxState, TxState --> Eth_FifoStateType Eth_HandleType -- TxBufIdxMap --> Eth_BufIdxMap Eth_FifoStateType -- RxInfo --> Eth_RxInfoType Eth_FifoStateType -- FirstDesc, LastDesc, DescList --> Eth_DMADescType Eth_BufIdxMap -- DescPtr --> Eth_DMADescType Eth_BufIdxMap -- NextPtr --> Eth_BufIdxMap Eth_RxInfoType -- DescPtr --> Eth_DMADescType </pre>
Data Fields	• uint8 TxRingCnt Enabled TX channel. • uint8 RxRingCnt Enabled RX channel. • Eth_FifoStateType TxState[ETH_FEATURE_MAX_CHANNEL_COUNT] TX Descriptors state. • Eth_FifoStateType RxState[ETH_FEATURE_MAX_CHANNEL_COUNT] RX Descriptors state. • Eth_BufIdxMapType TxBufIdxMap[ETH_MAX_BUF_IDX_SIZE] TX buffer ID map. • uint32 MaxTxBufCount Max TX buffers count, constant after Eth_Init.

2.6.10 Eth_FifoStateType

Structure	Eth_FifoStateType
Description	Eth FIFO state type.

Diagram	<pre> graph TD Eth_FifoStateType -- RxInfo --> Eth_RxInfoType Eth_FifoStateType -- FirstDesc, LastDesc, DescList --> Eth_DMADescType Eth_FifoStateType -- Head, Tail --> Eth_BufIdxMap Eth_DMADescType -- DescPtr --> Eth_BufIdxMap Eth_BufIdxMap -- NextPtr --> Eth_BufIdxMap </pre>
Data Fields	• <code>Eth_DMADescType *DescList</code> DMA descriptor list start address. • <code>uint16 BufferLen</code> DMA descriptor buffer length. • <code>uint16 RingSize</code> Number of DMA descriptors in <code>DescList</code>. • <code>uint32 BufIdxStart</code> Start <code>BufIdx</code> of this FIFO. • <code>uint32 BufIdxCurr</code> Current index of this FIFO. • <code>Eth_DMADescType *FirstDesc</code> First descriptor used by ETH driver. • <code>Eth_DMADescType *LastDesc</code> Last descriptor used by ETH driver. • <code>Eth_BufIdxMapType *Head</code> Head of transmit queue. • <code>Eth_BufIdxMapType *Tail</code> Tail of transmit queue. • <code>Eth_RxInfoType RxInfo</code> Information of the last received frame.

2.6.11 Eth_BufIdxMapType

Structure	<code>Eth_BufIdxMapType</code>
Description	ETH <code>BufIdx</code> map.
Diagram	<pre> graph TD Eth_BufIdxMap -- DescPtr --> Eth_DMADescType Eth_BufIdxMap -- NextPtr --> Eth_BufIdxMap </pre>
Data Fields	• <code>uint8 *BufferPtr</code> TX buffers bound to <code>BufIdx</code>, constant after <code>Eth_Init</code>. • <code>uint32 BufIdx</code> TX buffers index, constant after <code>Eth_Init</code>. • <code>uint8 FifIdx</code>

	Fifoldx for BufIdx, constant after Eth_Init. • boolean Locked TX Buffer locked by driver. • boolean TxConfirmation confirm transmit state or not. • uint8 TimeOut confirm transmit timeout. • boolean TsEnable Enable TX time stamping. • boolean TsValid Time stamp valid flag. • uint16 TimeStampSecHi Time stamp seconds high 16 bits. • uint32 TimeStampSec Time stamp seconds low 32 bits. • uint32 TimeStampNs Time stamp nanoseconds. • Eth_DMADescType *DescPtr Buffer related DMA descriptor. • struct Eth_BufIdxMap *NextPtr Next Eth_BufIdxMapType structure.
--	--

2.6.12 Eth_RxInfoType

Structure	Eth_RxInfoType
Description	Eth received frame information type.
Diagram	N/A
Data Fields	• uint8 *DataPtr Associated data pointer • uint32 Nanoseconds Nanoseconds part of the timestamp • uint32 Seconds Lower 32 bits of the 48 bits Seconds field • uint16 SecondsHi Upper 16 bits of the 48 bits Seconds field • boolean TimeStampValid Time stamp valid flag.

2.6.13 Eth_DrvBufType

Structure	Eth_DrvBufType
Description	Eth driver buffer type.
Diagram	N/A
Data Fields	• uint8 *DataPtr Payload data of the buffer. • uint16 Length Data length of current buffer. • uint16 TotalLen Total length of the buffer chain, TotalLen is only valid for the first buffer.

- struct Eth_DrvBuf *NextPtr
Pointer of next buffer.

2.6.14 EthDemEventStatsType

Structure	EthDemEventStatsType
Description	Eth dem event stats type.
Diagram	N/A
Data Fields	• uint32 RxFramesLost • uint32 RxCrcError • uint32 RxUndersizeError • uint32 RxOversizeError • uint32 RxAlignmentError • uint32 TxSingleCollisionError • uint32 TxMultipleCollisionError • uint32 TxLateCollisionError

2.6.15 Eth_TimeStampType

Structure	Eth_TimeStampType
Description	Timestamp value type.
Diagram	N/A
Data Fields	• uint32 nanoseconds Nanoseconds part of the timestamp. • uint32 seconds Lower 32 bits of the 48 bits Seconds field. • uint16 secondsHi Upper 16 bits of the 48 bits Seconds field.

2.6.16 Eth_TimeIntDiffType

Structure	Eth_TimeIntDiffType
Description	Variables of this type are used to express time differences.
Diagram	<pre> graph TD Eth_TimeIntDiffType[Eth_TimeIntDiffType] --> Eth_TimeStampType_struct[Eth_TimeStampType_struct] Eth_TimeIntDiffType -.-> diff Eth_TimeStampType[Eth_TimeStampType] Eth_TimeStampType --> Eth_TimeStampType_struct </pre>
Data Fields	• Eth_TimeStampType diff Time difference. • boolean sign Positive (True) Or negative (False) time.

2.6.17 Eth_RateRatioType

Structure	Eth_RateRatioType
Description	Variables of this type are used to express frequency ratios.
Diagram	<pre> graph TD Eth_RateRatioType[Eth_RateRatioType] Eth_RateRatioType_struct[Eth_RateRatioType_struct] Eth_TimeIntDiffType[Eth_TimeIntDiffType] Eth_TimeIntDiffType_struct[Eth_TimeIntDiffType_struct] Eth_TimeStampType[Eth_TimeStampType] Eth_TimeStampType_struct[Eth_TimeStampType_struct] Eth_RateRatioType --> Eth_RateRatioType_struct Eth_RateRatioType --> Eth_TimeIntDiffType Eth_TimeIntDiffType --> Eth_TimeIntDiffType_struct Eth_TimeIntDiffType --> Eth_TimeStampType Eth_TimeStampType --> Eth_TimeStampType_struct </pre>
Data Fields	- Eth_TimeIntDiffType IngressTimeStampDelta IngressTimeStampSync2 - IngressTimeStampSync1. - Eth_TimeIntDiffType OriginTimeStampDelta OriginTimeStampSync2[FUP2] - OriginTimeStampSync1[FUP1].

2.6.18 Eth_MacVlanType

Structure	Eth_MacVlanType
Description	Eth_MacVlanType type definition.
Diagram	N/A
Data Fields	- uint8 MacAddr[6U] Specifies the MAC address [0..255,0..255,0..255,0..255,0..255,0..255]. - uint16 VlanId Specifies the VLAN address 0..65535. - uint32 SwitchPort Specifies the ports of the switch as bit mask. (0x00000001->Port0,0x80000001->Port31+Port0).

2.6.19 Eth_CounterType

Structure	Eth_CounterType
Description	Counter values according to IETF RFC 1757, RFC 1643 and RFC 2233.
Diagram	N/A
Data Fields	- uint32 DropPktBufOverrun Dropped packets due to buffer overrun. - uint32 DropPktCrc Dropped packets due to CRC errors. - uint32 UndersizePkt Number of undersize packets which were less than 64 octets long (excluding framing bits, but including FCS octets) and were otherwise well formed. - uint32 OversizePkt

	Number of oversize packets which are longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed. • uint32 AlgnmtErr Number of alignment errors, i.e. packets which are received and are not an integral number of octets in length and do not pass the CRC. • uint32 SqeTestErr SQE test error according to IETF RFC1643 dot3StatsSQETestErrors. • uint32 DisclnbdPkt The number of inbound packets which were chosen to be discarded even though no errors had been detected to prevent their being deliverable to a higher-layer protocol. One possible reason for discarding such a packet could be to free up buffer space. • uint32 ErrlnbdPkt Total number of erroneous inbound packets. • uint32 DiscOtbdPkt The number of outbound packets which were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One possible reason for discarding such a packet could be to free up buffer space. • uint32 ErrOtbdPkt Total number of erroneous outbound packets. • uint32 SnglCollPkt Single collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by exactly one. • uint32 MultCollPkt Multiple collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by more than one collision. • uint32 DfrdPkt Number of deferred transmission: A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy. • uint32 LatCollPkt Number of late collisions: The number of times that a collision is detected on a particular interface later than 512 bittimes into the transmission of a packet. • uint32 HwDepCtr0 Hardware dependent counter value. • uint32 HwDepCtr1 Hardware dependent counter value. • uint32 HwDepCtr2 Hardware dependent counter value. • uint32 HwDepCtr3 Hardware dependent counter value.
--	---

2.6.20 Eth_RxStatsType

Structure	Eth_RxStatsType
Description	Reception statistics values according to IETF RFC 2819.
Diagram	N/A
Data Fields	• uint32 RxStatsDropEvents The total number of events in which packets were dropped by the probe due to lack of resources. • uint32 RxStatsOctets The total number of octets of data (including those in bad packets) received on the network

	(excluding framing bits but including FCS octets). • uint32 RxStatsPkts The total number of packets (including bad packets, broadcast packets, and multicast packets) received. • uint32 RxStatsBroadcastPkts The total number of good packets received that were directed to the broadcast address. • uint32 RxStatsMulticastPkts The total number of good packets received that were directed to a multicast address. • uint32 RxStatsCrcAlignErrors The total number of packets received that had a length of between 64 and 1518 octets that had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a non-integral number of octets (Alignment Error). • uint32 RxStatsUndersizePkts The total number of packets received that were less than 64 octets long (excluding framing bits, but including FCS octets) and were otherwise well formed. • uint32 RxStatsOversizePkts The total number of packets received that were longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed. • uint32 RxStatsFragments The total number of packets received that were less than 64 octets in length (excluding framing bits but including FCS octets) and had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a non-integral number of octets (Alignment Error). • uint32 RxStatsJabbers The total number of packets received that were longer than 1518 octets, and had either a bad Frame Check Sequence (FCS) with an integral number of octets (FCS Error) or a bad FCS with a nonintegral number of octets (Alignment Error). • uint32 RxStatsCollisions The best estimate of the total number of collisions on this Ethernet segment. • uint32 RxStatsPkts64Octets The total number of packets (including bad packets) received that were 64 octets in length. • uint32 RxStatsPkts65to127Octets The total number of packets (including bad packets) received that were between 65 and 127 octets in length. • uint32 RxStatsPkts128to255Octets The total number of packets (including bad packets) received that were between 128 and 255 octets in length. • uint32 RxStatsPkts256to511Octets The total number of packets (including bad packets) received that were between 256 and 511 octets in length. • uint32 RxStatsPkts512to1023Octets The total number of packets (including bad packets) received that were between 512 and 1023 octets in length. • uint32 RxStatsPkts1024to1518Octets The total number of packets (including bad packets) received that were between 1024 and 1518 octets in length. • uint32 RxUnicastFrames The number of subnetwork-unicast packets delivered to a higher-layer protocol.
--	---

2.6.21 Eth_TxStatsType

Structure	Eth_TxStatsType
Description	Statistic values for transmission.
Diagram	N/A
Data Fields	• uint32 TxNumberOfOctets The total number of octets transmitted out of the interface, including framing characters. • uint32 TxNUcastPkts The total number of packets that higherlevel protocols requested be transmitted to a non-unicast (i.e., a subnetworkbroadcast or subnetwork-multicast) address, including those that were discarded or not sent. • uint32 TxUniCastPkts The total number of packets that higherlevel protocols requested be transmitted to a subnetwork-unicast address, including those that were discarded or not sent.

2.6.22 Eth_TxErrorCounterValuesType

Structure	Eth_TxErrorCounterValuesType
Description	Statistics of error counter values for transmission.
Diagram	N/A
Data Fields	• uint32 TxDroppedNoErrorPkts The number of outbound packets which were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One possible reason for discarding such a packet could be to free up buffer space. • uint32 TxDroppedErrorPkts transmitted because of errors. • uint32 TxDeferredTrans A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy. The count represented by an instance of this object does not include frames involved in collisions. • uint32 TxSingleCollision A count of successfully transmitted frames on a particular interface for which transmission is inhibited by exactly one collision. A frame that is counted by an instance of this object is also counted by the corresponding instance of either the TxUniCastPkts and TxNUcastPkts and is not counted by the corresponding instance of the TxMultipleCollision object. • uint32 TxMultipleCollision A count of successfully transmitted frames on a particular interface for which transmission is inhibited by more than one collision. A frame that is counted by an instance of this object is also counted by the corresponding instance of either the TxUniCastPkts and TxNUcastPkts and is not counted by the corresponding instance of the TxSingleCollision object. • uint32 TxLateCollision The number of times that a collision is detected on a particular interface later than 512 bit-times into the transmission of a packet. Five hundred and twelve bit-times corresponds to 51.2 microseconds on a 10 Mbit/s system. A (late) collision included in a count represented by an instance of this object is also considered as a (generic) collision for purposes of other collision-related statistics. • uint32 TxExcessiveCollision A count of frames for which transmission on a particular interface fails due to excessive collisions.

2.6.23 EthSwt_MgmtInfoType

Structure	EthSwt_MgmtInfoType
Description	Type for holding the management information received/transmitted on Switches (ports).
Diagram	N/A
Data Fields	- uint8 SwitchIdx Switch index. - uint8 SwitchPortIdx Port index of the switch.

2.6.24 EthSwt_PortMirrorCfgType

Structure	EthSwt_PortMirrorCfgType
Description	Port mirroring configurations type.
Diagram	N/A
Data Fields	- uint8 srcMacAddrFilter[6U] Specifies the source MAC address [0..255,0..255,0..255,0..255,0..255,0..255] that should be mirrored.If set to 0,0,0,0,0,0, no source MAC address filtering shall take place. - uint8 dstMacAddrFilter[6U] Specifies the destination MAC address [0..255,0..255,0..255,0..255,0..255,0..255] that should be mirrored.If set to 0,0,0,0,0,0, no destination MAC address filtering shall take place. - uint16 VlanIdFilter Specifies the VLAN address 0..4094 that should be mirrored. If set to 65535, no VLAN filtering shall take place. - uint8 MirroringPacketDivider Divider if only a subset of received frames should be mirrored. E.g. MirroringPacketDivider = 2 means every second frames is mirrored. - uint8 MirroringMode Specifies the mode how the mirrored traffic should be tagged : 0x00 == No VLAN retagging; 0x01 == VLAN retagging;0x03 == VLAN Double tagging. - uint32 TrafficDirectionIngressBitMask Specifies the bit mask of Ethernet switch ingress port traffic direction to be mirrored. The bit mask is calculated depending of the values of Eth SwtPortIdx. (e.g. set EthSwtPortIdx == 2 => TrafficDirectionIngressBit Mask = 0b0000 0000 0000 0000 0000 0000 0000 0100). 0b0 == enable ingress port mirroring 0b1 == disable ingress port mirroring. Example: TrafficDirectionIngressBitMask = 0b0000 0000 0000 0000 0000 0000 0000 0100 => <u>Ingress</u> traffic mirroring is enabled of Ethernet switch port with EthSwtPortIdx=2. - uint32 TrafficDirectionEgressBitMask Specifies the bit mask of Ethernet switch egress port traffic direction to be mirrored. The bit mask is calculated depending of the values of Eth SwtPortIdx. (e.g. set EthSwtPortIdx == 2 => TrafficDirectionEgressBitMask Mask = 0b0000 0000 0000 0000 0000 0000 0000 0100). 0b0 == enable egress port mirroring 0b1 == disable egress port mirroring. Example: TrafficDirectionEgressBitMask = 0b0000 0000 0000 0000 0000 0000 0000 0001 => <u>Ingress</u> traffic mirroring is enabled of Ethernet switch port with EthSwtPortIdx=0. - uint8 CapturePortIdx Specifies the Ethernet switch port which capture the mirrored traffic. - uint16 ReTaggingVlanId Specifies the VLAN address 0..4094 which shall be used for re-tagging if MirroringMode is set to 0x01 (VLAN re-tagging). If the value is set to 65535, the value shall be ignored, because the VLAN

	address for re-tagging is provided by the Ethernet switch configuration. uint16 DoubleTaggingVlanId Specifies the VLAN address 0..4094 which shall be used for double-tagging if MirroringMode is set to 0x02 (VLAN double tagging).If the value is set to 65535, the value shall be ignored, because the VLAN address for double tagging is provided by the Ethernet switch configuration.
--	--

2.6.25 EthSwt_MgmtObjectValidType

Structure	EthSwt_MgmtObjectValidType
Description	Switch management object valid type. Will be set from EthSwt and marks EthSwt_MgmtObject as valid or not. So the upper layer will be able to detect inconsistencies.
Diagram	N/A
Data Fields	- Std_ReturnType IngressTimestampValid IngressTimestampValid shall be set to E_NOT_OK if ingress timestamp is not available. - Std_ReturnType EgressTimestampValid EgressTimestampValid shall be set to E_NOT_OK if egress timestamp is not available. - Std_ReturnType MgmtInfoValid MgmtInfoValid shall be set to E_NOT_OK if MgmtInfo is not available (e.g. timeout).

2.6.26 EthSwt_MgmtObjectType

Structure	EthSwt_MgmtObjectType
Description	Switch management object type. Provides information about all <u>struct</u> member elements. The ownership gives information whether EthSwt has finished its activities in providing all <u>struct</u> member elements.
Diagram	N/A
Data Fields	- EthSwt_MgmtObjectValidType Validation The validation information for the mgmt_obj. - Eth_TimeStampType IngressTimestamp The ingress timestamp value out of the switch. - Eth_TimeStampType EgressTimestamp The egress timestamp value out of the switch. - EthSwt_MgmtInfoType MgmtInfo Received/Transmitted Management information of the switches. - EthSwt_MgmtOwner Ownership The ownership of MgmtObj.

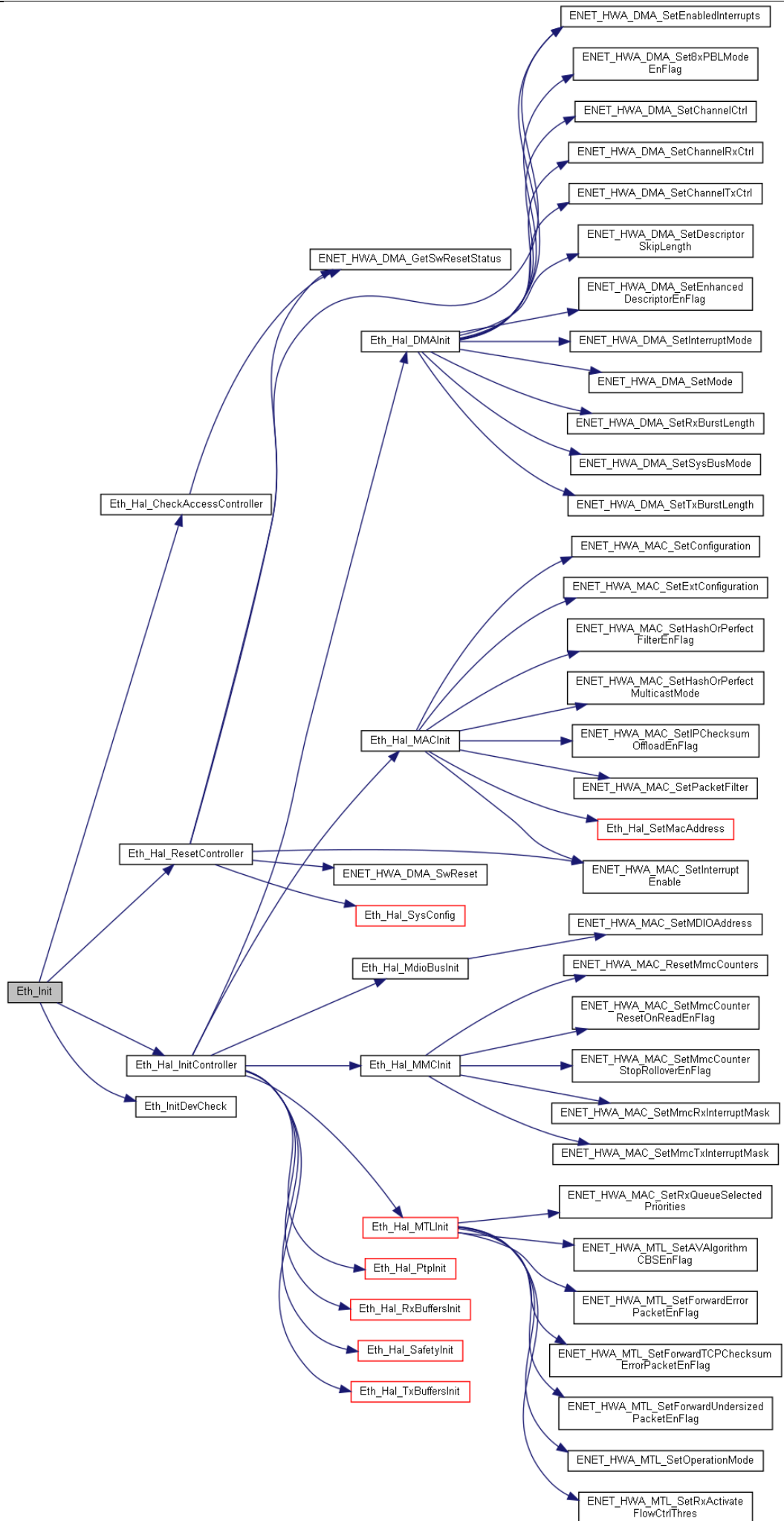
2.7 API Functions

2.7.1 Functions in Eth.h

2.7.1.1 void Eth_Init(const Eth_ConfigType* CfgPtr);

Function	void Eth_Init(const Eth_ConfigType* CfgPtr)
Description	This function is AUTOSAR standard API which is used to initialize ENET controller.

Diagram



Parameters	Parameter	Description
	CfgPtr	Points to the implementation specific structure.
Returns	N/A	

2.7.1.2 Std_ReturnType Eth_SetControllerMode(uint8 CtrlIdx, Eth_ModeType CtrlMode)

Function	Std_ReturnType Eth_SetControllerMode(uint8 CtrlIdx, Eth_ModeType CtrlMode)	
Description	Enables / Disables Rx/Tx communication of the indexed controller.	
Diagram	<pre> graph LR Eth_SetControllerMode --> Eth_EcucPartitionCheck Eth_SetControllerMode --> Eth_Hal_SetControllerMode Eth_SetControllerMode --> Eth_SetControllerModeDev_Check[Eth_SetControllerModeDev Check] Eth_Hal_SetControllerMode --> Eth_Hal_GetControllerMode Eth_Hal_SetControllerMode --> Eth_Hal_ReleaseTxBuffers Eth_Hal_SetControllerMode --> Eth_Hal_Start Eth_Hal_Start --> ENET_HWA_MAC_GetConfiguration Eth_Hal_Start --> ENET_HWA_DMA_ClearTxProcess_StoppedFlag[ENET_HWA_DMA_ClearTxProcess StoppedFlag] Eth_Hal_Start --> ENET_HWA_DMA_EnableInterrupt Eth_Hal_Start --> ENET_HWA_DMA_StartRx Eth_Hal_Start --> ENET_HWA_DMA_StartTx Eth_Hal_Start --> ENET_HWA_DMA_ClearRxProcess_StoppedFlag[ENET_HWA_DMA_ClearRxProcess StoppedFlag] Eth_Hal_Start --> ENET_HWA_MAC_SetRxEnFlag Eth_Hal_Start --> ENET_HWA_MAC_SetTxEnFlag Eth_Hal_Start --> ENET_HWA_MTL_FlushTxQueue Eth_Hal_Start --> ENET_HWA_DMA_SetRxPacket_FlushEnFlag[ENET_HWA_DMA_SetRxPacket FlushEnFlag] Eth_Hal_Start --> ENET_HWA_DMA_StopRx Eth_Hal_Start --> ENET_HWA_DMA_StopTx Eth_Hal_Start --> ENET_HWA_DMA_SetEnabledInterrupts Eth_SetControllerModeDev_Check --> Eth_Hal_Stop Eth_Hal_Stop --> ENET_HWA_MAC_SetRxEnFlag Eth_Hal_Stop --> ENET_HWA_MAC_SetTxEnFlag Eth_Hal_Stop --> ENET_HWA_MTL_FlushTxQueue Eth_Hal_Stop --> ENET_HWA_DMA_SetRxPacket_FlushEnFlag Eth_Hal_Stop --> ENET_HWA_DMA_StopRx Eth_Hal_Stop --> ENET_HWA_DMA_StopTx Eth_Hal_Stop --> ENET_HWA_DMA_SetEnabledInterrupts </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlMode	ETH_MODE_DOWN: Disable Rx/Tx communication of the Ethernet controller ETH_MODE_ACTIVE: Enable Rx/Tx communication of the Ethernet controller.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.3 Std_ReturnType Eth_GetControllerMode(uint8 CtrlIdx, Eth_ModeType* CtrlModePtr)

Function	Std_ReturnType Eth_GetControllerMode(uint8 CtrlIdx, Eth_ModeType* CtrlModePtr)	
Description	Obtains the state of the indexed controller.	
Diagram	<pre> graph LR Eth_GetControllerMode --> Eth_EcucPartitionCheck Eth_GetControllerMode --> Eth_GetControllerModeDev_Check[Eth_GetControllerModeDev Check] Eth_GetControllerMode --> Eth_Hal_GetControllerMode Eth_Hal_GetControllerMode --> ENET_HWA_MAC_GetConfiguration </pre>	
Parameters	Parameter	Description
	CtrlModePtr	Pointer to the Eth_ModeType variable to be updated with the controller mode.

	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlModePtr	ETH_MODE_DOWN: the controller is disabled. ETH_MODE_ACTIVE: the controller is enabled.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.4 void Eth_GetPhysAddr(uint8 CtrlIdx,uint8* PhysAddrPtr)

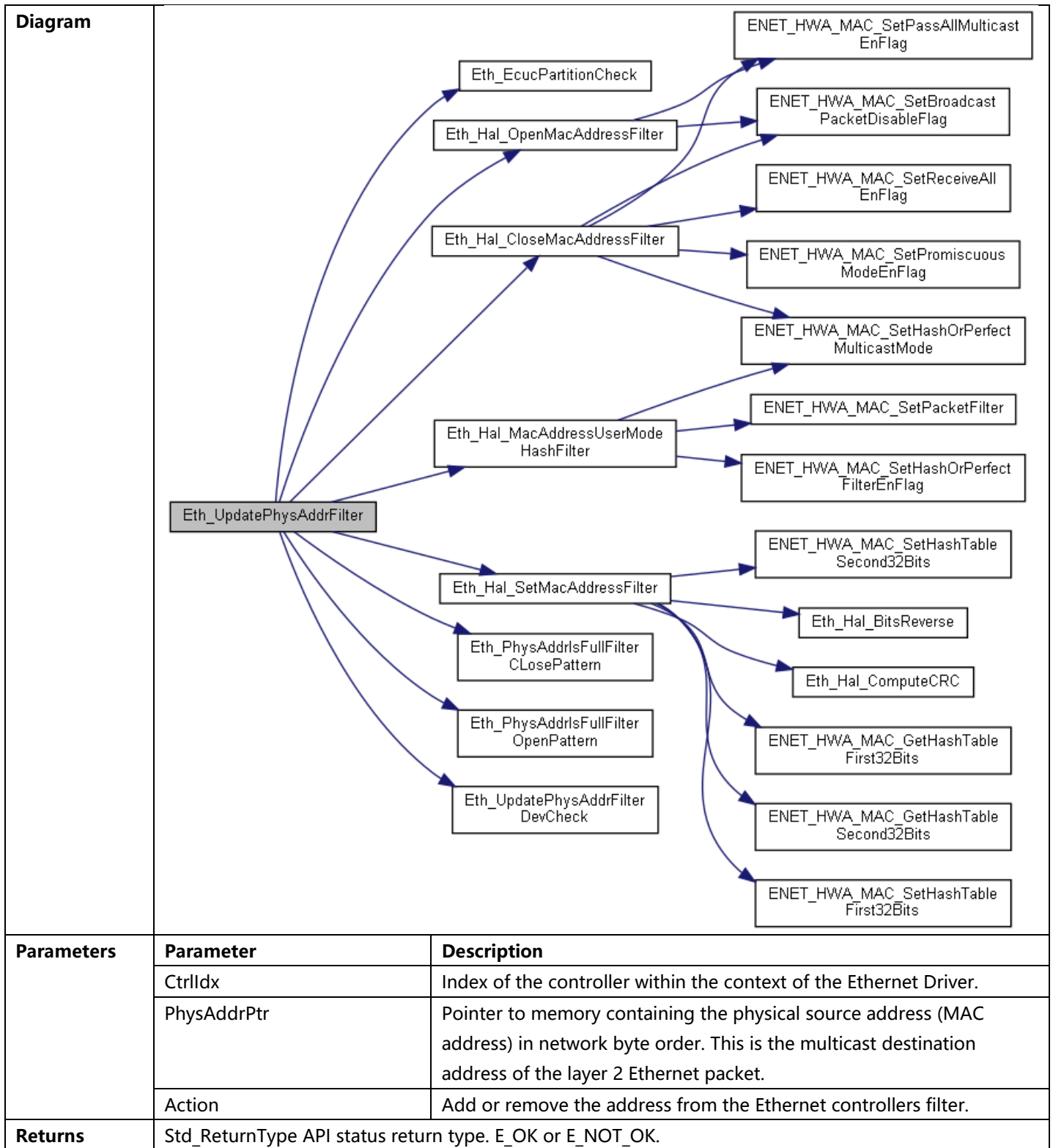
Function	void Eth_GetPhysAddr(uint8 CtrlIdx,uint8* PhysAddrPtr)	
Description	Obtains the physical source address used by the indexed controller.	
Diagram	<pre> graph LR Eth_GetPhysAddr --> Eth_EcuPartitionCheck Eth_GetPhysAddr --> Eth_GetPhysAddrDevCheck Eth_GetPhysAddr --> Eth_Hal_GetMacAddress Eth_Hal_GetMacAddress --> ENET_HWA_MAC_GetMacAddr0High Eth_Hal_GetMacAddress --> ENET_HWA_MAC_GetMacAddr0Low </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	PhysAddrPtr	Physical source address (MAC address) in network byte order.
Returns	N/A	

2.7.1.5 void Eth_SetPhysAddr(uint8 CtrlIdx,const uint8* PhysAddrPtr)

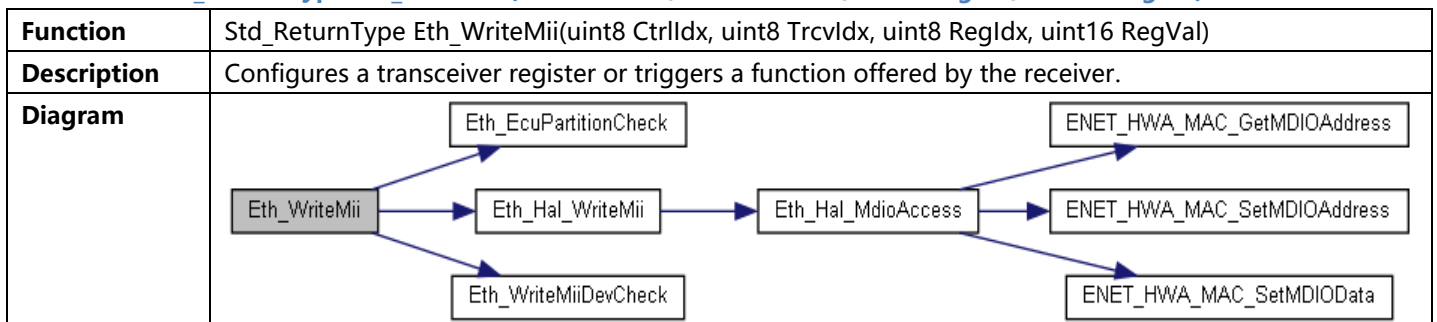
Function	void Eth_SetPhysAddr(uint8 CtrlIdx,const uint8* PhysAddrPtr)	
Description	Sets the physical source address used by the indexed controller.	
Diagram	<pre> graph LR Eth_SetPhysAddr --> Eth_EcuPartitionCheck Eth_SetPhysAddr --> Eth_Hal_SetMacAddress Eth_SetPhysAddr --> Eth_SetPhysAddrDevCheck Eth_Hal_SetMacAddress --> ENET_HWA_MAC_SetAddr0High Eth_Hal_SetMacAddress --> ENET_HWA_MAC_SetMacAddr0Low </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	PhysAddrPtr	Pointer to memory containing the physical source address (MAC address) in network byte order.
Returns	N/A	

2.7.1.6 Std_ReturnType Eth_UpdatePhysAddrFilter(uint8 CtrlIdx,const uint8* PhysAddrPtr,Eth_FilterActionType Action)

Function	Std_ReturnType Eth_UpdatePhysAddrFilter(uint8 CtrlIdx, const uint8* PhysAddrPtr, Eth_FilterActionType Action)
Description	Update the physical source address to/from the indexed controller filter. If the Ethernet Controller is not capable to do the filtering, the software has to do this.



2.7.1.7 Std_ReturnType Eth_WriteMii(uint8 CtrlIdx, uint8 TrcvIdx, uint8 RegIdx, uint16 RegVal)



Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver on the MII.
	RegIdx	Index of the transceiver register on the MII.
	RegVal	Value to be written into the indexed register .
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.8 Std_ReturnType Eth_ReadMii(uint8 CtrlIdx, uint8 TrcvIdx, uint8 RegIdx, uint16* RegValPtr)

Function	Std_ReturnType Eth_ReadMii(uint8 CtrlIdx, uint8 TrcvIdx, uint8 RegIdx, uint16* RegValPtr)	
Description	Reads a transceiver register.	
Diagram	<pre> graph LR Eth_ReadMii --> Eth_EcuPartitionCheck Eth_ReadMii --> Eth_Hal_ReadMii Eth_ReadMii --> Eth_ReadMiiDevCheck Eth_Hal_ReadMii --> ENET_HWA_MAC_GetMDIOData Eth_Hal_ReadMii --> ENET_HWA_MAC_GetMDIOAddress ENET_HWA_MAC_GetMDIOData --> Eth_Hal_MdioAccess ENET_HWA_MAC_GetMDIOAddress --> Eth_Hal_MdioAccess Eth_Hal_MdioAccess --> ENET_HWA_MAC_SetMDIOAddress Eth_Hal_MdioAccess --> ENET_HWA_MAC_SetMDIOData </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver on the MII.
	RegIdx	Index of the transceiver register on the MII.
	RegValPtr	Filled with the register content of the indexed register.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.9 Std_ReturnType Eth_WriteMmd(uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx, uint16 RegVal)

Function	Std_ReturnType Eth_WriteMmd(uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx, uint16 RegVal)	
Description	Configures a transceiver register via MII clause 45 protocol.	
Diagram	<pre> graph LR Eth_WriteMmd --> Eth_EcuPartitionCheck Eth_WriteMmd --> Eth_Hal_WriteMmd Eth_WriteMmd --> Eth_WriteMmdDevCheck Eth_Hal_WriteMmd --> ENET_HWA_MAC_GetMDIOAddress Eth_Hal_WriteMmd --> ENET_HWA_MAC_SetMDIOAddress ENET_HWA_MAC_GetMDIOAddress --> Eth_Hal_MdioAccess ENET_HWA_MAC_SetMDIOAddress --> Eth_Hal_MdioAccess Eth_Hal_MdioAccess --> ENET_HWA_MAC_SetMDIOData </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver on the MII.
	Mmd	(MMD)Device address.
	RegIdx	Index of the transceiver register on the MII.
	RegVal	Value to be written into the indexed register .
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

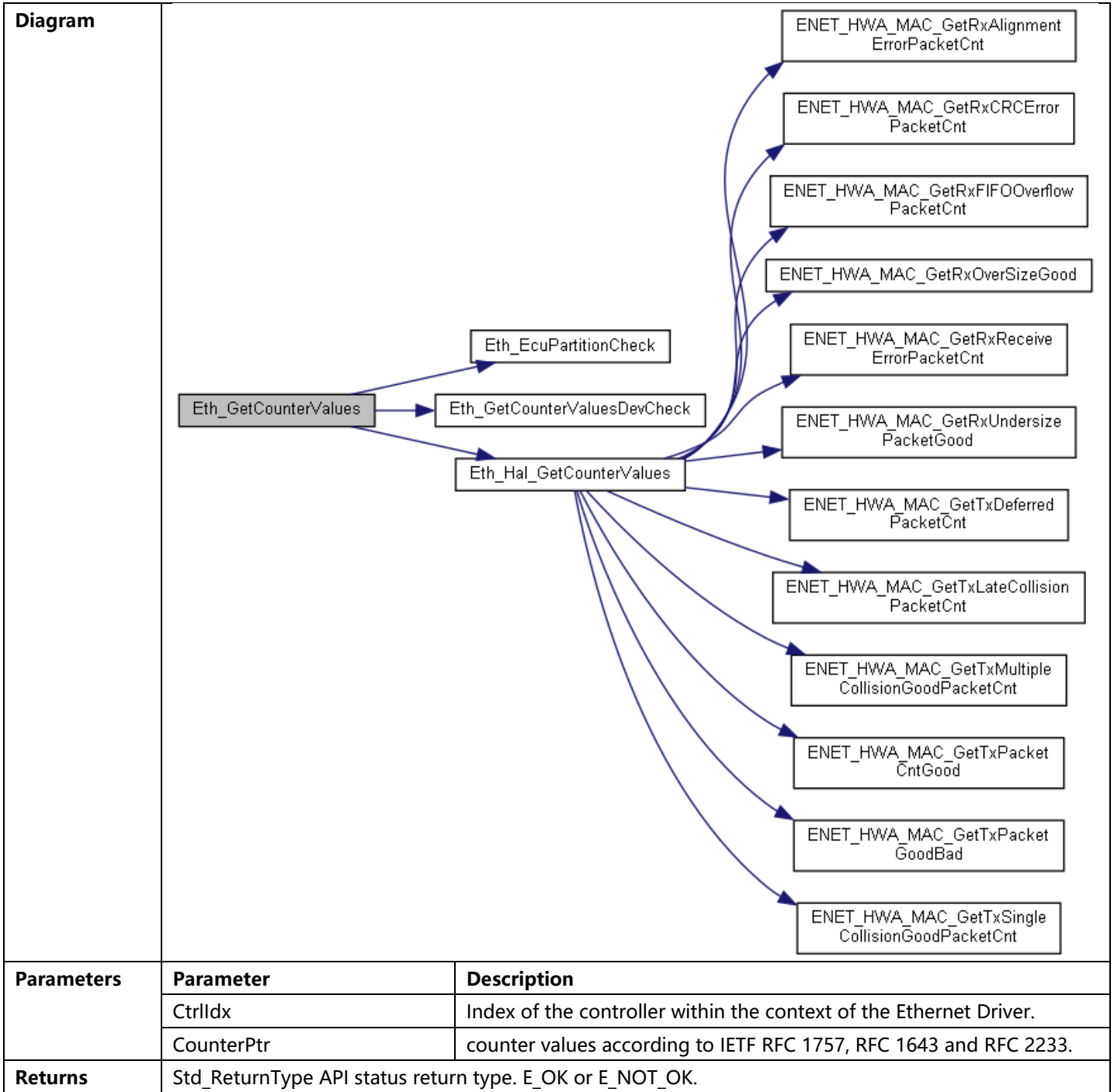
2.7.1.10 Std_ReturnType Eth_ReadMmd(uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd, uint16 RegIdx, uint16* RegValPtr)

Function	Std_ReturnType Eth_ReadMmd(uint8 CtrlIdx, uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx, uint16* RegValPtr)	
Description	Read a transceiver register via MII clause 45 protocol.	

Diagram	<pre>graph LR; Eth_ReadMmd[Eth_ReadMmd] --> Eth_EcucPartitionCheck[Eth_EcucPartitionCheck]; Eth_ReadMmd --> Eth_Hal_ReadMmd[Eth_Hal_ReadMmd]; Eth_ReadMmd --> Eth_ReadMmdDevCheck[Eth_ReadMmdDevCheck]; Eth_Hal_ReadMmd --> ENET_HWA_MAC_GetMDIOData[ENET_HWA_MAC_GetMDIOData]; Eth_Hal_ReadMmd --> Eth_Hal_MdioAccess[Eth_Hal_MdioAccess]; Eth_Hal_MdioAccess --> ENET_HWA_MAC_GetMDIOAddress[ENET_HWA_MAC_GetMDIOAddress]; Eth_Hal_MdioAccess --> ENET_HWA_MAC_SetMDIOAddress[ENET_HWA_MAC_SetMDIOAddress]; Eth_Hal_MdioAccess --> ENET_HWA_MAC_SetMDIOData[ENET_HWA_MAC_SetMDIOData];</pre>													
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>CtrlIdx</td><td>Index of the controller within the context of the Ethernet Driver.</td></tr><tr><td>TrcvIdx</td><td>Index of the transceiver on the MII.</td></tr><tr><td>Mmd</td><td>(MMD)Device address.</td></tr><tr><td>RegIdx</td><td>Index of the transceiver register on the MII.</td></tr><tr><td>RegValPtr</td><td>Filled with the register content of the indexed register.</td></tr></table>	Parameter	Description	CtrlIdx	Index of the controller within the context of the Ethernet Driver.	TrcvIdx	Index of the transceiver on the MII.	Mmd	(MMD)Device address.	RegIdx	Index of the transceiver register on the MII.	RegValPtr	Filled with the register content of the indexed register.	
Parameter	Description													
CtrlIdx	Index of the controller within the context of the Ethernet Driver.													
TrcvIdx	Index of the transceiver on the MII.													
Mmd	(MMD)Device address.													
RegIdx	Index of the transceiver register on the MII.													
RegValPtr	Filled with the register content of the indexed register.													
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.													

2.7.1.11 Std_ReturnType Eth_GetCounterValues(uint8 CtrlIdx, Eth_CounterType* CounterPtr)

Function	Std_ReturnType Eth_GetCounterValues(uint8 CtrlIdx, Eth_CounterType* CounterPtr)
Description	Reads a list with drop counter values of the corresponding controller. The meaning of these values is described at Eth_CounterType.

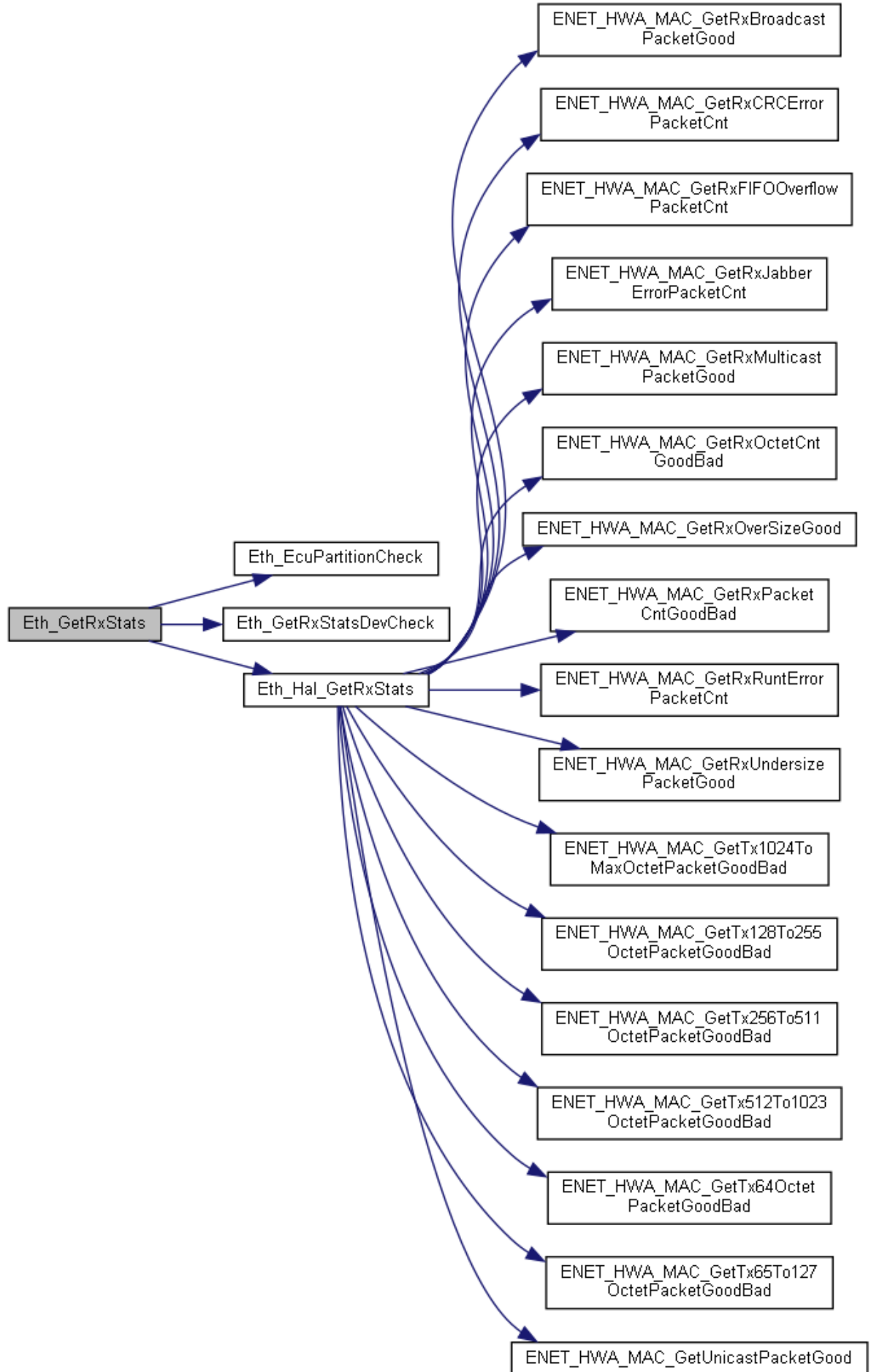


2.7.1.12 Std_ReturnType Eth_GetRxStats(uint8 CtrlIdx, Eth_RxStatsType* RxStats)

Function	Std_ReturnType Eth_GetRxStats(uint8 CtrlIdx, Eth_RxStatsType* RxStats)
Description	Returns the following list according to IETF RFC2819, where the maximal possible value shall denote an invalid value, e.g. if this counter is not available: 1) etherStatsDropEvents 2) etherStatsOctets 3) etherStatsPkts 4) etherStatsBroadcastPkts 5) etherStatsMulticastPkts 6) etherStatsCrcAlignErrors 7) etherStatsUndersizePkts 8) etherStatsOversizePkts 9) etherStatsFragments

	10) etherStatsJabbers
	11) etherStatsCollisions
	12) etherStatsPkts64Octets
	13) etherStatsPkts65to127Octets
	14) etherStatsPkts128to255Octets
	15) etherStatsPkts256to511Octets
	16) etherStatsPkts512to1023Octets
	17) etherStatsPkts1024to1518Octets

Diagram



Parameters

Parameter	Description
CtrlIdx	Index of the controller within the context of the Ethernet Driver.

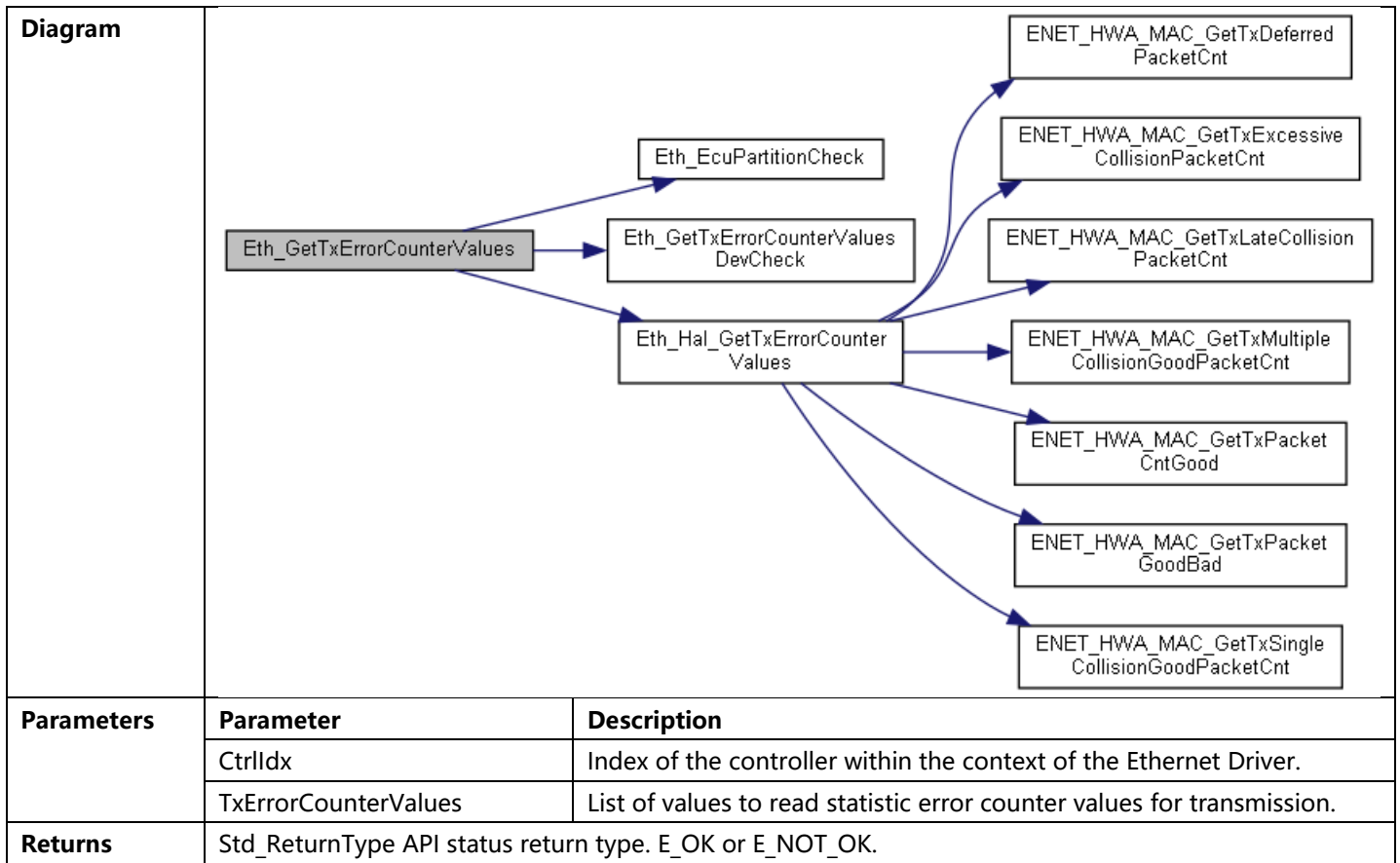
	RxStats	List of values according to IETF RFC 2819 (Remote Network Monitoring Management Information Base).
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.13 Std_ReturnType Eth_GetTxStats(uint8 CtrlIdx, Eth_TxStatsType* TxStats)

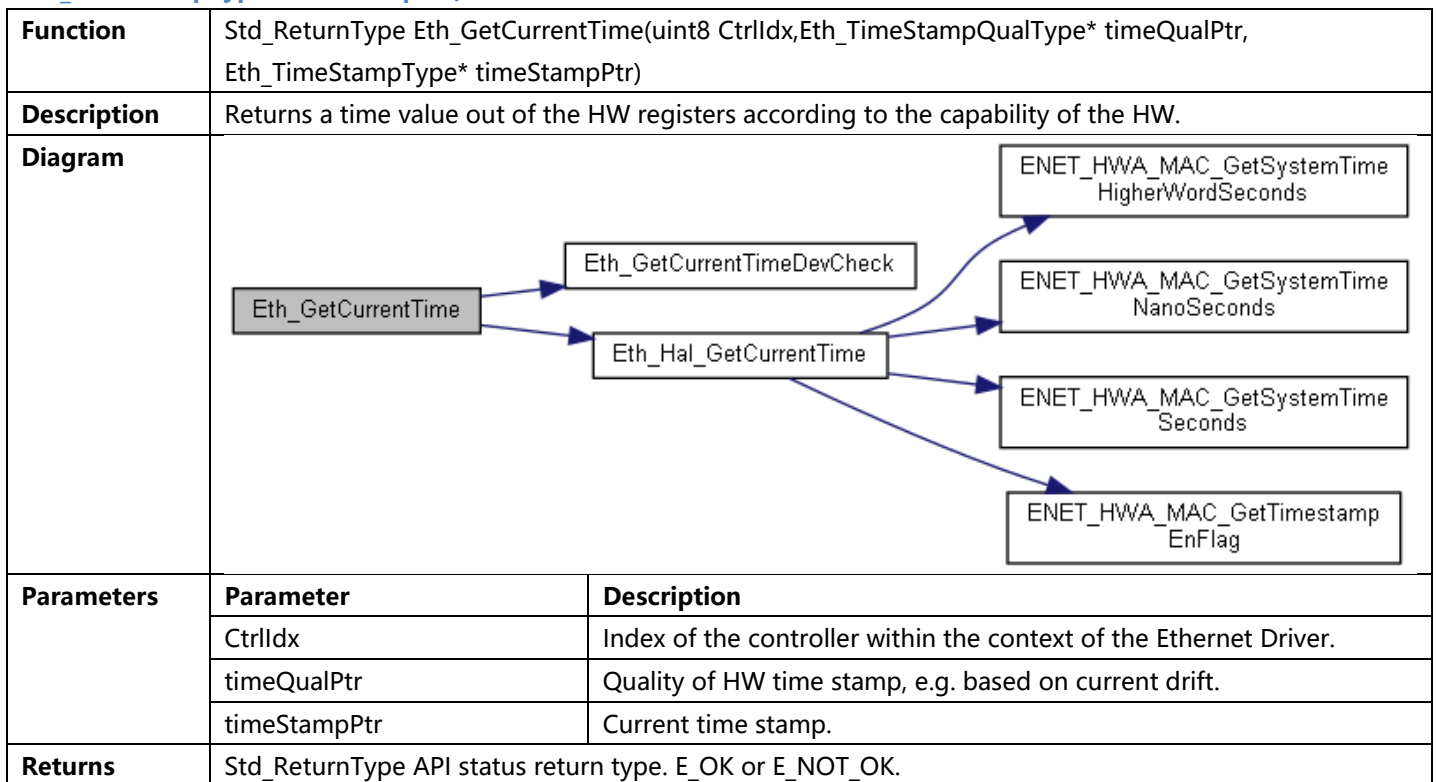
Function	Std_ReturnType Eth_GetTxStats(uint8 CtrlIdx, Eth_TxStatsType* TxStats)	
Description	Returns the list of Transmission Statistics out of IETF RFC1213 defined with Eth_TxStatsType, where the maximal possible value shall denote an invalid value, e.g. this counter is not available.	
Diagram	<pre> graph LR Eth_GetTxStats --> Eth_EcuPartitionCheck Eth_GetTxStats --> Eth_GetTxStatsDevCheck Eth_GetTxStats --> Eth_Hal_GetTxStats Eth_Hal_GetTxStats --> ENET_HWA_MAC_GetTxBroadcastPacketGoodBad Eth_Hal_GetTxStats --> ENET_HWA_MAC_GetTxMulticastPacketGoodBad Eth_Hal_GetTxStats --> ENET_HWA_MAC_GetTxOctetCntGoodBad Eth_Hal_GetTxStats --> ENET_HWA_MAC_GetTxUnicastPacketGoodBad </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TxStats	List of values to read statistic values for transmission.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.14 Std_ReturnType Eth_GetTxErrorCounterValues(uint8 CtrlIdx, Eth_TxErrorCounterValuesType* TxErrorCounterValues)

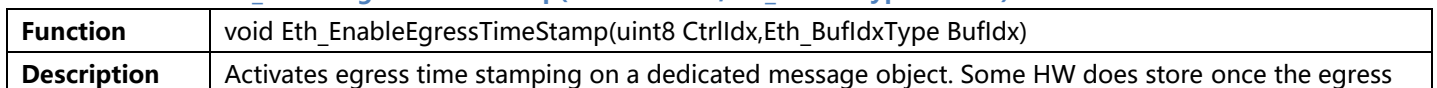
Function	Std_ReturnType Eth_GetTxErrorCounterValues(uint8 CtrlIdx, Eth_TxErrorCounterValuesType* TxErrorCounterValues)	
Description	Returns the list of Transmission Error Counters out of IETF RFC1213 and RFC1643 defined with Eth_TxErrorCounterValuesType, where the maximal possible value shall denote an invalid value, e.g. this counter is not available.	



2.7.1.15 Std_ReturnType Eth_GetCurrentTime(uint8 CtrlIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)



2.7.1.16 void Eth_EnableEgressTimeStamp(uint8 CtrlIdx, Eth_BufIdxType BufIdx)



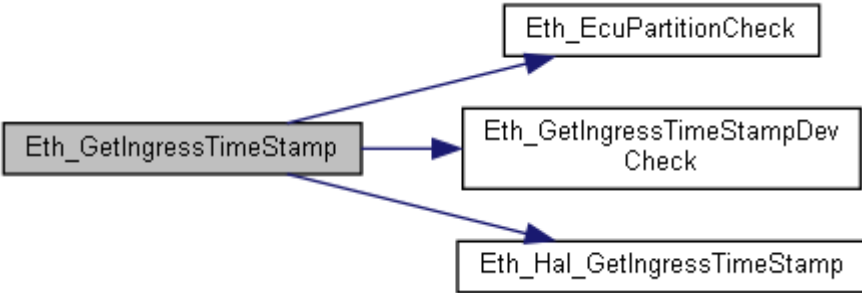
	time stamp marker and some HW needs it always before transmission. There will be no "disable" functionality, due to the fact, that the message type is always "time stamped" by network design.	
Diagram	<pre> graph LR A[Eth_EnableEgressTimeStamp] --> B[Eth_EcuPartitionCheck] A --> C[Eth_EnableEgressTimeStampDevCheck] A --> D[Eth_Hal_EnableEgressTimeStamp] C --> E[Eth_Hal_TxBuffersCount] </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BuIdx	Index of the message buffer, where Application expects egress time stamping.
Returns	N/A	

2.7.1.17 void Eth_GetEgressTimeStamp(uint8 CtrlIdx, Eth_BuIdxType BuIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)

Function	void Eth_GetEgressTimeStamp(uint8 CtrlIdx, Eth_BuIdxType BuIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)	
Description	Reads back the egress time stamp on a dedicated message object. It must be called within the TxConfirmation() function.	
Diagram	<pre> graph LR A[Eth_GetEgressTimeStamp] --> B[Eth_EcuPartitionCheck] A --> C[Eth_GetEgressTimeStampDevCheck] A --> D[Eth_Hal_GetEgressTimeStamp] C --> E[Eth_Hal_TxBuffersCount] </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BuIdx	Index of the message buffer, where Application expects egress time stamping.
	timeQualPtr	Quality of HW time stamp, e.g. based on current drift.
	timeStampPtr	Current time stamp.
Returns	N/A	

2.7.1.18 void Eth_GetIngressTimeStamp(uint8 CtrlIdx, const Eth_DataType* DataPtr, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)

Function	void Eth_GetIngressTimeStamp(uint8 CtrlIdx, const Eth_DataType* DataPtr, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)	
Description	Reads back the ingress time stamp on a dedicated message object. It must be called within the RxIndication() function.	

Diagram	 <pre>graph LR; A[Eth_GetIngressTimeStamp] --> B[Eth_EcuPartitionCheck]; A --> C[Eth_GetIngressTimeStampDev Check]; A --> D[Eth_Hal_GetIngressTimeStamp];</pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	DataPtr	Pointer to the message buffer, where Application expects ingress time stamping.
	timeQualPtr	Quality of HW time stamp, e.g. based on current drift.
	timeStampPtr	Current time stamp.
Returns	N/A	

2.7.1.19 Std_ReturnType Eth_SetCorrectionTime(uint8 CtrlIdx, const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)

Function	Std_ReturnType Eth_SetCorrectionTime(uint8 CtrlIdx, const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)									
Description	This API allows the Time Slave node to correct the local time and the local Eth reference clock ratio. If the secondsHi field of timeOffsetPtr or rateRatioPtr is nonzero value, that is, the time difference or time ratio is too large, this API will return E_NOT_OK, please call Eth_Hal_SetGlobalTime first.									
Diagram	<pre>graph LR; Eth_SetCorrectionTime --> Eth_EcuPartitionCheck; Eth_SetCorrectionTime --> Eth_Hal_SetCorrectionTime; Eth_SetCorrectionTime --> Eth_SetCorrectionTimeDev_Check[Eth_SetCorrectionTimeDev Check]; Eth_Hal_SetCorrectionTime --> ENET_HWA_MAC_GetTimestampAddend; Eth_Hal_SetCorrectionTime --> Eth_Hal_PTPUpdateAddend; Eth_Hal_PTPUpdateAddend --> ENET_HWA_MAC_SetTimestampAddend; Eth_Hal_PTPUpdateAddend --> ENET_HWA_MAC_UpdateAddend_Register[ENET_HWA_MAC_UpdateAddend Register]; Eth_Hal_PTPUpdateAddend --> ENET_HWA_MAC_GetTimestampUpdateStatus; Eth_Hal_PTPUpdateSystemTime --> ENET_HWA_MAC_SetSystemTimeNanoSecondsUpdateValue; Eth_Hal_PTPUpdateSystemTime --> ENET_HWA_MAC_SetSystemTimeSecondsUpdateValue; Eth_Hal_PTPUpdateSystemTime --> ENET_HWA_MAC_SetTimeUpdateOperation; Eth_Hal_PTPUpdateSystemTime --> ENET_HWA_MAC_UpdateTimestamp;</pre>									
Parameters	<table><thead><tr><th>Parameter</th><th>Description</th></tr></thead><tbody><tr><td>CtrlIdx</td><td>Index of the controller within the context of the Ethernet Driver.</td></tr><tr><td>timeOffsetPtr</td><td>The global time offset value.</td></tr><tr><td>rateRatioPtr</td><td>The global time ratio bias.</td></tr></tbody></table>	Parameter	Description	CtrlIdx	Index of the controller within the context of the Ethernet Driver.	timeOffsetPtr	The global time offset value.	rateRatioPtr	The global time ratio bias.	
Parameter	Description									
CtrlIdx	Index of the controller within the context of the Ethernet Driver.									
timeOffsetPtr	The global time offset value.									
rateRatioPtr	The global time ratio bias.									
Returns	E_OK: Time synchronization is succeeded. E_NOT_OK: Time synchronization is failed.									

2.7.1.20 Std_ReturnType Eth_SetGlobalTime(uint8 CtrlIdx, const Eth_TimeStampType* timeStampPtr)

Function	Std_ReturnType Eth_SetGlobalTime(uint8 CtrlIdx, const Eth_TimeStampType* timeStampPtr)
-----------------	--

Description	This API allows the user to set the local time in hardware.	
Diagram	<pre> graph LR Eth_SetGlobalTime --> Eth_EcuPartitionCheck Eth_SetGlobalTime --> Eth_Hal_SetGlobalTime Eth_SetGlobalTime --> Eth_SetGlobalTimeDevCheck Eth_Hal_SetGlobalTime --> Eth_Hal_PTPlnitSystemTime Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_GetTimestamp_InitStatus Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_InitTimestamp Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_SetSystemTime_HigherWordSeconds Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_SetSystemTime_NanoSecondsUpdateValue Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_SetSystemTime_SecondsUpdateValue Eth_Hal_PTPlnitSystemTime --> ENET_HWA_MAC_SetTimeUpdate_Operation </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	timeStampPtr	The global time value to set.
Returns	E_OK: Set global time successfully. E_NOT_OK: Set global time failed.	

2.7.1.21 BufReq_ReturnType Eth_ProvideTxBuffer(uint8 CtrlIdx,uint8 Priority,Eth_BufIdxType* BufIdxPtr,uint8** BufPtr,uint16* LenBytePtr)

Function	BufReq_ReturnType Eth_ProvideTxBuffer(uint8 CtrlIdx,uint8 Priority,Eth_BufIdxType* BufIdxPtr,uint8** BufPtr,uint16* LenBytePtr)	
Description	Provides access to a transmit buffer of the FIFO related to the specified priority.	
Diagram	<pre> graph LR Eth_ProvideTxBuffer --> Eth_EcuPartitionCheck Eth_ProvideTxBuffer --> Eth_Hal_ProvideTxBuffer Eth_ProvideTxBuffer --> Eth_Hal_TxPriorityToFifoid Eth_ProvideTxBuffer --> Eth_ProvideTxBufferPDevCheck Eth_Hal_ProvideTxBuffer --> Eth_Hal_AllocTxBuffer </pre>	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	Priority	Frame priority for transmit buffer FIFO selection.
	BufIdxPtr	Index to the granted buffer resource. To be used for subsequent requests.
	BufPtr	Pointer to the granted buffer.
	LenBytePtr	n: desired length in bytes, out: granted length in bytes.
Returns	BufReq_ReturnType : BUFREQ_OK: success BUFREQ_E_NOT_OK: development error detected BUFREQ_E_BUSY: all buffers in use BUFREQ_E_OVFL: requested buffer too large	

2.7.1.22 Std_ReturnType Eth_Transmit(uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8* PhysAddrPtr)

Function	Std_ReturnType Eth_Transmit(uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8* PhysAddrPtr)	
Description	Triggers transmission of a previously filled transmit buffer.	
Diagram		
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BufIdx	Index of the buffer resource.
	FrameType	Ethernet frame type.
	TxConfirmation	Activates transmission confirmation.
	LenByte	Data length in byte.
	PhysAddrPtr	Physical target address (MAC address) in network byte order.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

2.7.1.23 void Eth_Receive(uint8 CtrlIdx, uint8 Fifoldx, Eth_RxStatusType* RxStatusPtr)

Function	void Eth_Receive(uint8 CtrlIdx, uint8 Fifoldx, Eth_RxStatusType* RxStatusPtr)	
Description	Receive a frame from the related fifo.	
Diagram		
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	Fifoldx	Specifies the related fifo.
	RxStatusPtr	Indicates whether a frame has been received and if so, whether more frames are available for the related fifo.
Returns	N/A	

2.7.1.24 void Eth_TxConfirmation(uint8 CtrlIdx)

Function	void Eth_TxConfirmation(uint8 CtrlIdx)	
Description	Triggers frame transmission confirmation.	

Diagram		
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	N/A	

2.7.1.25 Std_ReturnType Eth_TimeAwareShaperInit(uint8 CtrlIdx)

Function	Std_ReturnType Eth_TimeAwareShaperInit(uint8 CtrlIdx)	
Description	Initialize the time-aware shaper on the egress side.	
Diagram		
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	

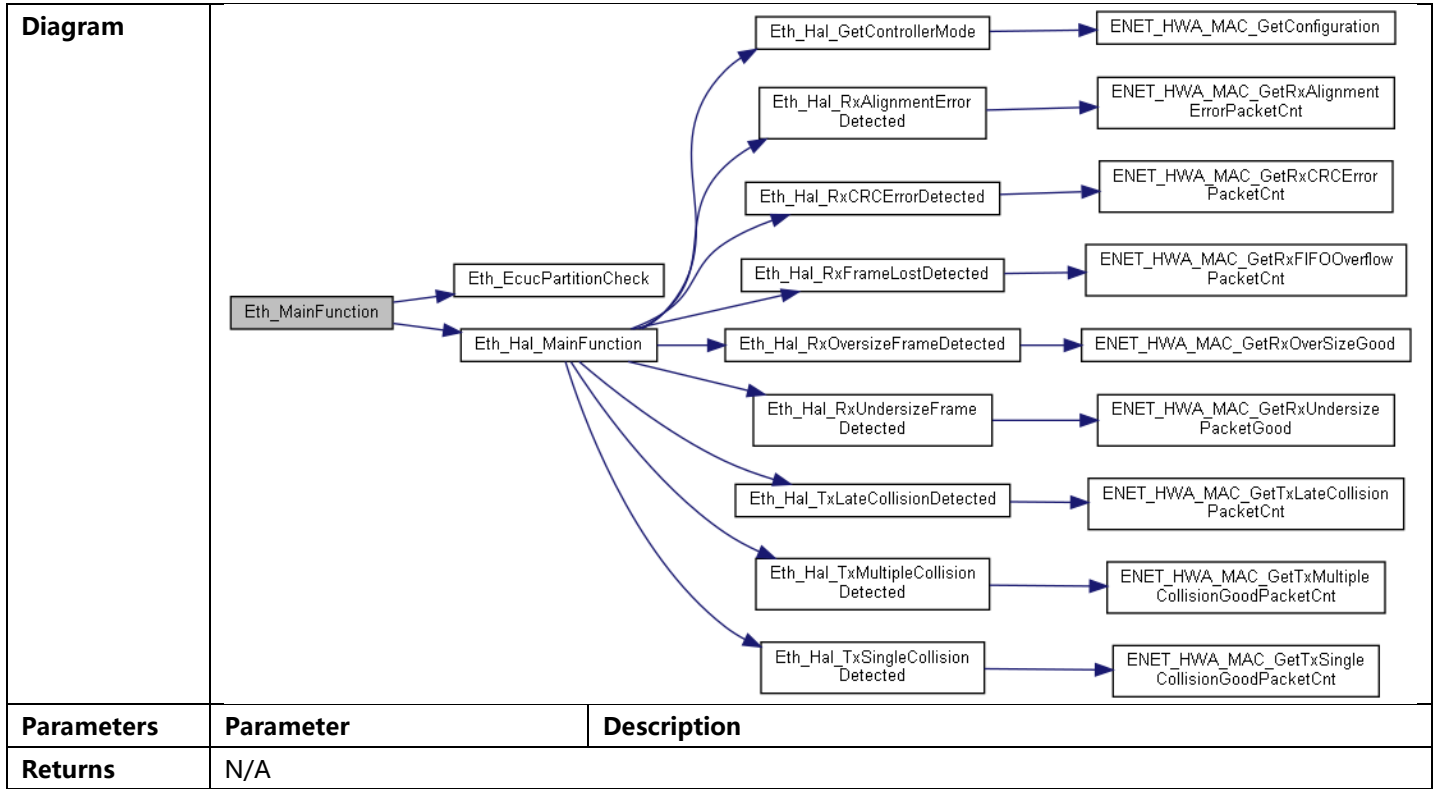
2.7.1.26 void Eth_GetVersionInfo(Std_VersionInfoType* VersionInfoPtr)

Function	void Eth_GetVersionInfo(Std_VersionInfoType* VersionInfoPtr)	
Description	Returns the version information of this module.	
Diagram	N/A	
Parameters	Parameter	Description
	VersionInfoPtr	Version information of this module.
Returns	N/A	

2.7.2 Functions in SchM_Eth.h

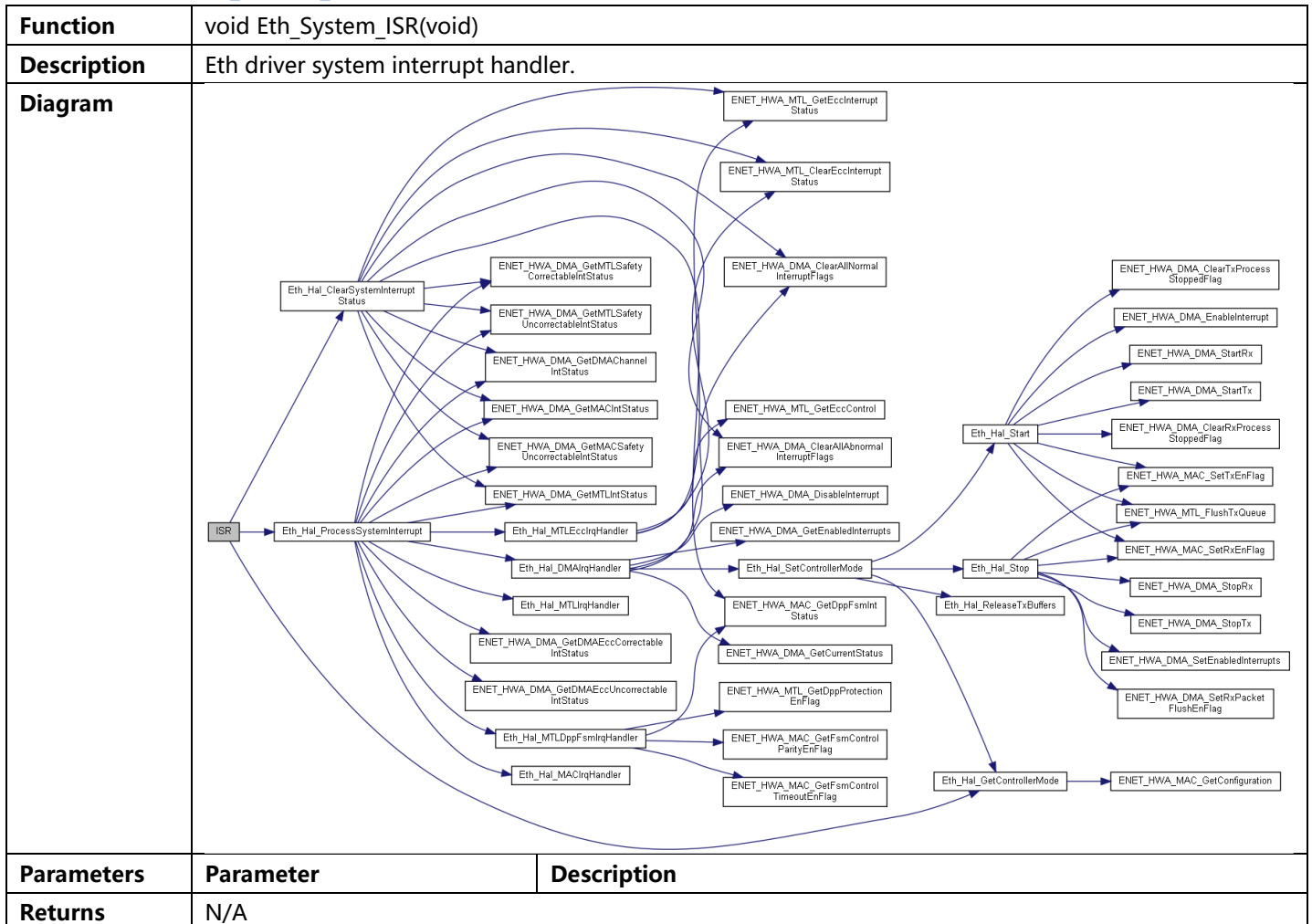
2.7.2.1 void Eth_MainFunction(void)

Function	void Eth_MainFunction(void)	
Description	The function checks for controller errors and lost frames. Used for polling state changes. Calls EthIf_CtrlModelIndication when the controller mode changed.	



2.7.3 Functions in Eth_Irq.c

2.7.3.1 void Eth_System_ISR(void)



2.7.3.2 void Eth_Rx_Channel0_ISR (void)

Function	void Eth_Rx_Channel0_ISR (void)	
Description	Eth driver DMA RX channel 0 interrupt handler.	
Diagram		
Parameters	Parameter	Description
Returns	N/A	

2.7.3.3 void Eth_Tx_Channel0_ISR (void)

Function	void Eth_Tx_Channel0_ISR (void)	
Description	ENET driver DMA TX channel 0 interrupt handler.	
Diagram		
Parameters	Parameter	Description
Returns	N/A	

2.7.3.4 void Eth_Rx_Channel1_ISR (void)

Function	void Eth_Rx_Channel1_ISR (void)	
Description	Eth driver DMA RX channel 1 interrupt handler.	
Diagram		
Parameters	Parameter	Description
Returns	N/A	

2.7.3.5 void Eth_Tx_Channel1_ISR (void)

Function	void Eth_Tx_Channel1_ISR (void)	
Description	ENET driver DMA TX channel 1 interrupt handler.	
Diagram		
Parameters	Parameter	Description
Returns	N/A	

2.7.4 Local Functions in Eth.c

2.7.4.1 LOCAL_INLINE boolean Eth_InitDevCheck (const Eth_ConfigType* CfgPtr)

Function	LOCAL_INLINE boolean Eth_InitDevCheck (const Eth_ConfigType* CfgPtr)	
Description	Check the development errors of Eth_Init.	
Parameters	Parameter	Description
	CfgPtr	Points to the implementation specific structure.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_Init ().	

2.7.4.2 LOCAL_INLINE boolean Eth_SetControllerModeDevCheck(uint8 CtrlIdx, uint8 CoreId)

Function	LOCAL_INLINE boolean Eth_SetControllerModeDevCheck(uint8 CtrlIdx, uint8 CoreId)	
Description	Check the development errors of Eth_SetControllerMode.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_SetControllerMode ().	

2.7.4.3 LOCAL_INLINE boolean Eth_GetControllerModeDevCheck(uint8 CtrlIdx, uint8 CoreId, Eth_ModeType* CtrlModePtr)

Function	LOCAL_INLINE boolean Eth_GetControllerModeDevCheck(uint8 CtrlIdx, uint8 CoreId, Eth_ModeType* CtrlModePtr)	
Description	Check the development errors of Eth_GetControllerMode.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	CtrlModePtr	Pointer to memory containing the controller mode.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetControllerMode ().	

2.7.4.4 LOCAL_INLINE boolean Eth_GetPhysAddrDevCheck(uint8 CtrlIdx, uint8 CoreId, uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_GetPhysAddrDevCheck(uint8 CtrlIdx, uint8 CoreId, uint8* PhysAddrPtr)	
Description	Check the development errors of Eth_GetPhysAddr.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetPhysAddr ().	

2.7.4.5 LOCAL_INLINE boolean Eth_SetPhysAddrDevCheck(uint8 CtrlIdx, uint8 CoreId, const uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_SetPhysAddrDevCheck(uint8 CtrlIdx, uint8 CoreId, const uint8* PhysAddrPtr)	
Description	Check the development errors of Eth_SetPhysAddr.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_SetPhysAddr ().	

2.7.4.6 LOCAL_INLINE boolean Eth_UpdatePhysAddrFilterDevCheck(uint8 CtrlIdx, uint8 CoreId, const uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_UpdatePhysAddrFilterDevCheck(uint8 CtrlIdx, uint8 CoreId, const uint8* PhysAddrPtr)	
Description	Check the development errors of Eth_UpdatePhysAddrFilter.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_UpdatePhysAddrFilter ().	

2.7.4.7 LOCAL_INLINE boolean Eth_WriteMiiDevCheck(uint8 CtrlIdx, uint8 CoreId)

Function	LOCAL_INLINE boolean Eth_WriteMiiDevCheck(uint8 CtrlIdx, uint8 CoreId)	
Description	Check the development errors of Eth_WriteMii.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_WriteMii ().	

2.7.4.8 LOCAL_INLINE boolean Eth_ReadMiiDevCheck(uint8 CtrlIdx, uint8 CoreId, uint16* RegValPtr)

Function	LOCAL_INLINE boolean Eth_ReadMiiDevCheck(uint8 CtrlIdx, uint8 CoreId, uint16* RegValPtr)	
Description	Check the development errors of Eth_ReadMii.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	RegValPtr	Pointer to memory containing the register value.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_ReadMii ().	

2.7.4.9 LOCAL_INLINE boolean Eth_WriteMmdDevCheck(uint8 CtrlIdx,uint8 CoreId)

Function	LOCAL_INLINE boolean Eth_WriteMmdDevCheck(uint8 CtrlIdx,uint8 CoreId)	
Description	Check the development errors of Eth_WriteMmd.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_WriteMmd ().	

2.7.4.10 LOCAL_INLINE boolean Eth_ReadMmdDevCheck(uint8 CtrlIdx,uint8 CoreId,uint16* RegValPtr)

Function	LOCAL_INLINE boolean Eth_ReadMmdDevCheck(uint8 CtrlIdx,uint8 CoreId,uint16* RegValPtr)	
Description	Check the development errors of Eth_ReadMmd.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	RegValPtr	Pointer to memory containing the register value.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_ReadMmd ().	

2.7.4.11 LOCAL_INLINE boolean Eth_GetCounterValuesDevCheck(uint8 CtrlIdx,uint8 CoreId, Eth_CounterType* CounterPtr)

Function	LOCAL_INLINE boolean Eth_GetCounterValuesDevCheck(uint8 CtrlIdx,uint8 CoreId, Eth_CounterType* CounterPtr)	
Description	Check the development errors of Eth_GetCounterValues.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	CounterPtr	Pointer to memory containing the counter values.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetCounterValues ().	

2.7.4.12 LOCAL_INLINE boolean Eth_GetRxStatsDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_RxStatsType* RxStats)

Function	LOCAL_INLINE boolean Eth_GetRxStatsDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_RxStatsType* RxStats)	
Description	Check the development errors of Eth_GetRxStats.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	RxStats	Pointer to memory containing the RxStats values.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetRxStats ().	

2.7.4.13 LOCAL_INLINE boolean Eth_GetTxStatsDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TxStatsType* TxStats)

Function	LOCAL_INLINE boolean Eth_GetTxStatsDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TxStatsType* TxStats)	
Description	Check the development errors of Eth_GetTxStats.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	TxStats	Pointer to memory containing the TxStats values.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetTxStats ().	

2.7.4.14 LOCAL_INLINE boolean Eth_GetTxErrorCounterValuesDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TxErrorCounterValuesType* TxErrorCounterValues)

Function	LOCAL_INLINE boolean Eth_GetTxErrorCounterValuesDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TxErrorCounterValuesType* TxErrorCounterValues)	
Description	Check the development errors of Eth_GetTxErrorCounterValues.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	TxErrorCounterValues	Pointer to memory containing the Tx error counter values.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetTxErrorCounterValues ().	

2.7.4.15 LOCAL_INLINE boolean Eth_GetCurrentTimeDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TimeStampQualType* timeQualPtr,Eth_TimeStampType* timeStampPtr)

Function	LOCAL_INLINE boolean Eth_GetCurrentTimeDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_TimeStampQualType* timeQualPtr,Eth_TimeStampType* timeStampPtr)	
Description	Check the development errors of Eth_GetCurrentTime.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	timeQualPtr	Pointer to memory containing the quality of HW time stamp.
	timeStampPtr	Pointer to memory containing the HW time stamp.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetCurrentTime ().	

2.7.4.16 LOCAL_INLINE boolean Eth_EnableEgressTimeStampDevCheck(uint8 CtrlIdx,uint8CoreId,Eth_BufIdxType BufIdx)

Function	LOCAL_INLINE boolean Eth_EnableEgressTimeStampDevCheck(uint8 CtrlIdx,uint8CoreId,Eth_BufIdxType BufIdx)	
Description	Check the development errors of Eth_EnableEgressTimeStamp.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.

	BuflDx	Index of the message buffer.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_EnableEgressTimeStamp ().	

2.7.4.17 LOCAL_INLINE boolean Eth_GetEgressTimeStampDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_BuflDxType BuflDx,Eth_TimeStampQualType* timeQualPtr,const Eth_TimeStampType* timeStampPtr)

Function	LOCAL_INLINE boolean Eth_GetEgressTimeStampDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_BuflDxType BuflDx,Eth_TimeStampQualType* timeQualPtr,const Eth_TimeStampType* timeStampPtr)	
Description	Check the development errors of Eth_GetEgressTimeStamp.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	BuflDx	Index of the message buffer.
	timeQualPtr	Pointer to memory containing the quality of HW time stamp.
	timeStampPtr	Pointer to memory containing the HW time stamp.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetEgressTimeStamp ().	

2.7.4.18 LOCAL_INLINE boolean Eth_GetIngressTimeStampDevCheck(uint8 CtrlIdx,uint8 CoreId,const Eth_DataType* DataPtr,Eth_TimeStampQualType* timeQualPtr,const Eth_TimeStampType* timeStampPtr)

Function	LOCAL_INLINE boolean Eth_GetIngressTimeStampDevCheck(uint8 CtrlIdx,uint8 CoreId,const Eth_DataType* DataPtr,Eth_TimeStampQualType* timeQualPtr,const Eth_TimeStampType* timeStampPtr)	
Description	Check the development errors of Eth_GetIngressTimeStamp.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	DataPtr	Pointer to the message buffer.
	timeQualPtr	Pointer to memory containing the quality of HW time stamp.
	timeStampPtr	Pointer to memory containing the HW time stamp.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetIngressTimeStamp ().	

2.7.4.19 LOCAL_INLINE boolean Eth_SetCorrectionTimeDevCheck (uint8 CtrlIdx,uint8 CoreId, const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)

Function	LOCAL_INLINE boolean Eth_SetCorrectionTimeDevCheck (uint8 CtrlIdx,uint8 CoreId,const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)	
Description	Check the development errors of Eth_SetCorrectionTime.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	timeOffsetPtr	Pointer to memory containing the time stamp difference.
	rateRatioPtr	Pointer to memory containing the time rate ratio difference.

Returns	TRUE: No development error detected. FALSE: Development error detected.
Referenced By	Eth_SetCorrectionTime ().

2.7.4.20 LOCAL_INLINE boolean Eth_SetGlobalTimeDevCheck (uint8 CtrlIdx,uint8 CoreId, const Eth_TimeStampType* timeStampPtr)

Function	LOCAL_INLINE boolean Eth_SetGlobalTimeDevCheck (uint8 CtrlIdx,uint8 CoreId, const Eth_TimeStampType* timeStampPtr	
Description	Check the development errors of Eth_SetGlobalTime.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	timeStampPtr	Pointer to memory containing the time stamp to set.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_SetGlobalTime ().	

2.7.4.21 LOCAL_INLINE boolean Eth_ProvideTxBufferDevCheck(uint8 CtrlIdx,uint8 CoreId,uint8 Fifoldx, Eth_BufIdxType* BufIdxPtr,uint8** BufPtr,uint16* LenBytePtr)

Function	LOCAL_INLINE boolean Eth_ProvideTxBufferDevCheck(uint8 CtrlIdx,uint8 CoreId,uint8 Fifoldx, Eth_BufIdxType* BufIdxPtr,uint8** BufPtr,uint16* LenBytePtr)	
Description	Check the development errors of Eth_ProvideTxBuffer.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	Fifoldx	Index of TX FIFO.
	BufIdxPtr	Pointer to memory containing the Buffer index.
	BufPtr	Pointer to memory containing the TX buffer.
	LenBytePtr	Pointer to memory containing the buffer length.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_ProvideTxBuffer ().	

2.7.4.22 LOCAL_INLINE boolean Eth_TransmitDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_BufIdxType BufIdx,const uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_TransmitDevCheck(uint8 CtrlIdx,uint8 CoreId,Eth_BufIdxType BufIdx,const uint8* PhysAddrPtr)	
Description	Check the development errors of Eth_Transmit.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	BufIdx	Index of TX buffer.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_Transmit ().	

2.7.4.23 LOCAL_INLINE boolean Eth_ReceiveDevCheck(uint8 CtrlIdx,uint8 CoreId,uint8 Fifoldx,Eth_RxStatusType* RxStatusPtr)

Function	LOCAL_INLINE boolean Eth_ReceiveDevCheck(uint8 CtrlIdx,uint8 CoreId,uint8 Fifoldx,Eth_RxStatusType* RxStatusPtr)	
Description	Check the development errors of Eth_Receive.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
	Fifoldx	Index of RX FIFO.
	RxStatusPtr	Pointer to memory containing the receive status.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_Receive ().	

2.7.4.24 LOCAL_INLINE boolean Eth_TxConfirmationDevCheck(uint8 CtrlIdx,uint8 CoreId)

Function	LOCAL_INLINE boolean Eth_TxConfirmationDevCheck(uint8 CtrlIdx,uint8 CoreId)	
Description	Check the development errors of Eth_TxConfirmation.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_TxConfirmation ().	

2.7.4.25 LOCAL_INLINE boolean Eth_TimeAwareShaperInitDevCheck(uint8 CtrlIdx, uint8 CoreId)

Function	LOCAL_INLINE boolean Eth_TimeAwareShaperInitDevCheck(uint8 CtrlIdx, uint8 CoreId)	
Description	Check the development errors of Eth_TimeAwareShaperInit.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CoreId	Partition ID of current processor.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_TimeAwareShaperInit().	

2.7.4.26 LOCAL_INLINE boolean Eth_GetVersionInfoDevCheck(Std_VersionInfoType* VersionInfoPtr)

Function	LOCAL_INLINE boolean Eth_GetVersionInfoDevCheck(Std_VersionInfoType* VersionInfoPtr)	
Description	Check the development errors of Eth_GetVersionInfo.	
Parameters	Parameter	Description
	VersionInfoPtr	Pointer to memory containing the version information.
Returns	TRUE: No development error detected. FALSE: Development error detected.	
Referenced By	Eth_GetVersionInfo ().	

2.7.4.27 **LOCAL_INLINE** boolean Eth_EcucPartitionCheck(const Eth_CtrlCfgType *CtrlCfgPtr, uint8 CoreId, uint8 Sid)

Function	LOCAL_INLINE boolean Eth_EcucPartitionCheck(const Eth_CtrlCfgType *CtrlCfgPtr, uint8 CoreId, uint8 Sid)	
Description	Check the development errors of Eth_GetVersionInfo.	
Parameters	Parameter	Description
	CtrlCfgPtr	Pointer to the controller configuration.
	CoreId	Partition ID of current processor.
	Sid	Service ID.
Returns	TRUE: ECU partition check success. FALSE: ECU partition check failed.	
Referenced By	All function in Eth.h except Eth_GetCurrentTime	

2.7.4.28 **LOCAL_INLINE** boolean Eth_PhysAddrIsFullFilterOpenPattern(const uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_PhysAddrIsFullFilterOpenPattern(const uint8* PhysAddrPtr)	
Description	Check the physical address is full filter open pattern (FF:FF:FF: FF:FF:FF).	
Parameters	Parameter	Description
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: The physical address is full filter open pattern. FALSE: The physical address is not full filter open pattern.	
Referenced By	Eth_UpdatePhysAddrFilter ().	

2.7.4.29 **LOCAL_INLINE** boolean Eth_PhysAddrIsFullFilterClosePattern(const uint8* PhysAddrPtr)

Function	LOCAL_INLINE boolean Eth_PhysAddrIsFullFilterClosePattern(const uint8* PhysAddrPtr)	
Description	Check the physical address is full filter close pattern (00:00:00: 00:00:00).	
Parameters	Parameter	Description
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	TRUE: The physical address is full filter close pattern. FALSE: The physical address is not full filter close pattern.	
Referenced By	Eth_UpdatePhysAddrFilter ().	

2.8 Hardware Functions

2.8.1 Functions in Eth_Hal.h

2.8.1.1 uint32 Eth_Hal_TxBuffersCount(uint8 CtrlIdx)

Function	uint32 Eth_Hal_TxBuffersCount(uint8 CtrlIdx)	
Description	Get the max TX buffer count of the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	The max buffer count.	
Referenced By	Eth_EnableEgressTimeStampDevCheck (),Eth_GetEgressTimeStampDevCheck(),Eth_TransmitDevCheck().	

2.8.1.2 uint8 Eth_Hal_TxPriorityToFifold(const Eth_CtrlCfgType *CtrlCfgPtr,uint8 Priority)

Function	uint8 Eth_Hal_TxPriorityToFifold(const Eth_CtrlCfgType *CtrlCfgPtr,uint8 Priority)	
Description	Get the FIFO ID corresponding to the priority.	
Parameters	Parameter	Description

	CtrlCfgPtr	Pointer to the controller configuration.
	Priority	Priority of The TX FIFO.
Returns	The index of TX FIFO.	
Referenced By	Eth_EnableEgressTimeStampDevCheck (),Eth_GetEgressTimeStampDevCheck(),Eth_TransmitDevCheck()	

2.8.1.3 void Eth_Hal_ResetController(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_ResetController(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Reset the controller indicated by CtrlIdx.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_Init()	

2.8.1.4 boolean Eth_Hal_CheckAccessController(uint8 CtrlIdx)

Function	boolean Eth_Hal_CheckAccessController(uint8 CtrlIdx)	
Description	Check whether the controller is accessible.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	TRUE: The controller is accessible. FALSE: The controller is not accessible.	
Referenced By	Eth_Init()	

2.8.1.5 Std_ReturnType Eth_Hal_InitController(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	Std_ReturnType Eth_Hal_InitController(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Initialize the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	E_OK The controller is initialized successfully. E_NOT_OK The controller is initialized failed.	
Referenced By	Eth_Init()	

2.8.1.6 Std_ReturnType Eth_Hal_SetControllerMode(uint8 CtrlIdx, Eth_ModeType CtrlMode, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	Std_ReturnType Eth_Hal_SetControllerMode(uint8 CtrlIdx, Eth_ModeType CtrlMode, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Set the state of the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlMode	The controller mode.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	Std_ReturnType E_OK or E_NOT_OK.	
Referenced By	Eth_SetControllerMode ()	

2.8.1.7 Eth_ModeType Eth_Hal_GetControllerMode(uint8 CtrlIdx)

Function	Eth_ModeType Eth_Hal_GetControllerMode(uint8 CtrlIdx)	
Description	Get the state of the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	The state of the indexed controller.	
Referenced By	Eth_GetControllerMode ()	

2.8.1.8 void Eth_Hal_GetMacAddress(uint8 CtrlIdx,uint8* PhysAddrPtr)

Function	void Eth_Hal_GetMacAddress(uint8 CtrlIdx,uint8* PhysAddrPtr)	
Description	Get the MAC address of the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	N/A	
Referenced By	Eth_GetMacAddress ()	

2.8.1.9 void Eth_Hal_SetMacAddress(uint8 CtrlIdx,const uint8* PhysAddrPtr)

Function	void Eth_Hal_SetMacAddress(uint8 CtrlIdx,const uint8* PhysAddrPtr)	
Description	Set the MAC address of the indexed controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	PhysAddrPtr	Pointer to memory containing the physical address.
Returns	N/A	
Referenced By	Eth_SetMacAddress ()	

2.8.1.10 void Eth_Hal_OpenMacAddressFilter(uint8 CtrlIdx)

Function	void Eth_Hal_OpenMacAddressFilter(uint8 CtrlIdx)	
Description	Completely open the filter. All multicast and broadcast frames will be forwarded to application.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	N/A	
Referenced By	Eth_UpdatePhysAddrFilter ()	

2.8.1.11 void Eth_Hal_CloseMacAddressFilter(uint8 CtrlIdx)

Function	void Eth_Hal_CloseMacAddressFilter(uint8 CtrlIdx)	
Description	Reduce the filter to the controller' s unique unicast MAC address.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	N/A	
Referenced By	Eth_UpdatePhysAddrFilter ()	

2.8.1.12 void Eth_Hal_MacAddressUserModeHashFilter(uint8 CtrlIdx,uint32 MacFilter)

Function	void Eth_Hal_MacAddressUserModeHashFilter(uint8 CtrlIdx,uint32 MacFilter)	
Description	Set the filter to user configuration hash filter mode.	

Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	MacFilter	MAC packet filter register value.
Returns	N/A	
Referenced By	Eth_UpdatePhysAddrFilter ()	

2.8.1.13 void Eth_Hal_SetMacAddressFilter(uint8 CtrlIdx,const uint8* PhysAddrPtr, Eth_FilterActionType Action)

Function	void Eth_Hal_SetMacAddressFilter(uint8 CtrlIdx,const uint8* PhysAddrPtr, Eth_FilterActionType Action)	
Description	Set the MAC address filter.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	PhysAddrPtr	Pointer to memory containing the physical address.
	Action	Add or remove from the MAC address filter.
Returns	N/A	
Referenced By	Eth_UpdatePhysAddrFilter ()	

2.8.1.14 Std_ReturnType Eth_Hal_WriteMii(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal)

Function	Std_ReturnType Eth_Hal_WriteMii(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16 RegVal)	
Description	Write the specified transceiver register through the MII interface.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver.
	RegIdx	Index of the register.
	RegVal	Value to write.
Returns	E_OK: Write successfully. E_NOT_OK: Write failed.	
Referenced By	Eth_WriteMii ()	

2.8.1.15 Std_ReturnType Eth_Hal_ReadMii(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16* RegValPtr)

Function	Std_ReturnType Eth_Hal_ReadMii(uint8 CtrlIdx,uint8 TrcvIdx,uint8 RegIdx,uint16* RegValPtr)	
Description	Read the specified transceiver register through the MII interface.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver.
	RegIdx	Index of the register.
	RegValPtr	Pointer to memory to save register value.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_ReadMii ()	

2.8.1.16 Std_ReturnType Eth_Hal_WriteMmd(uint8 CtrlIdx,uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx,uint16 RegVal)

Function	Std_ReturnType Eth_Hal_WriteMmd(uint8 CtrlIdx,uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx,uint16 RegVal)	
Description	Write the specified transceiver register through the MII interface via clause 45.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver.

	Mmd	(MMD)Device address.
	RegIdx	Index of the register.
	RegVal	Value to write.
Returns	E_OK: Write successfully. E_NOT_OK: Write failed.	
Referenced By	Eth_WriteMmd ()	

2.8.1.17 Std_ReturnType Eth_Hal_ReadMmd(uint8 CtrlIdx,uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx,uint16* RegValPtr)

Function	Std_ReturnType Eth_Hal_ReadMmd(uint8 CtrlIdx,uint8 TrcvIdx, uint8 Mmd,uint16 RegIdx,uint16* RegValPtr)	
Description	Read the specified transceiver register through the MII interface via clause 45.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TrcvIdx	Index of the transceiver.
	Mmd	(MMD)Device address.
	RegIdx	Index of the register.
	RegValPtr	Pointer to memory to save register value.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_ReadMmd ()	

2.8.1.18 Std_ReturnType Eth_Hal_GetCounterValues(uint8 CtrlIdx,Eth_CounterType* CounterPtr)

Function	Std_ReturnType Eth_Hal_GetCounterValues(uint8 CtrlIdx,Eth_CounterType* CounterPtr)	
Description	Reads a list with drop counter values of the corresponding controller.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CounterPtr	Pointer to memory containing the counter values.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_GetCounterValues ()	

2.8.1.19 Std_ReturnType Eth_Hal_GetRxStats(uint8 CtrlIdx,Eth_RxStatsType* RxStats)

Function	Std_ReturnType Eth_Hal_GetRxStats(uint8 CtrlIdx,Eth_RxStatsType* RxStats)	
Description	Read the RX statistics counter values.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	RxStats	Pointer to memory containing the RxStats values.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_GetRxStats ()	

2.8.1.20 Std_ReturnType Eth_Hal_GetTxStats(uint8 CtrlIdx,Eth_TxStatsType* TxStats)

Function	Std_ReturnType Eth_Hal_GetTxStats(uint8 CtrlIdx,Eth_TxStatsType* TxStats)	
Description	Read the TX statistics counter values.	
Parameters	Parameter	Description

	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TxStats	Pointer to memory containing the TxStats values.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_GetTxStats ()	

2.8.1.21 Std_ReturnType Eth_Hal_GetTxErrorCounterValues(uint8 CtrlIdx, Eth_TxErrorCounterValuesType* TxErrorCounterValues)

Function	Std_ReturnType Eth_Hal_GetTxErrorCounterValues(uint8 CtrlIdx, Eth_TxErrorCounterValuesType* TxErrorCounterValues)	
Description	Read the transmission error counter values.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	TxErrorCounterValues	Pointer to memory containing the TX error counter values.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_GetTxErrorCounterValues ()	

2.8.1.22 Std_ReturnType Eth_Hal_GetCurrentTime(uint8 CtrlIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)

Function	Std_ReturnType Eth_Hal_GetCurrentTime(uint8 CtrlIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)	
Description	Get current PTP system time.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	timeQualPtr	Pointer to memory containing the quality of HW timestamp.
	timeStampPtr	Pointer to memory containing the HW timestamp.
Returns	E_OK: Read successfully. E_NOT_OK: Read failed.	
Referenced By	Eth_GetCurrentTime ()	

2.8.1.23 void Eth_Hal_EnableEgressTimeStamp(uint8 CtrlIdx, Eth_BufIdxType BufIdx)

Function	void Eth_Hal_EnableEgressTimeStamp(uint8 CtrlIdx, Eth_BufIdxType BufIdx)	
Description	Enable egress time stamping on a dedicated message.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BufIdx	Index of the message buffer.
Returns	N/A	
Referenced By	Eth_EnableEgressTimeStamp ()	

2.8.1.24 void Eth_Hal_GetEgressTimeStamp(uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)

Function	void Eth_Hal_GetEgressTimeStamp(uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_TimeStampQualType* timeQualPtr, Eth_TimeStampType* timeStampPtr)	
Description	Get PTP time stamp of egress frame.	
Parameters	Parameter	Description

	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BufIdx	Index of the message buffer.
	timeQualPtr	Pointer to memory containing the quality of HW timestamp.
	timeStampPtr	Pointer to memory containing the HW timestamp.
Returns	N/A	
Referenced By	Eth_GetEgressTimeStamp ()	

2.8.1.25 void Eth_Hal_GetIngressTimeStamp(uint8 CtrlIdx,const Eth_DataType* DataPtr, Eth_TimeStampQualType* timeQualPtr,Eth_TimeStampType* timeStampPtr)

Function	void Eth_Hal_GetIngressTimeStamp(uint8 CtrlIdx,const Eth_DataType* DataPtr, Eth_TimeStampQualType* timeQualPtr,Eth_TimeStampType* timeStampPtr)	
Description	Get PTP time stamp of ingressed frame.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	DataPtr	Pointer to the ingressed frame buffer.
	timeQualPtr	Pointer to memory containing the quality of HW timestamp.
	timeStampPtr	Pointer to memory containing the HW timestamp.
Returns	N/A	
Referenced By	Eth_GetIngressTimeStamp ()	

2.8.1.26 Std_ReturnType Eth_Hal_SetCorrectionTime(uint8 CtrlIdx,const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)

Function	Std_ReturnType Eth_Hal_SetCorrectionTime(uint8 CtrlIdx,const Eth_TimeIntDiffType* timeOffsetPtr, const Eth_RateRatioType* rateRatioPtr)	
Description	Set PTP system time correction.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	timeOffsetPtr	The PTP time offset value.
	rateRatioPtr	The PTP time ratio bias.
Returns	E_OK: Set correction successfully. E_NOT_OK: Set correction failed.	
Referenced By	Eth_SetCorrectionTime ()	

2.8.1.27 Std_ReturnType Eth_Hal_SetGlobalTime(uint8 CtrlIdx,const Eth_TimeStampType* timeStampPtr)

Function	Std_ReturnType Eth_Hal_SetGlobalTime(uint8 CtrlIdx,const Eth_TimeStampType* timeStampPtr)	
Description	Set PTP system time.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	timeStampPtr	The time stamp value.
Returns	E_OK: Set time successfully. E_NOT_OK: Set time failed.	
Referenced By	Eth_SetGlobalTime ()	

2.8.1.28 BufReq_ReturnType Eth_Hal_ProvideTxBuffer(uint8 CtrlIdx,uint8 FifIdx,Eth_BufIdxType* BufIdxPtr, uint8** BufPtr,uint16* LenBytePtr)

Function	BufReq_ReturnType Eth_Hal_ProvideTxBuffer(uint8 CtrlIdx,uint8 FifIdx,Eth_BufIdxType* BufIdxPtr,
-----------------	---

	uint8** BufPtr,uint16* LenBytePtr)	
Description	Provides access to a transmit buffer of the FIFO related to the specified priority.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	FifIdx	Index of the transmit FIFO.
	BuflIdxPtr	Index to the granted buffer resource. To be used for subsequent requests.
	BufPtr	Pointer to the granted buffer.
	LenBytePtr	In: desired length in bytes, out: granted length in bytes.
Returns	BufReq_ReturnType : BUFREQ_OK: success BUFREQ_E_NOT_OK: development error detected BUFREQ_E_BUSY: all buffers in use BUFREQ_E_OVFL: requested buffer too large	
Referenced By	Eth_ProvideTxBuffer ()	

2.8.1.29 Std_ReturnType Eth_Hal_Transmit(uint8 CtrlIdx,Eth_BuflIdxType BuflIdx,Eth_FrameType FrameType, boolean TxConfirmation,uint16 LenByte,const uint8* PhysAddrPtr, boolean TxIntEnable)

Function	Std_ReturnType Eth_Hal_Transmit(uint8 CtrlIdx,Eth_BuflIdxType BuflIdx,Eth_FrameType FrameType, boolean TxConfirmation,uint16 LenByte,const uint8* PhysAddrPtr, boolean TxIntEnable)	
Description	Triggers transmission of a previously filled transmit buffer.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	BuflIdx	Index of the buffer resource.
	FrameType	Ethernet frame type.
	TxConfirmation	Activates transmission confirmation.
	LenByte	Data length in byte.
	PhysAddrPtr	Physical target address (MAC address) in network byte order.
	TxIntEnable	Tx interrupt enable.
Returns	E_OK: Set correction successfully. E_NOT_OK: Set correction failed.	
Referenced By	Eth_Transmit ()	

2.8.1.30 void Eth_Hal_ReportTransmission(uint8 CtrlIdx,uint8 FifIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_ReportTransmission(uint8 CtrlIdx,uint8 FifIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Report transmission to EthIf layer.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	FifIdx	Index of TX FIFO.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_TxConfirmation () , Eth_Hal_ProcessTxInterrupt()	

2.8.1.31 Eth_RxStatusType Eth_Hal_ReportReception(uint8 CtrlIdx,uint8 FifIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	Eth_RxStatusType Eth_Hal_ReportReception(uint8 CtrlIdx,uint8 FifIdx, const Eth_CtrlCfgType
-----------------	--

	*CtrlCfgPtr)	
Description	Report reception to EthIf layer.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	FifIdx	Index of RX FIFO.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	ETH_RECEIVED: Ethernet frame has been received, no further frames available. ETH_NOT_RECEIVED: Ethernet frame has not been received, no further frames available. ETH_RECEIVED_MORE_DATA_AVAILABLE: Ethernet frame has been received, more frames are available.	
Referenced By	Eth_Receive (), Eth_Hal_ProcessRxInterrupt()	

2.8.1.32 Std_ReturnType Eth_Hal_TimeAwareShaperInit(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	Std_ReturnType Eth_Hal_TimeAwareShaperInit(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Initialize the time-aware shaper configurations on the egress side.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	Std_ReturnType API status return type. E_OK or E_NOT_OK.	
Referenced By	Eth_TimeAwareShaperInit()	

2.8.1.33 void Eth_Hal_MainFunction(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_MainFunction(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Eth driver scheduled main function.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_MainFunction ()	

2.8.1.34 void Eth_Hal_ProcessSystemInterrupt(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_ProcessSystemInterrupt(uint8 CtrlIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Process the system interrupt event.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_System_ISR ()	

2.8.1.35 void Eth_Hal_ProcessTxInterrupt(uint8 CtrlIdx, uint8 ChIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_ProcessTxInterrupt(uint8 CtrlIdx, uint8 ChIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Process the transmission interrupt event.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.

	ChIdx	Index of the channel.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_Tx_Channel0_ISR (),Eth_Tx_Channel1_ISR ()	

2.8.1.36 void Eth_Hal_ProcessRxInterrupt(uint8 CtrlIdx,uint8 ChIdx, const Eth_CtrlCfgType *CtrlCfgPtr)

Function	void Eth_Hal_ProcessRxInterrupt(uint8 CtrlIdx,uint8 ChIdx, const Eth_CtrlCfgType *CtrlCfgPtr)	
Description	Process the reception interrupt event.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	ChIdx	Index of the channel.
	CtrlCfgPtr	Pointer to the controller configuration.
Returns	N/A	
Referenced By	Eth_Rx_Channel0_ISR (),Eth_Rx_Channel1_ISR ()	

2.8.1.37 void Eth_Hal_ClearSystemInterruptStatus(uint8 CtrlIdx)

Function	void Eth_Hal_ClearSystemInterruptStatus(uint8 CtrlIdx)	
Description	Clear the system interrupt status.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
Returns	N/A	
Referenced By	Eth_System_ISR ()	

2.8.1.38 void Eth_Hal_ClearTxInterruptStatus(uint8 CtrlIdx,uint8 ChIdx)

Function	void Eth_Hal_ClearTxInterruptStatus(uint8 CtrlIdx,uint8 ChIdx)	
Description	Clear the transmission interrupt status.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	ChIdx	Index of the channel.
Returns	N/A	
Referenced By	Eth_Tx_Channel0_ISR (),Eth_Tx_Channel1_ISR ()	

2.8.1.39 void Eth_Hal_ClearRxInterruptStatus(uint8 CtrlIdx,uint8 ChIdx)

Function	void Eth_Hal_ClearRxInterruptStatus(uint8 CtrlIdx,uint8 ChIdx)	
Description	Clear the reception interrupt status.	
Parameters	Parameter	Description
	CtrlIdx	Index of the controller within the context of the Ethernet Driver.
	ChIdx	Index of the channel.
Returns	N/A	
Referenced By	Eth_Rx_Channel0_ISR (),Eth_Rx_Channel1_ISR ()	

2.8.2 Functions in Eth_Hw.h

2.8.2.1 LOCAL_INLINE void ENET_HWA_MAC_SetPhyMiiMode(SCM_Type *const pScm, ENET_MAC_MiiModeType eMode)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetPhyMiiMode(SCM_Type *const pScm,
-----------------	--

	ENET_MAC_MiiModeType eMode)	
Description	Set phy interface mii mode.	
Parameters	Parameter	Description
	pScm	The base address of the SCM instance.
	eMode	Phy interface mode.
Returns	N/A	
Referenced By	Eth_Hal_SysConfig ()	

2.8.2.2 LOCAL_INLINE uint32 ENET_HWA_MAC_GetConfiguration(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetConfiguration(const ENET_Type *const pEnet)	
Description	Get the MAC Configuration Register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MAC Configuration Register value.	
Referenced By	Eth_Hal_GetControllerMode ()	

2.8.2.3 LOCAL_INLINE void ENET_HWA_MAC_SetConfiguration(ENET_Type *const pEnet, uint32 u32Config)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetConfiguration(ENET_Type *const pEnet, uint32 u32Config)	
Description	Set the MAC Configuration Register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Config	Configuration value.
Returns	N/A	
Referenced By	Eth_Hal_MACInit ()	

2.8.2.4 LOCAL_INLINE void ENET_HWA_MAC_SetIPChecksumOffloadEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetIPChecksumOffloadEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set checksum offload function enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_MACInit ()	

2.8.2.5 LOCAL_INLINE void ENET_HWA_MAC_SetTxEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTxEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set transmitter enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_Start (),Eth_Hal_Stop()	

2.8.2.6 LOCAL_INLINE void ENET_HWA_MAC_SetRxEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetRxEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set receiver enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_Start (),Eth_Hal_Stop()	

2.8.2.7 LOCAL_INLINE void ENET_HWA_MAC_SetExtConfiguration(ENET_Type *const pEnet, uint32 u32Configuration)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetExtConfiguration(ENET_Type *const pEnet, uint32 u32Configuration)	
Description	Set the MAC extended configuration register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Configuration	Configuration value.
Returns	N/A	
Referenced By	Eth_Hal_MACInit ()	

2.8.2.8 LOCAL_INLINE void ENET_HWA_MAC_SetPacketFilter(ENET_Type *const pEnet, uint32 u32FilterCtrl)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetPacketFilter(ENET_Type *const pEnet, uint32 u32FilterCtrl)	
Description	Set the MAC packet filter register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32FilterCtrl	Filter control value.
Returns	N/A	
Referenced By	Eth_Hal_MACInit ()	

2.8.2.9 LOCAL_INLINE void ENET_HWA_MAC_SetReceiveAllEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetReceiveAllEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set receive all packets enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_CloseMacAddressFilter ()	

2.8.2.10 LOCAL_INLINE void ENET_HWA_MAC_SetHashOrPerfectFilterEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetHashOrPerfectFilterEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set hash or perfect filter enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.

Returns	N/A
Referenced By	Eth_Hal_MACInit ()

2.8.2.11 LOCAL_INLINE void ENET_HWA_MAC_SetBroadcastPacketDisableFlag(ENET_Type *const pEnet, uint8 bDisable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetBroadcastPacketDisableFlag(ENET_Type *const pEnet, uint8 bDisable)	
Description	Set broadcast packets disable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bDisable	1U disable , 0U enable.
Returns	N/A	
Referenced By	Eth_Hal_OpenMacAddressFilter(),Eth_Hal_CloseMacAddressFilter ()	

2.8.2.12 LOCAL_INLINE void ENET_HWA_MAC_SetPassAllMulticastEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetPassAllMulticastEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set pass all the received multicast packets enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_OpenMacAddressFilter(),Eth_Hal_CloseMacAddressFilter ()	

2.8.2.13 LOCAL_INLINE void ENET_HWA_MAC_SetHashOrPerfectMulticastMode(ENET_Type *const pEnet, ENET_MAC_HashOrPerfectModeType eMode)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetHashOrPerfectMulticastMode(ENET_Type *const pEnet, ENET_MAC_HashOrPerfectModeType eMode)	
Description	Set hash or perfect multicast filter mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eMode	Hash or perfect multicast filter mode.
Returns	N/A	
Referenced By	Eth_Hal_MACInit ()	

2.8.2.14 LOCAL_INLINE void ENET_HWA_MAC_SetPromiscuousModeEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetPromiscuousModeEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set promiscuous mode enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_CloseMacAddressFilter ()	

2.8.2.15 LOCAL_INLINE uint32 ENET_HWA_MAC_GetHashTableFirst32Bits(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetHashTableFirst32Bits(const ENET_Type *const pEnet)	
Description	Get the hash table first 32 bits value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Hash table first 32 bits value.	
Referenced By	Eth_Hal_SetMacAddressFilter ()	

2.8.2.16 LOCAL_INLINE void ENET_HWA_MAC_SetHashTableFirst32Bits(ENET_Type *const pEnet, uint32 u32Hash0)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetHashTableFirst32Bits(ENET_Type *const pEnet, uint32 u32Hash0)	
Description	Set the hash table first 32 bits value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Hash0	Hash table first 32 bits value.
Returns	N/A	
Referenced By	Eth_Hal_SetMacAddressFilter ()	

2.8.2.17 LOCAL_INLINE uint32 ENET_HWA_MAC_GetHashTableSecond32Bits(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetHashTableSecond32Bits(const ENET_Type *const pEnet)	
Description	Get the hash table second 32 bits value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Hash table second 32 bits value.	
Referenced By	Eth_Hal_SetMacAddressFilter ()	

2.8.2.18 LOCAL_INLINE void ENET_HWA_MAC_SetHashTableSecond32Bits(ENET_Type *const pEnet, uint32 u32Hash1)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetHashTableSecond32Bits(ENET_Type *const pEnet, uint32 u32Hash1)	
Description	Set the hash table second 32 bits value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Hash1	Hash table second 32 bits value.
Returns	N/A	
Referenced By	Eth_Hal_SetMacAddressFilter ()	

2.8.2.19 LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueEnStatus(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_QueueEnableType eEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueEnStatus(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_QueueEnableType eEnable)	
Description	Set Receive queue enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of the queue.

	eEnable	Receive queue enable mode.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.20 LOCAL_INLINE void ENET_HWA_MAC_SetRxQueueSelectedPriorities(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue,uint8 u8SelPrio)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetRxQueueSelectedPriorities(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue,uint8 u8SelPrio)	
Description	Set receive queue priorities.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of the queue.
	u8SelPrio	Priority of the queue.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.21 LOCAL_INLINE void ENET_HWA_MAC_SetInterruptEnable(ENET_Type *const pEnet, uint32 u32IntEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetInterruptEnable(ENET_Type *const pEnet, uint32 u32IntEnable)	
Description	Set the MAC interrupts enable register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eEnable	Interrupts enable.
Returns	N/A	
Referenced By	Eth_Hal_MACInit (),Eth_Hal_ResetController()	

2.8.2.22 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampInterruptEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampInterruptEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set timestamp interrupt enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eEnable	1U Enable , 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit (),Eth_Hal_ResetController()	

2.8.2.23 LOCAL_INLINE uint32 ENET_HWA_MAC_GetDppFsmIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetDppFsmIntStatus(const ENET_Type *const pEnet)	
Description	Get MAC_DPP_FSM_Interrupt_Status register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MAC_DPP_FSM_Interrupt_Status register value.	
Referenced By	Eth_Hal_MTLDppFsmIrqHandler(),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.24 LOCAL_INLINE boolean ENET_HWA_MAC_GetFsmControlParityEnFlag(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_MAC_GetFsmControlParityEnFlag(const ENET_Type *const pEnet)	
Description	Get FSM state parity feature enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: FSM state parity is enabled	
	FALSE: FSM state parity is disabled	
Referenced By	Eth_Hal_MTLDPpFsmIrqHandler()	

2.8.2.25 LOCAL_INLINE void ENET_HWA_MAC_SetFsmControlParityEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetFsmControlParityEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set FSM state parity feature enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable , 0U disable
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit()	

2.8.2.26 LOCAL_INLINE boolean ENET_HWA_MAC_GetFsmControlTimeoutEnFlag(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_MAC_GetFsmControlTimeoutEnFlag(const ENET_Type *const pEnet)	
Description	Get FSM timeout feature enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: FSM timeout is enabled	
	FALSE: FSM timeout is disabled	
Referenced By	Eth_Hal_MTLDPpFsmIrqHandler()	

2.8.2.27 LOCAL_INLINE void ENET_HWA_MAC_SetFsmControlTimeoutEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetFsmControlTimeoutEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set FSM timeout feature enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable , 0U disable
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit()	

2.8.2.28 LOCAL_INLINE void ENET_HWA_MAC_SetFsmActTimerNormalTimeout(ENET_Type *const pEnet, uint8 u8Timeout)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetFsmActTimerNormalTimeout(ENET_Type *const pEnet, uint8 u8Timeout)	
Description	Set normal mode FSM and other interface timeouts.	

Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u8Timeout	0: Timer disabled 1: 1 us, 2: 1.024 ms 3: 16.384 ms 4: 65.536 ms
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit()	

2.8.2.29 LOCAL_INLINE void ENET_HWA_MAC_SetFsmActTimerCsrClockCycle(ENET_Type *const pEnet, uint16 u16Cycle)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetFsmActTimerCsrClockCycle(ENET_Type *const pEnet, uint16 u16Cycle)	
Description	Set the CSR clocks required to generate 1 us tic.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u16Cycle	CSR clock cycles.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit()	

2.8.2.30 LOCAL_INLINE uint32 ENET_HWA_MAC_GetMDIOAddress(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetMDIOAddress(const ENET_Type *const pEnet)	
Description	Get MAC MDIO address register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MDIO address register value	
Referenced By	Eth_Hal_MdioAccess ()	

2.8.2.31 LOCAL_INLINE void ENET_HWA_MAC_SetMDIOAddress(ENET_Type *const pEnet, uint32 u32MDIOAddr)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMDIOAddress(ENET_Type *const pEnet, uint32 u32MDIOAddr)	
Description	Set MAC MDIO address register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32MDIOAddr	Register value.
Returns	N/A	
Referenced By	Eth_Hal_MdioBusInit (),Eth_Hal_MdioAccess ()	

2.8.2.32 LOCAL_INLINE uint32 ENET_HWA_MAC_GetMDIOData(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetMDIOData(const ENET_Type *const pEnet)	
Description	Get MDIO data register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MDIO data register value.	
Referenced By	Eth_Hal_ReadMii ()	

2.8.2.33 LOCAL_INLINE void ENET_HWA_MAC_SetMDIOData(ENET_Type *const pEnet, uint32 u32Data)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMDIOData(ENET_Type *const pEnet, uint32 u32Data)	
Description	Set MDIO data register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Data	Register value.
Returns	N/A	
Referenced By	Eth_Hal_MdioAccess ()	

2.8.2.34 LOCAL_INLINE void ENET_HWA_MAC_SetFramePreemptionEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetFramePreemptionEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable TX frame preemption.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U Enable , 0U Disable.
Returns	N/A	
Referenced By	Eth_Hal_TimeAwareShaperInit()	

2.8.2.35 LOCAL_INLINE uint32 ENET_HWA_MAC_GetMacAddr0High(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetMacAddr0High(const ENET_Type *const pEnet)	
Description	Get MAC address0 high register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MAC address0 high register value.	
Referenced By	Eth_Hal_GetMacAddress ()	

2.8.2.36 LOCAL_INLINE void ENET_HWA_MAC_SetAddr0High(ENET_Type *const pEnet, uint32 u32AddrHigh)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetAddr0High(ENET_Type *const pEnet, uint32 u32AddrHigh)	
Description	Set MAC address0 high register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32AddrHigh	MAC address0 high register value.
Returns	N/A	
Referenced By	Eth_Hal_SetMacAddress ()	

2.8.2.37 LOCAL_INLINE uint32 ENET_HWA_MAC_GetMacAddr0Low(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetMacAddr0Low(const ENET_Type *const pEnet)	
Description	Get MAC address0 low register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MAC address0 low register value.	
Referenced By	Eth_Hal_GetMacAddress ()	

2.8.2.38 LOCAL_INLINE void ENET_HWA_MAC_SetMacAddr0Low(ENET_Type *const pEnet, uint32 u32AddrLow)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMacAddr0Low(ENET_Type *const pEnet, uint32 u32AddrLow)	
Description	Set MAC address0 low register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32AddrLow	MAC address0 low register value.
Returns	N/A	
Referenced By	Eth_Hal_SetMacAddress ()	

2.8.2.39 LOCAL_INLINE void ENET_HWA_MAC_SetMmcCounterResetOnReadEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMmcCounterResetOnReadEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set MMC counters reset on read enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_MMCInit ()	

2.8.2.40 LOCAL_INLINE void ENET_HWA_MAC_SetMmcCounterStopRolloverEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMmcCounterStopRolloverEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set MMC counters stop rollover enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_MMCInit ()	

2.8.2.41 LOCAL_INLINE void ENET_HWA_MAC_ResetMmcCounters(ENET_Type *const pEnet)

Function	LOCAL_INLINE void ENET_HWA_MAC_ResetMmcCounters(ENET_Type *const pEnet)	
Description	Reset all MMC counters.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	N/A	
Referenced By	Eth_Hal_MMCInit ()	

2.8.2.42 LOCAL_INLINE void ENET_HWA_MAC_SetMmcRxInterruptMask(ENET_Type *const pEnet, uint32 u32Mask)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMmcRxInterruptMask(ENET_Type *const pEnet, uint32 u32Mask)	
Description	Set MMC interrupts mask of receive statistics counters.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.

	u32Mask	MMC interrupts mask value.
Returns	N/A	
Referenced By	Eth_Hal_MMCInit ()	

2.8.2.43 LOCAL_INLINE void ENET_HWA_MAC_SetMmcTxInterruptMask(ENET_Type *const pEnet, uint32 u32Mask)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetMmcTxInterruptMask(ENET_Type *const pEnet, uint32 u32Mask)	
Description	Set MMC interrupts mask of transmit statistics counters.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Mask	MMC interrupts mask value.
Returns	N/A	
Referenced By	Eth_Hal_MMCInit ()	

2.8.2.44 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxOctetCntGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxOctetCntGoodBad(const ENET_Type *const pEnet)	
Description	Get transmitted bytes of good and bad packets.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Transmitted bytes of good and bad packets.	
Referenced By	Eth_Hal_GetTxStats ()	

2.8.2.45 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets transmitted.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets transmitted.	
Referenced By	Eth_Hal_GetTxErrorCounterValues (),Eth_Hal_GetCounterValues()	

2.8.2.46 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxUnicastPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxUnicastPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad unicast packets transmitted.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad unicast packets transmitted.	
Referenced By	Eth_Hal_GetTxStats ()	

2.8.2.47 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxMulticastPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxMulticastPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad multicast packets transmitted.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad multicast packets transmitted.	

Referenced By	Eth_Hal_GetTxStats ()
----------------------	-----------------------

2.8.2.48 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxBroadcastPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxBroadcastPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad broadcast packets transmitted.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad broadcast packets transmitted.	
Referenced By	Eth_Hal_GetTxStats ()	

2.8.2.49 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxSingleCollisionGoodPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxSingleCollisionGoodPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of successfully transmitted packets after a single collision in half-duplex mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of successfully transmitted packets after a single collision in half-duplex mode.	
Referenced By	Eth_Hal_TxSingleCollisionDetected (),Eth_Hal_GetCounterValues(),Eth_Hal_GetTxErrorCounterValues()	

2.8.2.50 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxMultipleCollisionGoodPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxMultipleCollisionGoodPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of successfully transmitted packets after a multiple collision in half-duplex mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of successfully transmitted packets after a multiple collision in half-duplex mode.	
Referenced By	Eth_Hal_TxMultipleCollisionDetected (),Eth_Hal_GetCounterValues(),Eth_Hal_GetTxErrorCounterValues()	

2.8.2.51 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxDeferredPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxDeferredPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of successfully transmitted packets after a deferral in half-duplex mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of successfully transmitted packets after a deferral in half-duplex mode.	
Referenced By	Eth_Hal_GetCounterValues(),Eth_Hal_GetTxErrorCounterValues()	

2.8.2.52 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxLateCollisionPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxLateCollisionPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets with transmit late collision error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets with transmit late collision error.	
Referenced By	Eth_Hal_TxLateCollisionDetected (),Eth_Hal_GetCounterValues(),Eth_Hal_GetTxErrorCounterValues()	

2.8.2.53 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxExcessiveCollisionPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxExcessiveCollisionPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets with transmit excessive collision errors.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets with transmit excessive collision errors.	
Referenced By	Eth_Hal_GetTxErrorCounterValues ()	

2.8.2.54 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxPacketCntGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTxPacketCntGood(const ENET_Type *const pEnet)	
Description	Get the number of good packets transmitted.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good packets transmitted.	
Referenced By	Eth_Hal_GetCounterValues(),Eth_Hal_GetTxErrorCounterValues ()	

2.8.2.55 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxPacketCntGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxPacketCntGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.56 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxOctetCntGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxOctetCntGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of bytes received in good and bad packets.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of bytes received in good and bad packets.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.57 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxBroadcastPacketGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxBroadcastPacketGood(const ENET_Type *const pEnet)	
Description	Get the number of good broadcast packets received.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good broadcast packets received.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.58 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxMulticastPacketGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxMulticastPacketGood(const ENET_Type *const pEnet)	
Description	Get the number of good multicast packets received.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good multicast packets received.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.59 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxCRCErrorPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxCRCErrorPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets received with CRC error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with CRC error.	
Referenced By	Eth_Hal_RxCRCErrorDetected(),Eth_Hal_GetCounterValues(),Eth_Hal_GetRxStats ()	

2.8.2.60 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxAlignmentErrorPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxAlignmentErrorPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets received with alignment error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with alignment error.	
Referenced By	Eth_Hal_RxAlignmentErrorDetected (),Eth_Hal_GetCounterValues()	

2.8.2.61 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxRunErrorPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxRunErrorPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets received with runt error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with runt error.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.62 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxJabberErrorPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxJabberErrorPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets received with jabber error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with jabber error.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.63 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxUndersizePacketGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxUndersizePacketGood(const ENET_Type *const pEnet)	
Description	Get the number of packets received with length less than 64 bytes, without any errors.	

Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with length less than 64 bytes, without any errors.	
Referenced By	Eth_Hal_RxUndersizeFrameDetected (),Eth_Hal_GetCounterValues(),Eth_Hal_GetRxStats ()	

2.8.2.64 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxOverSizeGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxOverSizeGood(const ENET_Type *const pEnet)	
Description	Get the number of packets received without errors and with length greater than maxsize.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received without errors and with length greater than maxsize.	
Referenced By	Eth_Hal_RxOversizeFrameDetected (),Eth_Hal_GetCounterValues (),Eth_Hal_GetRxStats ()	

2.8.2.65 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx64OctetPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx64OctetPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length 64 bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length 64 bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.66 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx65To127OctetPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx65To127OctetPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length between 65 and 127 bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length between 65 and 127 bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.67 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx128To255OctetPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx128To255OctetPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length between 128 and 255 bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length between 128 and 255 bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.68 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx256To511OctetPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx256To511OctetPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length between 256 and 511bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length between 256 and 511bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.69 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx512To1023OctetPacketGoodBad(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx512To1023OctetPacketGoodBad(const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length between 512 and 1023 bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length between 512 and 1023 bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.70 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx1024ToMaxOctetPacketGoodBad (const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRx1024ToMaxOctetPacketGoodBad (const ENET_Type *const pEnet)	
Description	Get the number of good and bad packets received with length 1024 to maxsize bytes.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good and bad packets received with length 1024 to maxsize bytes.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.71 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxUnicastPacketGood(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxUnicastPacketGood(const ENET_Type *const pEnet)	
Description	Get the number of good unicast packets received.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of good unicast packets received.	
Referenced By	Eth_Hal_GetRxStats ()	

2.8.2.72 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxFIFOOverflowPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxFIFOOverflowPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of missed received packets because of FIFO overflow.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of missed received packets because of FIFO overflow.	

Referenced By	Eth_Hal_RxFrameLostDetected (),Eth_Hal_GetCounterValues(),Eth_Hal_GetRxStats()
----------------------	--

2.8.2.73 LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxReceiveErrorPacketCnt(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetRxReceiveErrorPacketCnt(const ENET_Type *const pEnet)	
Description	Get the number of packets received with receive error or Packet Extension error.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	The number of packets received with receive error or Packet Extension error.	
Referenced By	Eth_Hal_GetCounterValues ()	

2.8.2.74 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampControl(ENET_Type *const pEnet, uint32 u32Ctrl)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampControl(ENET_Type *const pEnet, uint32 u32Ctrl)	
Description	Set time stamp control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Ctrl	Control value.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.75 LOCAL_INLINE void ENET_HWA_MAC_SetlimestampRolloverCtrl(ENET_Type *const pEnet, ENET_MAC_TimestampRolloverType eCtrl)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetlimestampRolloverCtrl(ENET_Type *const pEnet, ENET_MAC_TimestampRolloverType eCtrl)	
Description	Set PTP timestamp rollover mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eCtrl	Rollover mode.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.76 LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetAddendRegisterUpdateStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetAddendRegisterUpdateStatus(const ENET_Type *const pEnet)	
Description	Get PTP addend register update status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	PTP addend register update status.	
Referenced By	Eth_Hal_PTPUpdateAddend ()	

2.8.2.77 LOCAL_INLINE void ENET_HWA_MAC_UpdateAddendRegister(ENET_Type *const pEnet)

Function	LOCAL_INLINE void ENET_HWA_MAC_UpdateAddendRegister(ENET_Type *const pEnet)	
Description	Initiate the addend register update operation.	

Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	N/A	
Referenced By	Eth_Hal_PTPUpdateAddend ()	

2.8.2.78 LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetTimestampUpdateStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetTimestampUpdateStatus(const ENET_Type *const pEnet)	
Description	Get timestamp update status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Timestamp update status	
Referenced By	Eth_Hal_PTPUpdateSystemTime ()	

2.8.2.79 LOCAL_INLINE void ENET_HWA_MAC_UpdateTimestamp(ENET_Type *const pEnet)

Function	LOCAL_INLINE void ENET_HWA_MAC_UpdateTimestamp(ENET_Type *const pEnet)	
Description	Initiate the timestamp update operation.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	N/A	
Referenced By	Eth_Hal_PTPUpdateSystemTime ()	

2.8.2.80 LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetTimestampInitStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE ENET_MAC_StatusType ENET_HWA_MAC_GetTimestampInitStatus(const ENET_Type *const pEnet)	
Description	Get timestamp initialize status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Timestamp initialize status.	
Referenced By	Eth_Hal_PTPInitSystemTime ()	

2.8.2.81 LOCAL_INLINE void ENET_HWA_MAC_InitTimestamp(ENET_Type *const pEnet)

Function	LOCAL_INLINE void ENET_HWA_MAC_InitTimestamp(ENET_Type *const pEnet)	
Description	Initialize PTP timestamp.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	N/A	
Referenced By	Eth_Hal_PTPInitSystemTime ()	

2.8.2.82 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampUpdateMethod(ENET_Type *const pEnet, ENET_MAC_TimestampUpdateMethodType eMethod)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampUpdateMethod(ENET_Type *const pEnet, ENET_MAC_TimestampUpdateMethodType eMethod)	
Description	Set timestamp update method (fine or coarse).	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eMethod	Update method.
Returns	N/A	
Referenced By	Eth_Hal_PtPlnit ()	

2.8.2.83 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set timestamp processing enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable or 0U disable timestamp processing.
Returns	N/A	
Referenced By	Eth_Hal_PtPlnit ()	

2.8.2.84 LOCAL_INLINE boolean ENET_HWA_MAC_GetTimestampEnFlag(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_MAC_GetTimestampEnFlag(const ENET_Type *const pEnet)	
Description	Get timestamp processing enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: Timestamp processing is enabled FALSE: Timestamp processing is disabled	
Referenced By	Eth_Hal_GetCurrentTime ()	

2.8.2.85 LOCAL_INLINE void ENET_HWA_MAC_SetSubSecondIncrementReg(ENET_Type *const pEnet, uint32 u32Increment)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetSubSecondIncrementReg(ENET_Type *const pEnet, uint32 u32Increment)	
Description	Set PTP sub-second increment register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Increment	Increment value.
Returns	N/A	
Referenced By	Eth_Hal_PtPlnit ()	

2.8.2.86 LOCAL_INLINE uint32 ENET_HWA_MAC_GetSystemTimeSeconds(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetSystemTimeSeconds(const ENET_Type *const pEnet)	
Description	Get PTP system time seconds.	
Parameters	Parameter	Description

	pEnet	The base address of the ENET instance.
Returns	PTP system time seconds.	
Referenced By	Eth_Hal_GetTransmitStatus(),Eth_Hal_ReadRxDescTimestamp(),Eth_Hal_GetCurrentTime()	

2.8.2.87 LOCAL_INLINE uint32 ENET_HWA_MAC_GetSystemTimeNanoSeconds(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetSystemTimeNanoSeconds(const ENET_Type *const pEnet)	
Description	Get PTP system time nanoseconds.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	PTP system time nanoseconds.	
Referenced By	Eth_Hal_GetCurrentTime ()	

2.8.2.88 LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeSecondsUpdateValue(ENET_Type *const pEnet, uint32 u32Seconds)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeSecondsUpdateValue(ENET_Type *const pEnet, uint32 u32Seconds)	
Description	Set PTP system time seconds update register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Seconds	Seconds update register value.
Returns	N/A	
Referenced By	Eth_Hal_PTPIInitSystemTime (),Eth_Hal_PTPUpdateSystemTime()	

2.8.2.89 LOCAL_INLINE void ENET_HWA_MAC_SetTimeUpdateOperation(ENET_Type *const pEnet, ENET_MAC_SystemTimeOperationType eOperation)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimeUpdateOperation(ENET_Type *const pEnet, ENET_MAC_SystemTimeOperationType eOperation)	
Description	Set PTP system time update operation,add or subtract time.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eOperation	Add or subtract time.
Returns	N/A	
Referenced By	Eth_Hal_PTPIInitSystemTime (),Eth_Hal_PTPUpdateSystemTime()	

2.8.2.90 LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeNanoSecondsUpdateValue(ENET_Type *const pEnet, uint32 u32NanoSeconds)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeNanoSecondsUpdateValue(ENET_Type *const pEnet, uint32 u32NanoSeconds)	
Description	Set PTP system time nanoseconds update value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32NanoSeconds	Nanoseconds update register value.
Returns	N/A	
Referenced By	Eth_Hal_PTPIInitSystemTime (),Eth_Hal_PTPUpdateSystemTime()	

2.8.2.91 LOCAL_INLINE uint32 ENET_HWA_MAC_GetTimestampAddend(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MAC_GetTimestampAddend(const ENET_Type *const pEnet)	
Description	Get PTP timestamp addend register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	Timestamp addend value.	
Referenced By	Eth_Hal_SetCorrectionTime ()	

2.8.2.92 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampAddend(ENET_Type *const pEnet, uint32 u32Time)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampAddend(ENET_Type *const pEnet, uint32 u32Time)	
Description	Set PTP timestamp addend register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Time	Timestamp addend value.
Returns	N/A	
Referenced By	Eth_Hal_PTUpdateAddend ()	

2.8.2.93 LOCAL_INLINE uint16 ENET_HWA_MAC_GetSystemTimeHigherWordSeconds(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint16 ENET_HWA_MAC_GetSystemTimeHigherWordSeconds(const ENET_Type *const pEnet)	
Description	Get PTP system time seconds bits[47:32] value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	PTP system time seconds bits[47:32] value.	
Referenced By	Eth_Hal_GetTransmitStatus (),Eth_Hal_ReadRxDescTimestamp (),Eth_Hal_GetCurrentTime()	

2.8.2.94 LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeHigherWordSeconds(ENET_Type *const pEnet, uint16 u16Seconds)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetSystemTimeHigherWordSeconds(ENET_Type *const pEnet, uint16 u16Seconds)	
Description	Set PTP system time seconds bits[47:32] value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u16Seconds	System time seconds bits[47:32] value.
Returns	N/A	
Referenced By	Eth_Hal_PTPIInitSystemTime ()	

2.8.2.95 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampIngressCorrectionInNanoseconds(ENET_Type *const pEnet, int32 s32NanoSec)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampIngressCorrectionInNanoseconds(ENET_Type *const pEnet, int32 s32NanoSec)	
-----------------	---	--

Description	Set timestamp ingress correction in nanoseconds, used for timestamp in ingress path.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	s32NanoSec	Correction value.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.96 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEgressCorrectionInNanoseconds(ENET_Type *const pEnet, sint32 s32NanoSec)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEgressCorrectionInNanoseconds(ENET_Type *const pEnet, sint32 s32NanoSec)	
Description	Set timestamp egress correction in nanoseconds, used for timestamp in egress path.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	s32NanoSec	Correction value.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.97 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampIngressCorrectionInSubNanoseconds(ENET_Type *const pEnet, sint8 s8SubNanoSec)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampIngressCorrectionInSubNanoseconds(ENET_Type *const pEnet, sint8 s8SubNanoSec)	
Description	Set timestamp ingress correction in sub-nanoseconds, used for timestamp in ingress path.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	s8SubNanoSec	Correction value.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.98 LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEgressCorrectionInSubNanoseconds(ENET_Type *const pEnet, sint8 s8SubNanoSec)

Function	LOCAL_INLINE void ENET_HWA_MAC_SetTimestampEgressCorrectionInSubNanoseconds(ENET_Type *const pEnet, sint8 s8SubNanoSec)	
Description	Set timestamp ingress correction in sub-nanoseconds, used for timestamp in ingress path.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	s8SubNanoSec	Correction value.
Returns	N/A	
Referenced By	Eth_Hal_PtpInit ()	

2.8.2.99 LOCAL_INLINE void ENET_HWA_MTL_SetOperationMode(ENET_Type *const pEnet, uint32 u32OpMode)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetOperationMode(ENET_Type *const pEnet, uint32 u32OpMode)	
Description	Set MTL operation mode register value.	
Parameters	Parameter	Description

	pEnet	The base address of the ENET instance.
	u32OpMode	MTL operation mode.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.100 LOCAL_INLINE void ENET_HWA_MTL_SetRxDABasedDMAChannelSelectionEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxDABasedDMAChannelSelectionEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set DA-based DMA channel selection enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.101 LOCAL_INLINE void ENET_HWA_MTL_SetRxDMAChannelMap(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxDMAChannelMap(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_DMA_ChannelType eDMAChannel)	
Description	Set DMA channel mapped to eQueue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	eDMAChannel	Index of DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.102 LOCAL_INLINE uint32 ENET_HWA_MTL_GetEstControl(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MTL_GetEstControl(const ENET_Type *const pEnet)	
Description	Get enhancements to scheduled transmission control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	EST control register value.	
Referenced By	Eth_Hal_TimeAwareShaperInit()	

2.8.2.103 LOCAL_INLINE void ENET_HWA_MTL_SetEstControl(ENET_Type *const pEnet, uint32 u32EstControl)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEstControl(ENET_Type *const pEnet, uint32 u32EstControl)	
Description	Set enhancements to scheduled transmission control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32EstControl	EST control value.
Returns	N/A	

Referenced By	Eth_Hal_TimeAwareShaperInit()
----------------------	-------------------------------

2.8.2.104 LOCAL_INLINE void ENET_HWA_MTL_SetEstGateControlListAddress(ENET_Type *const pEnet, uint16 u16Address)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEstGateControlListAddress(ENET_Type *const pEnet, uint16 u16Address)	
Description	Set GCL address or GCL related registers address depend on GCRR bit.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u16Address	GCL address or GCL related registers address.
Returns	N/A	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.105 LOCAL_INLINE void ENET_HWA_MTL_SetGateControlReleatedRegistersEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetGateControlReleatedRegistersEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set gate control releated registers enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable or 0U disable
Returns	N/A	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.106 LOCAL_INLINE void ENET_HWA_MTL_SetGclOperation(ENET_Type *const pEnet, ENET_MTL_GclOperationType eOperation)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetGclOperation(ENET_Type *const pEnet, ENET_MTL_GclOperationType eOperation)	
Description	Set GCL operation mode (read or write).	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eOperation	read or write.
Returns	N/A	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.107 LOCAL_INLINE uint8 ENET_HWA_MTL_GetGclOperationBusyState(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint8 ENET_HWA_MTL_GetGclOperationBusyState(const ENET_Type *const pEnet)	
Description	Get GCL operation busy status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	1U: Start Read/Write operation is enabled,status is busy 0U: Start Read/Write operation is disabled,status is idle	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.108 LOCAL_INLINE void ENET_HWA_MTL_SetGclOperationStartFlag(ENET_Type *const pEnet,uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetGclOperationStartFlag(ENET_Type *const pEnet,uint8 bEnable)	
Description	Start GCL Read/Write operation.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U start operation, 0U no effect.
Returns	N/A	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.109 LOCAL_INLINE void ENET_HWA_MTL_SetGclData(ENET_Type *const pEnet,uint32 u32Data)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetGclData(ENET_Type *const pEnet,uint32 u32Data)	
Description	Set GCL data register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Data	GCL data value.
Returns	N/A	
Referenced By	Eth_Hal_GclWrite()	

2.8.2.110 LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionClassification(ENET_Type *const pEnet, uint8 u8value)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionClassification(ENET_Type *const pEnet, uint8 u8value)	
Description	Set the Classification of Queues.A bit indicates a queue, when '1' indicates the corresponding Queue must be classified as preemptable, when '0' Queue is classified as express.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u8value	Preemption classification value.
Returns	N/A	
Referenced By	Eth_Hal_TimeAwareShaperInit ()	

2.8.2.111 LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionReleaseAdvance(ENET_Type *const pEnet, uint16 u16value)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionReleaseAdvance(ENET_Type *const pEnet, uint16 u16value)	
Description	Set the maximum time in nanoseconds that can elapse between issuing a RELEASE to the MAC and the MAC being ready to resume transmission of preemptable frames.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u16value	Release Advance time.
Returns	N/A	
Referenced By	Eth_Hal_TimeAwareShaperInit ()	

2.8.2.112 LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionHoldAdvance(ENET_Type *const pEnet, uint16 u16value)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetPreemptionHoldAdvance(ENET_Type *const pEnet, uint16 u16value)	
Description	Set the maximum time in nanoseconds that can elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit any preemptable frame.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u16value	Hold Advance time.
Returns	N/A	
Referenced By	Eth_Hal_TimeAwareShaperInit ()	

2.8.2.113 LOCAL_INLINE uint32 ENET_HWA_MTL_GetEccControl(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MTL_GetEccControl(const ENET_Type *const pEnet)	
Description	Get the ECC Control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	ECC Control register value.	
Referenced By	Eth_Hal_MTLecclrqHandler()	

2.8.2.114 LOCAL_INLINE void ENET_HWA_MTL_SetEccControlEstEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEccControlEstEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable the ECC protection for EST Gate Control List memory.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.115 LOCAL_INLINE void ENET_HWA_MTL_SetEccControlRxFifoEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEccControlRxFifoEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable the ECC protection for MTL RX FIFO memory.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.116 LOCAL_INLINE void ENET_HWA_MTL_SetEccControlTxFifoEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEccControlTxFifoEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable the ECC protection for MTL TX FIFO memory.	
Parameters	Parameter	Description

	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.117 LOCAL_INLINE void ENET_HWA_MTL_SetEstMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetEstMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable correctable error interrupt for EST memory ECC protection.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.118 LOCAL_INLINE void ENET_HWA_MTL_SetRxMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable correctable error interrupt for Rx memory ECC protection.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.119 LOCAL_INLINE void ENET_HWA_MTL_SetTxMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxMemCorrectableErrIntEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Enable correctable error interrupt for Tx memory ECC protection.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.120 LOCAL_INLINE uint32 ENET_HWA_MTL_GetEcclInterruptStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE uint32 ENET_HWA_MTL_GetEcclInterruptStatus(const ENET_Type *const pEnet)	
Description	Get MTL ECC interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	MTL ECC interrupt status.	

Referenced By	Eth_Hal_MTLecclrqHandler(),Eth_Hal_ClearSystemInterruptStatus()
----------------------	---

2.8.2.121 LOCAL_INLINE void ENET_HWA_MTL_ClearEccInterruptStatus(ENET_Type *const pEnet, uint32 u32Status)

Function	LOCAL_INLINE void ENET_HWA_MTL_ClearEccInterruptStatus(ENET_Type *const pEnet, uint32 u32Status)	
Description	Clear MTL ECC interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Status	The status bits to be cleared.
Returns	N/A	
Referenced By	Eth_Hal_MTLecclrqHandler(),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.122 LOCAL_INLINE void ENET_HWA_MTL_SetDppOddParityEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetDppOddParityEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set the MTL Data Path Odd Parity enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.123 LOCAL_INLINE boolean ENET_HWA_MTL_GetDppProtectionEnFlag(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_MTL_GetDppProtectionEnFlag(const ENET_Type *const pEnet)	
Description	Get MTL Data Path Parity Protection enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: Data Path Parity Protection is enabled.	
	FALSE: Data Path Parity Protection is not enabled.	
Referenced By	Eth_Hal_MTLDppFsmIrqHandler ()	

2.8.2.124 LOCAL_INLINE void ENET_HWA_MTL_SetDppProtectionEnFlag(ENET_Type *const pEnet, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetDppProtectionEnFlag(ENET_Type *const pEnet, uint8 bEnable)	
Description	Set MTL Data Path Parity Protection enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	bEnable	1U enable, 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_SafetyInit ()	

2.8.2.125 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueSize(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8QueueSize)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueSize(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8QueueSize)
-----------------	---

Description	Set MTL transmit queue size.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u8QueueSize	queue size in blocks of 256 bytes,range(0~15).
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.126 LOCAL_INLINE void ENET_HWA_MTL_SetTxThreshold(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_TxThresholdType eThreshold)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxThreshold(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_TxThresholdType eThreshold)	
Description	Set the threshold level of the MTL Tx Queue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	eThreshold	Threshold level.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.127 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueEnStatus(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_QueueEnableType eEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueEnStatus(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_QueueEnableType eEnable)	
Description	Set the transmit queue enable mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	Queue enable mode.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.128 LOCAL_INLINE void ENET_HWA_MTL_FlushTxQueue(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue)

Function	LOCAL_INLINE void ENET_HWA_MTL_FlushTxQueue(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue)	
Description	Flush transmit queue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
Returns	N/A	
Referenced By	Eth_Hal_Start (),Eth_Hal_Stop()	

2.8.2.129 LOCAL_INLINE void ENET_HWA_MTL_SetAVAlgorithmCBSEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetAVAlgorithmCBSEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set AV algorithm for strict priority or credit based shaper.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U use credit based shaper. 0U use strict priority.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.130 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueQuantumOrWeight(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32QuantumWeight)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueQuantumOrWeight(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32QuantumWeight)	
Description	Set the quantum or weights for transmit algorithm (DWRR WRR WFQ).	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u32QuantumWeight	Quantum or weights.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.131 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueSendSlopeCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32SendSlopeCredit)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueSendSlopeCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32SendSlopeCredit)	
Description	Set the sendSlope credit value required for the credit-based shaper algorithm for the Queue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u32SendSlopeCredit	SendSlope credit for credit based shaper.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.132 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueHiCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32hiCredit)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueHiCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32hiCredit)	
Description	Set hiCredit value required for the credit-based shaper algorithm for the Queue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.

	u32HiCredit	HiCredit for credit based shaper.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.133 LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueLoCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32LoCredit)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetTxQueueLoCredit(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint32 u32LoCredit)	
Description	Set hiCredit value required for the credit-based shaper algorithm for the Queue.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u32LoCredit	LoCredit for credit based shaper.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.134 LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueSize(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8QueueSize)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueSize(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8QueueSize)	
Description	Set receive queue size.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u8QueueSize	Receive queues size in blocks of 256 bytes, range(0~15).
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.135 LOCAL_INLINE void ENET_HWA_MTL_SetRxDeactivateFlowCtrlThres(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxFlowCtrlThresholdType eThreshold)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxDeactivateFlowCtrlThres(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxFlowCtrlThresholdType eThreshold)	
Description	Set the threshold of receive queue at which the flow control is deactivated.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	eThreshold	Flow control deactivating threshold.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.136 LOCAL_INLINE void ENET_HWA_MTL_SetRxActivateFlowCtrlThres(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxFlowCtrlThresholdType eThreshold)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxActivateFlowCtrlThres(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxFlowCtrlThresholdType eThreshold)	
-----------------	---	--

Description	Set the threshold of receive queue at which the flow control is deactivated.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	eThreshold	Flow control activating threshold.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.137 LOCAL_INLINE void ENET_HWA_MTL_SetRxHwFlowCtrlEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxHwFlowCtrlEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set receive queue hardware flow control enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U enable or 0U disable hardware flow control.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.138 LOCAL_INLINE void ENET_HWA_MTL_SetForwardTCPChecksumErrorPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetForwardTCPChecksumErrorPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set forwarding of TCP/IP checksum error packets enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U forward or 0U drop TCP/IP checksum error packets.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.139 LOCAL_INLINE void ENET_HWA_MTL_SetForwardErrorPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetForwardErrorPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set forward error packets enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U forward or 0U drop error packets.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.140 LOCAL_INLINE void ENET_HWA_MTL_SetForwardUndersizedPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetForwardUndersizedPacketEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set forward undersized good packets enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U forward or 0U drop undersized packets.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.141 LOCAL_INLINE void ENET_HWA_MTL_SetRxThreshold(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxThresholdType eThreshold)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxThreshold(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, ENET_MTL_RxThresholdType eThreshold)	
Description	Set the receive queue threshold.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	eThreshold	Receive queue threshold.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.142 LOCAL_INLINE void ENET_HWA_MTL_SetRxQueuePacketArbitrationEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxQueuePacketArbitrationEnFlag(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 bEnable)	
Description	Set receive queue packet arbitration enable flag.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	bEnable	1U enable or 0U disable.
Returns	N/A	
Referenced By	Eth_Hal_MTLInit ()	

2.8.2.143 LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueWeight(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8Weight)

Function	LOCAL_INLINE void ENET_HWA_MTL_SetRxQueueWeight(ENET_Type *const pEnet, ENET_MTL_QueueType eQueue, uint8 u8Weight)	
Description	Set receive queue weight.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eQueue	Index of MTL queue.
	u8Weight	Receive queue weight.

Returns	N/A
Referenced By	Eth_Hal_MTLInit ()

2.8.2.144 LOCAL_INLINE void ENET_HWA_DMA_SetMode(ENET_Type *const pEnet, uint32 u32Mode)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetMode(ENET_Type *const pEnet, uint32 u32Mode)	
Description	Set DMA bus mode register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Mode	Index of MTL queue.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.145 LOCAL_INLINE ENET_DMA_StatusType ENET_HWA_DMA_GetSwResetStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE ENET_DMA_StatusType ENET_HWA_DMA_GetSwResetStatus(const ENET_Type *const pEnet)	
Description	Get DMA software reset status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	ENET_DMA_STATUS_IDLE DMA is idle ENET_DMA_STATUS_ACTIVE DMA is busy	
Referenced By	Eth_Hal_ResetController (),Eth_Hal_CheckAccessController()	

2.8.2.146 LOCAL_INLINE void ENET_HWA_DMA_SwReset(ENET_Type *const pEnet)

Function	LOCAL_INLINE void ENET_HWA_DMA_SwReset(ENET_Type *const pEnet)	
Description	Initiate DMA software reset.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	N/A	
Referenced By	Eth_Hal_ResetController ()	

2.8.2.147 LOCAL_INLINE void ENET_HWA_DMA_SetInterruptMode(ENET_Type *const pEnet, ENET_DMA_IntModeType eIntMode)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetInterruptMode(ENET_Type *const pEnet, ENET_DMA_IntModeType eIntMode)	
Description	Set DMA interrupt mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eIntMode	DMA interrupt mode.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.148 LOCAL_INLINE void ENET_HWA_DMA_SetSysBusMode(ENET_Type *const pEnet, uint32 u32Mode)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetSysBusMode(ENET_Type *const pEnet, uint32 u32Mode)	
Description	Set DMA sys bus mode register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	u32Mode	DMA sys bus mode.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.149 LOCAL_INLINE boolean ENET_HWA_DMA_GetMACIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetMACIntStatus(const ENET_Type *const pEnet)	
Description	Get MAC interrupt event in DMA interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: MAC interrupt event detected FALSE: MAC interrupt event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt ()	

2.8.2.150 LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLIntStatus(const ENET_Type *const pEnet)	
Description	Get MTL interrupt event in DMA interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: MTL interrupt event detected FALSE: MTL interrupt event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt ()	

2.8.2.151 LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAChannelIntStatus(const ENET_Type *const pEnet, ENET_DMA_ChannelType eChannel)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAChannelIntStatus(const ENET_Type *const pEnet, ENET_DMA_ChannelType eChannel)	
Description	Get DMA channel interrupt event in DMA interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eChannel	DMA channel.
Returns	TRUE: DMA interrupt event detected FALSE: DMA interrupt event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt ()	

2.8.2.152 LOCAL_INLINE boolean ENET_HWA_DMA_GetMACSafetyUncorrectableIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetMACSafetyUncorrectableIntStatus(const ENET_Type *const pEnet)	
Description	Get MAC safety uncorrectable error event in DMA safety interrupt status.	
Parameters	Parameter	Description

	pEnet	The base address of the ENET instance.
Returns	TRUE: MAC safety uncorrectable error event detected FALSE: MAC safety uncorrectable error event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt (),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.153 LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLSafetyUncorrectableIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLSafetyUncorrectableIntStatus(const ENET_Type *const pEnet)	
Description	Get MTL safety uncorrectable error event in DMA safety interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: MTL safety uncorrectable error event detected FALSE: MTL safety uncorrectable error event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt (),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.154 LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLSafetyCorrectableIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetMTLSafetyCorrectableIntStatus(const ENET_Type *const pEnet)	
Description	Get MTL safety correctable error event in DMA safety interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: MTL safety correctable error event detected FALSE: MTL safety correctable error event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt (),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.155 LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAEccUncorrectableIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAEccUncorrectableIntStatus(const ENET_Type *const pEnet)	
Description	Get DMA ECC uncorrectable error event in DMA safety interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: DMA safety uncorrectable error event detected FALSE: DMA safety uncorrectable error event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt (),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.156 LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAEccCorrectableIntStatus(const ENET_Type *const pEnet)

Function	LOCAL_INLINE boolean ENET_HWA_DMA_GetDMAEccCorrectableIntStatus(const ENET_Type *const pEnet)	
Description	Get DMA ECC correctable error event in DMA safety interrupt status.	

Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
Returns	TRUE: DMA safety correctable error event detected FALSE: DMA safety correctable error event not detected	
Referenced By	Eth_Hal_ProcessSystemInterrupt (),Eth_Hal_ClearSystemInterruptStatus()	

2.8.2.157 LOCAL_INLINE void ENET_HWA_DMA_SetChannelCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32ChannelCtrl)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetChannelCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32ChannelCtrl)	
Description	Set DMA channel control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32ChannelCtrl	Control value.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.158 LOCAL_INLINE void ENET_HWA_DMA_SetDescriptorSkipLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetDescriptorSkipLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)	
Description	Set DMA descriptor skip length.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u8Length	Descriptor skip length.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.159 LOCAL_INLINE void ENET_HWA_DMA_Set8xPBLModeEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_DMA_Set8xPBLModeEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)	
Description	Set DMA burst length multiply by 8 mode.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	bEnable	1U enable or 0U disable x8 mode.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.160 LOCAL_INLINE void ENET_HWA_DMA_SetChannelTxCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxCtrl)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetChannelTxCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxCtrl)		
Description	Set DMA transmit channel control register value.		
Parameters	Parameter	Description	
	pEnet	The base address of the ENET instance.	
	eDMAChannel	DMA channel.	
	u32TxCtrl	Control value.	
Returns	N/A		
Referenced By	Eth_Hal_DMAInit ()		

2.8.2.161 LOCAL_INLINE void ENET_HWA_DMA_SetEnhancedDescriptorEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetEnhancedDescriptorEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)		
Description	Set DMA enhanced descriptor enable flag.		
Parameters	Parameter	Description	
	pEnet	The base address of the ENET instance.	
	eDMAChannel	DMA channel.	
	bEnable	1U enable or 0U disable enhanced descriptor.	
Returns	N/A		
Referenced By	Eth_Hal_DMAInit ()		

2.8.2.162 LOCAL_INLINE void ENET_HWA_DMA_SetTxBurstLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetTxBurstLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)		
Description	Set DMA transmit programmable burst length.		
Parameters	Parameter	Description	
	pEnet	The base address of the ENET instance.	
	eDMAChannel	DMA channel.	
	u8Length	Transmit burst length.	
Returns	N/A		
Referenced By	Eth Hal DMAInit ()		

2.8.2.163 LOCAL_INLINE void ENET_HWA_DMA_StartTx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_StartTx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Start DMA transmission.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	

Referenced By	Eth_Hal_Start ()
----------------------	------------------

2.8.2.164 LOCAL_INLINE void ENET_HWA_DMA_StopTx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_StopTx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Stop DMA transmission.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_Stop ()	

2.8.2.165 LOCAL_INLINE void ENET_HWA_DMA_SetChannelRxCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxCtrl)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetChannelRxCtrl(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxCtrl)	
Description	Set DMA receive channel control register value.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32RxCtrl	Control value.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.166 LOCAL_INLINE void ENET_HWA_DMA_SetRxPacketFlushEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxPacketFlushEnFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 bEnable)	
Description	Set DMA transmit programmable burst length.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	bEnable	1U enable or 0U disable flush receive queues when RX DMA stopped.
Returns	N/A	
Referenced By	Eth_Hal_Stop ()	

2.8.2.167 LOCAL_INLINE void ENET_HWA_DMA_SetRxBurstLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxBurstLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint8 u8Length)	
Description	Set DMA receive programmable burst length.	
Parameters	Parameter	Description

	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u8Length	Receive burst length.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit ()	

2.8.2.168 LOCAL_INLINE void ENET_HWA_DMA_SetRxBufferSize(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16BufferSize)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxBufferSize(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16BufferSize)	
Description	Set DMA receive buffer size.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u16BufferSize	Receive buffer size.
Returns	N/A	
Referenced By	Eth_Hal_RxBuffersInit ()	

2.8.2.169 LOCAL_INLINE void ENET_HWA_DMA_StartRx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_StartRx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Start DMA reception.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_Start ()	

2.8.2.170 LOCAL_INLINE void ENET_HWA_DMA_StopRx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_StopRx(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Stop DMA reception.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_Stop ()	

2.8.2.171 LOCAL_INLINE void ENET_HWA_DMA_SetTxDescListHeadAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxAddr)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetTxDescListHeadAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxAddr)	
-----------------	--	--

Description	Set DMA channel Tx descriptor list header address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32TxAddr	Tx descriptor list header address.
Returns	N/A	
Referenced By	Eth_Hal_TxBuffersInit ()	

2.8.2.172 LOCAL_INLINE void ENET_HWA_DMA_SetRxDescListHeadAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxAddr)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxDescListHeadAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxAddr)	
Description	Set DMA channel Rx descriptor list header address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32RxAddr	Rx descriptor list header address.
Returns	N/A	
Referenced By	Eth_Hal_RxBuffersInit ()	

2.8.2.173 LOCAL_INLINE uint32 ENET_HWA_DMA_GetTxDescListTailAddr(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE uint32 ENET_HWA_DMA_GetTxDescListTailAddr(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Get DMA channel Tx descriptor list tail address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	Tx descriptor list tail address.	
Referenced By	Eth_Hal_SendFrame ()	

2.8.2.174 LOCAL_INLINE void ENET_HWA_DMA_SetTxDescListTailAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxAddr)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetTxDescListTailAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32TxAddr)	
Description	Set DMA channel Tx descriptor list tail address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32TxAddr	Tx descriptor list tail address.
Returns	N/A	
Referenced By	Eth_Hal_TxBuffersInit(),Eth_Hal_SendFrame()	

2.8.2.175 LOCAL_INLINE uint32 ENET_HWA_DMA_GetRxDescListTailAddr(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE uint32 ENET_HWA_DMA_GetRxDescListTailAddr(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Get DMA channel Rx descriptor list tail address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	Rx descriptor list tail address.	
Referenced By	Eth_Hal_ReadFrame()	

2.8.2.176 LOCAL_INLINE void ENET_HWA_DMA_SetRxDescListTailAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxAddr)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxDescListTailAddr(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32RxAddr)	
Description	Set DMA channel Rx descriptor list tail address.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32RxAddr	Rx descriptor list tail address.
Returns	N/A	
Referenced By	Eth_Hal_RxBuffersInit(), Eth_Hal_ReleaseRxDescriptors()	

2.8.2.177 LOCAL_INLINE void ENET_HWA_DMA_SetTxDescRingLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16Length)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetTxDescRingLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16Length)	
Description	Set DMA channel Tx descriptor ring length.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u16Length	Tx descriptor ring length.
Returns	N/A	
Referenced By	Eth_Hal_TxBuffersInit()	

2.8.2.178 LOCAL_INLINE void ENET_HWA_DMA_SetRxDescRingLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16Length)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetRxDescRingLength(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint16 u16Length)	
Description	Set DMA channel Rx descriptor ring length.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u16Length	Rx descriptor ring length.
Returns	N/A	

Referenced By	Eth_Hal_RxBuffersInit()
----------------------	-------------------------

2.8.2.179 LOCAL_INLINE uint32 ENET_HWA_DMA_GetEnabledInterrupts(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE uint32 ENET_HWA_DMA_GetEnabledInterrupts(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Get DMA channel enabled interrupts.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	DMA channel enabled interrupts.	
Referenced By	Enet_DMAIrqHandler(),Eth_Hal_ProcessTxInterrupt(),Eth_Hal_ProcessRxInterrupt()	

2.8.2.180 LOCAL_INLINE void ENET_HWA_DMA_SetEnabledInterrupts(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32IntEnable)

Function	LOCAL_INLINE void ENET_HWA_DMA_SetEnabledInterrupts(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, uint32 u32IntEnable)	
Description	Set DMA channel interrupts enable state.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	u32IntEnable	Interrupts to be enabled.
Returns	N/A	
Referenced By	Eth_Hal_DMAInit(),Eth_Hal_Stop(),Eth_Hal_ResetController()	

2.8.2.181 LOCAL_INLINE void ENET_HWA_DMA_EnableInterrupt(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, ENET_DMA_InterruptFlagType eIntFlag)

Function	LOCAL_INLINE void ENET_HWA_DMA_EnableInterrupt(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, ENET_DMA_InterruptFlagType eIntFlag)	
Description	Enable one of the DMA channel interrupts.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	eIntFlag	Interrupts defined in ENET_DMA_InterruptFlagType.
Returns	N/A	
Referenced By	Eth_Hal_Start()	

2.8.2.182 LOCAL_INLINE void ENET_HWA_DMA_DisableInterrupt(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, ENET_DMA_InterruptFlagType eIntFlag)

Function	LOCAL_INLINE void ENET_HWA_DMA_DisableInterrupt(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel, ENET_DMA_InterruptFlagType eIntFlag)	
Description	Disable one of the DMA channel interrupts.	

Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
	eIntFlag	Interrupts defined in ENET_DMA_InterruptFlagType.
Returns	N/A	
Referenced By	Enet_DMAIrqHandler()	

2.8.2.183 LOCAL_INLINE uint32 ENET_HWA_DMA_GetCurrentStatus(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE uint32 ENET_HWA_DMA_GetCurrentStatus(const ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Get DMA current status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	DMA channel status.	
Referenced By	Enet_DMAIrqHandler (),Eth_Hal_ProcessTxInterrupt (),Eth_Hal_ProcessRxInterrupt ()	

2.8.2.184 LOCAL_INLINE void ENET_HWA_DMA_ClearAllAbnormalInterruptFlags(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearAllAbnormalInterruptFlags(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Clear all DMA abnormal interrupts.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Enet_DMAIrqHandler ()	

2.8.2.185 LOCAL_INLINE void ENET_HWA_DMA_ClearAllNormalInterruptFlags(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearAllNormalInterruptFlags(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Clear all DMA normal interrupts.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Enet_DMAIrqHandler ()	

2.8.2.186 LOCAL_INLINE void ENET_HWA_DMA_ClearRxProcessStoppedFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearRxProcessStoppedFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
-----------------	--	--

Description	Clear DMA receive process stopped status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_Start ()	

2.8.2.187 LOCAL_INLINE void ENET_HWA_DMA_ClearRxInterruptFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearRxInterruptFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Clear DMA receive interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_ClearRxInterruptStatus (),Eth_Hal_ProcessRxInterrupt()	

2.8.2.188 LOCAL_INLINE void ENET_HWA_DMA_ClearTxProcessStoppedFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearTxProcessStoppedFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Clear DMA transmit process stopped status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_Start ()	

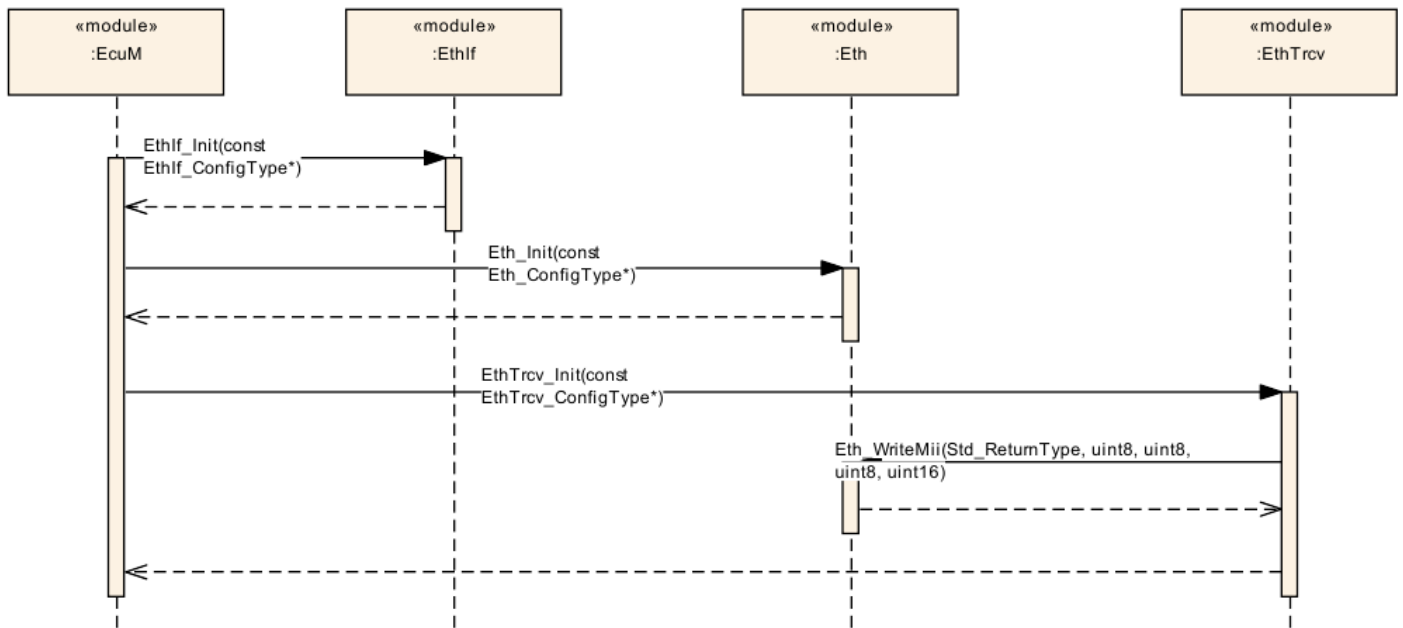
2.8.2.189 LOCAL_INLINE void ENET_HWA_DMA_ClearTxInterruptFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)

Function	LOCAL_INLINE void ENET_HWA_DMA_ClearTxInterruptFlag(ENET_Type *const pEnet, ENET_DMA_ChannelType eDMAChannel)	
Description	Clear DMA transmit interrupt status.	
Parameters	Parameter	Description
	pEnet	The base address of the ENET instance.
	eDMAChannel	DMA channel.
Returns	N/A	
Referenced By	Eth_Hal_ClearTxInterruptStatus (),Eth_Hal_ProcessTxInterrupt()	

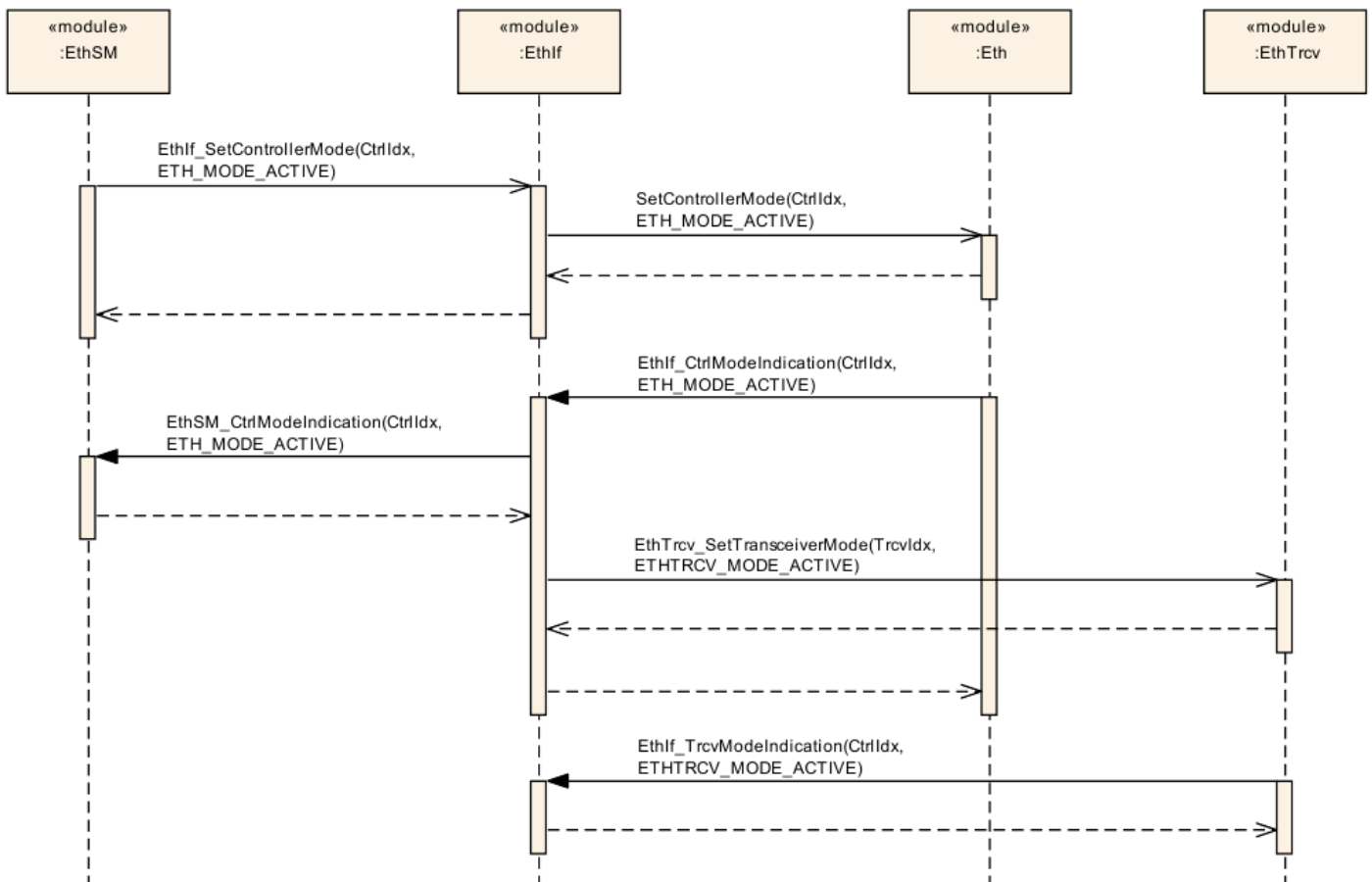
2.9 API Sequence Diagram

2.9.1 Initialization

The ECU State Manager (EcuM) is responsible for calling the initialization function.

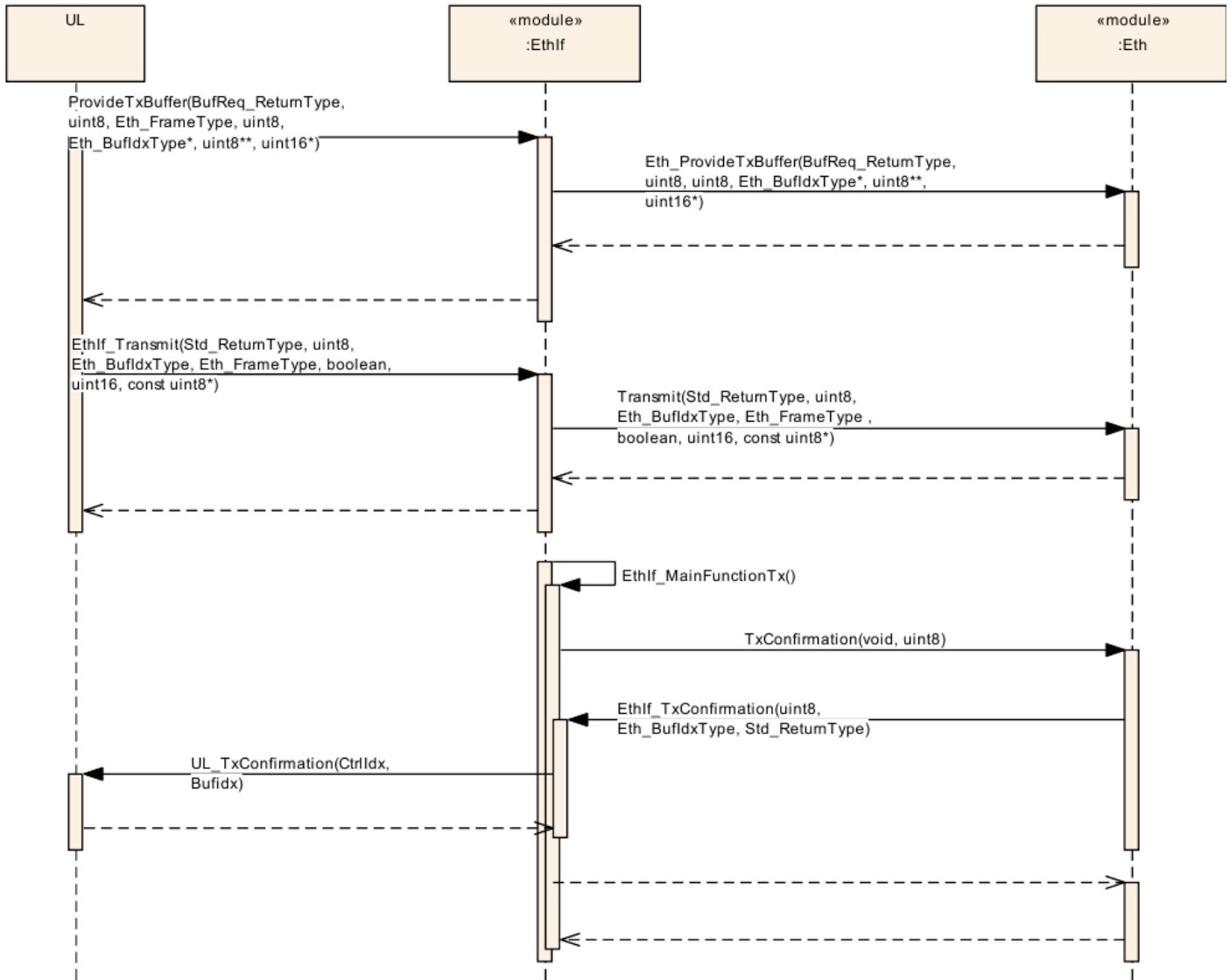


2.9.2 Communication Initialization

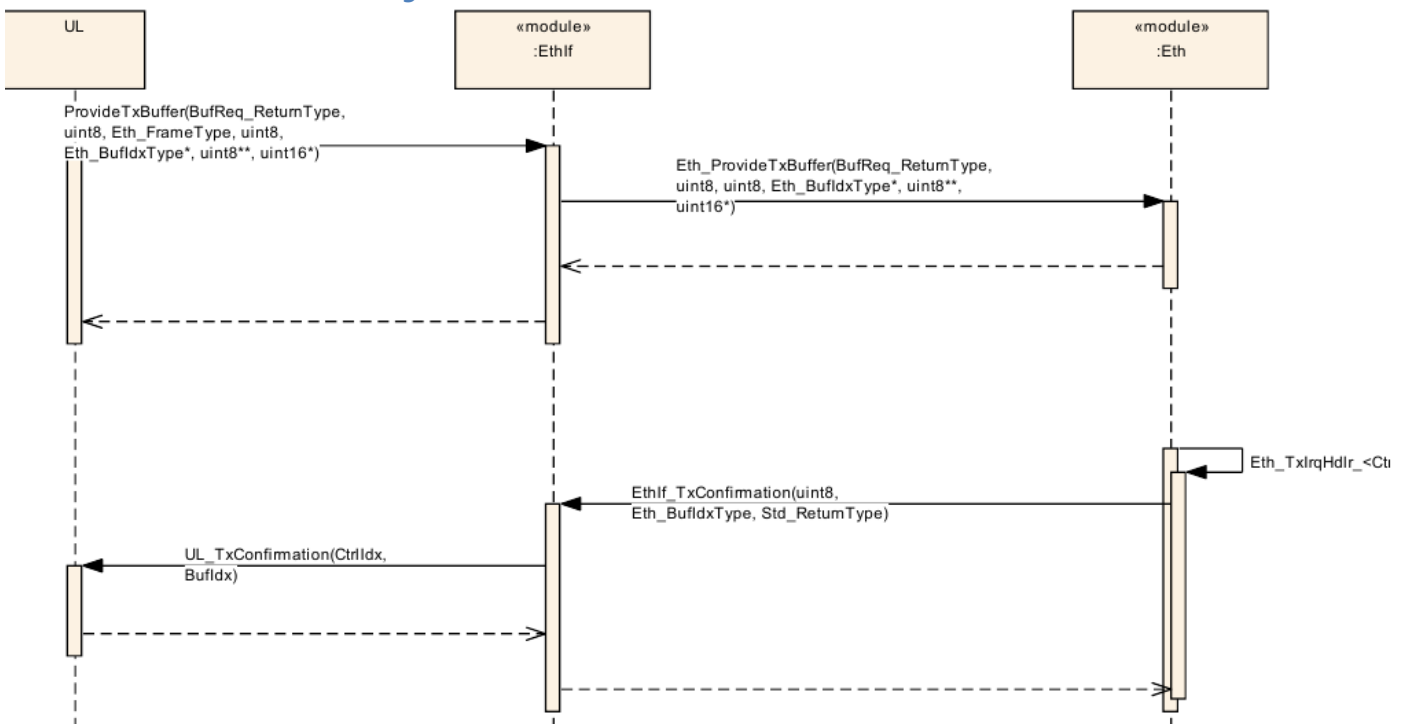


2.9.3 Data Transmission

2.9.3.1 Transmission in Polling Mode

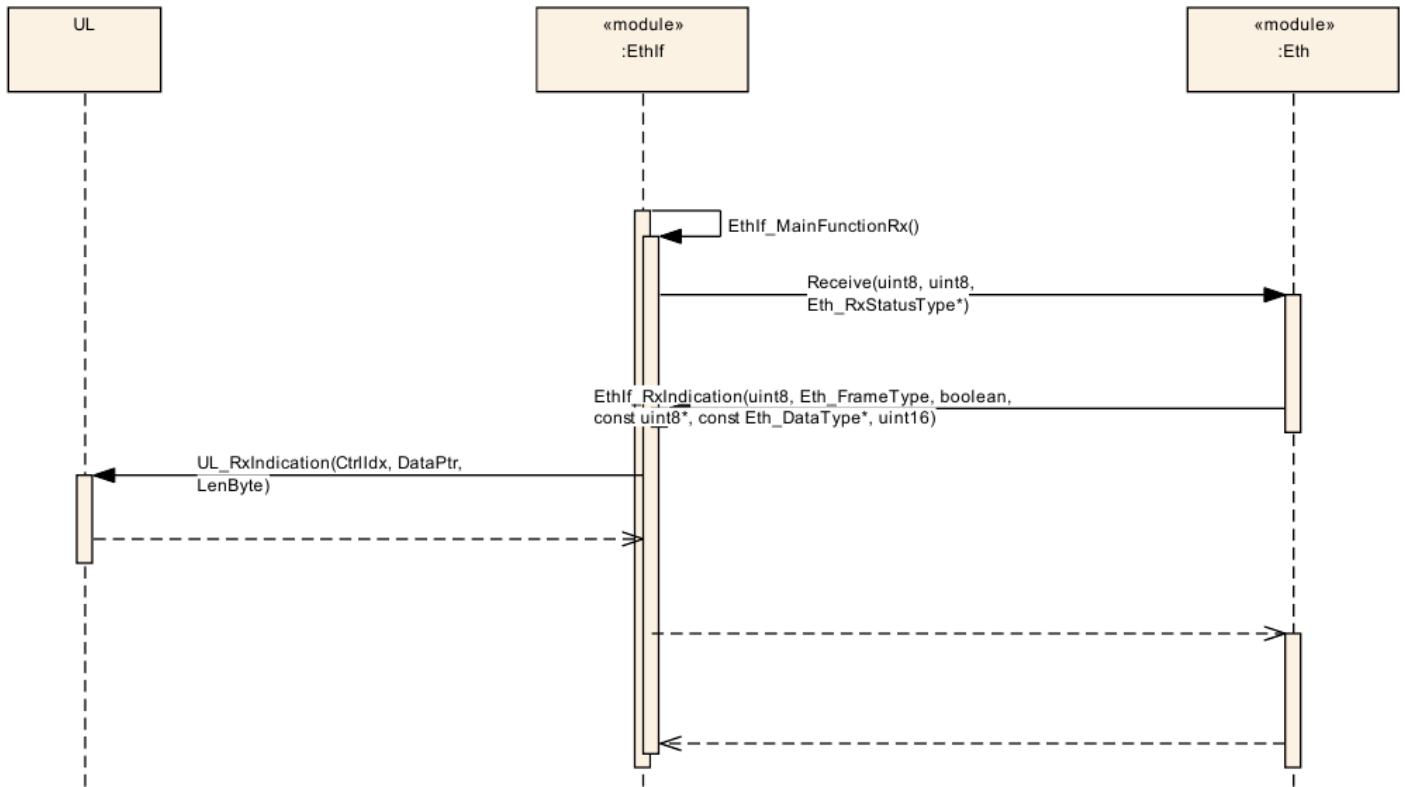


2.9.3.2 Transmission in Polling Mode

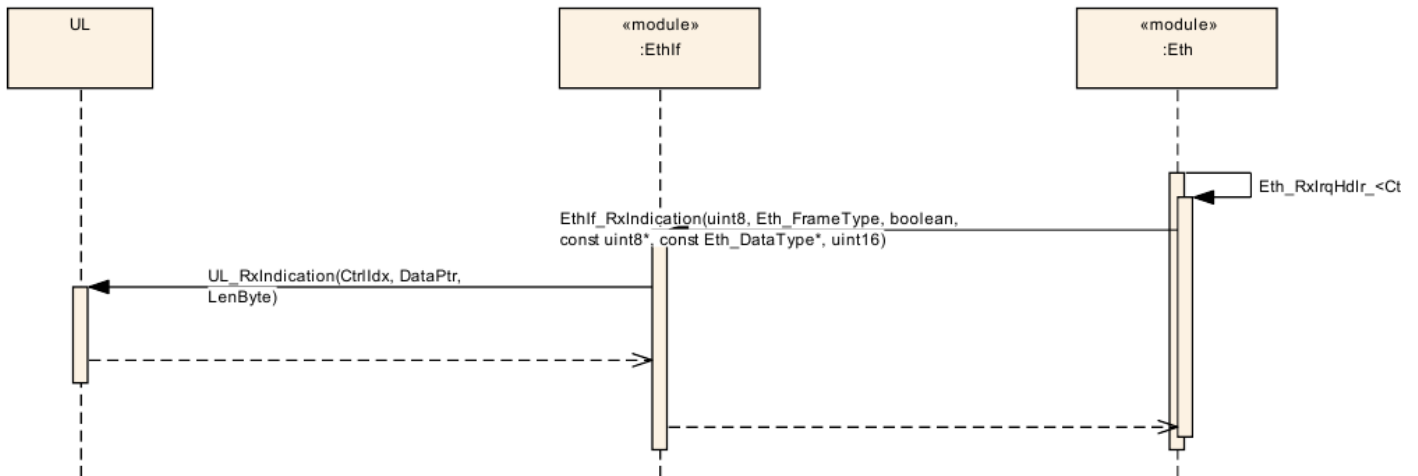


2.9.4 Data Reception

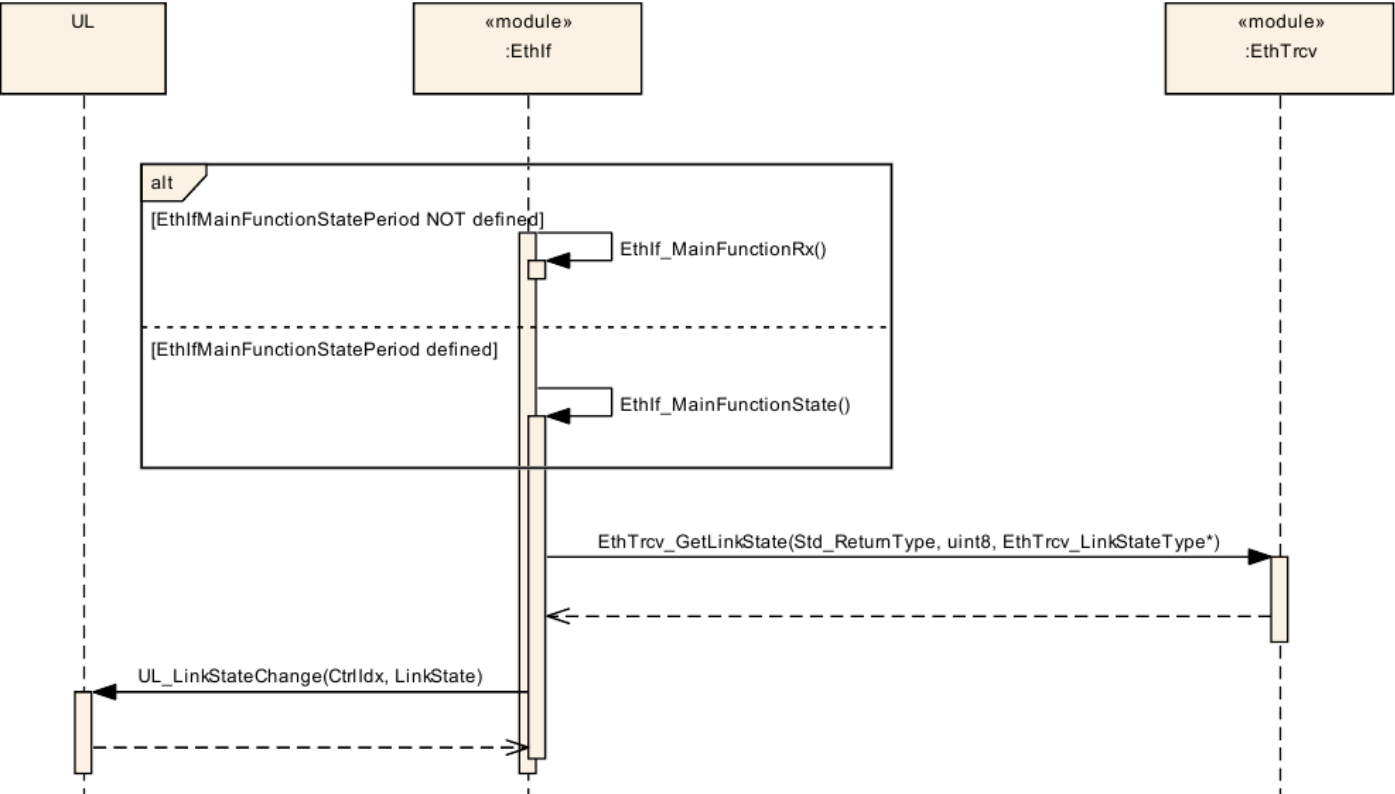
2.9.4.1 Reception in Polling Mode



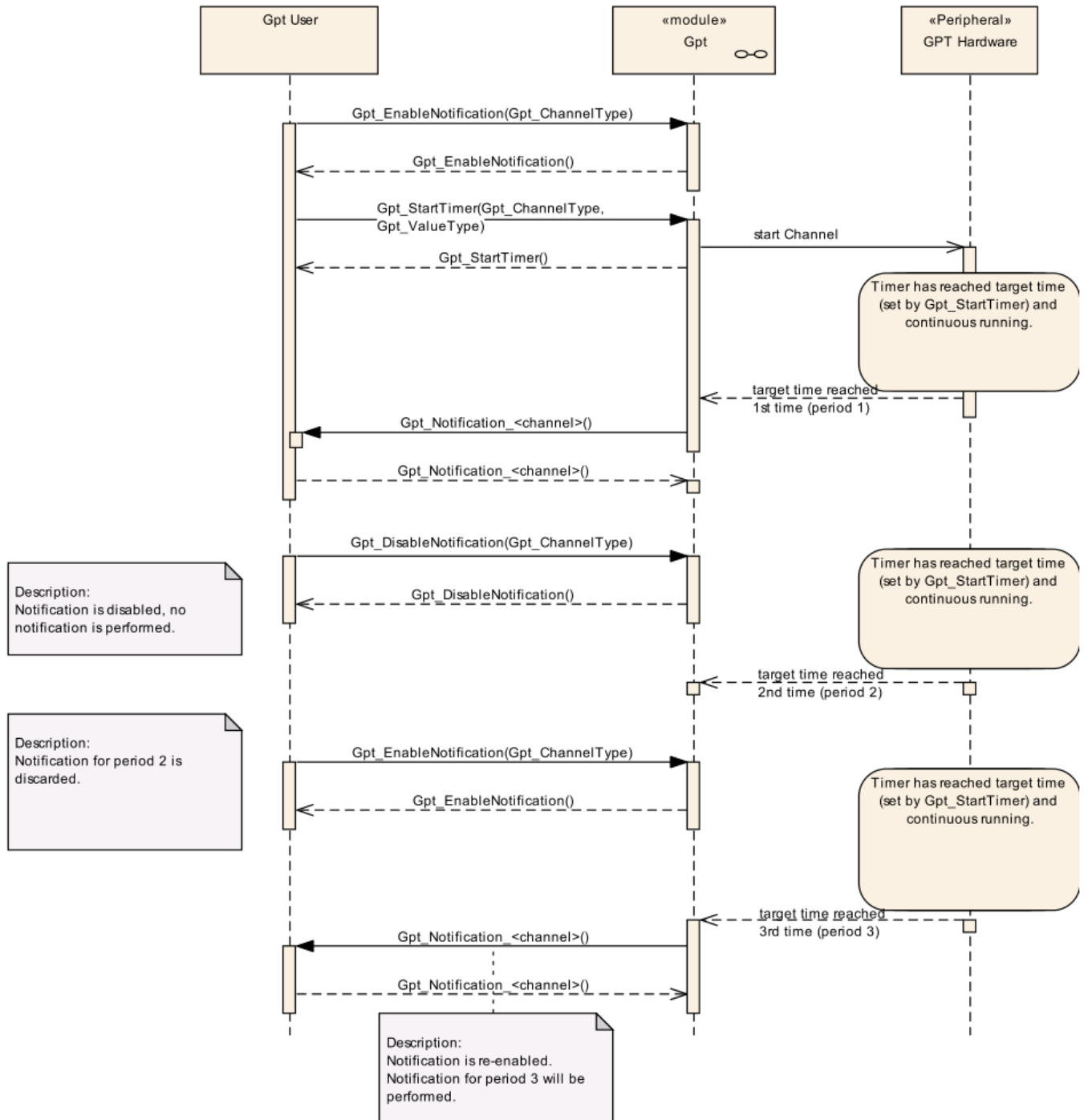
2.9.4.2 Reception in Interrupt Mode



2.9.5 Link State Change



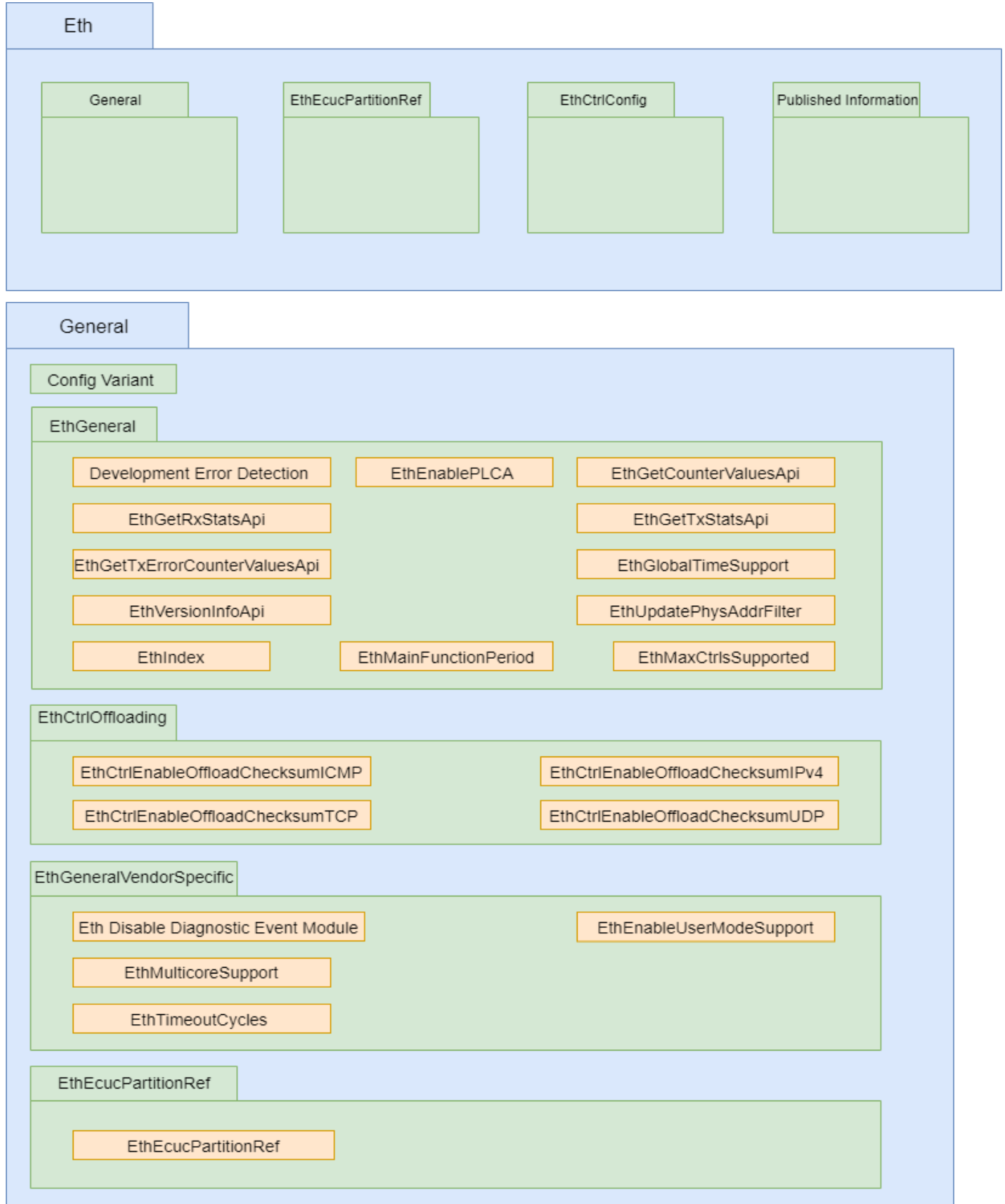
This notification is discarded and not made up again, when the notification is re-enabled.

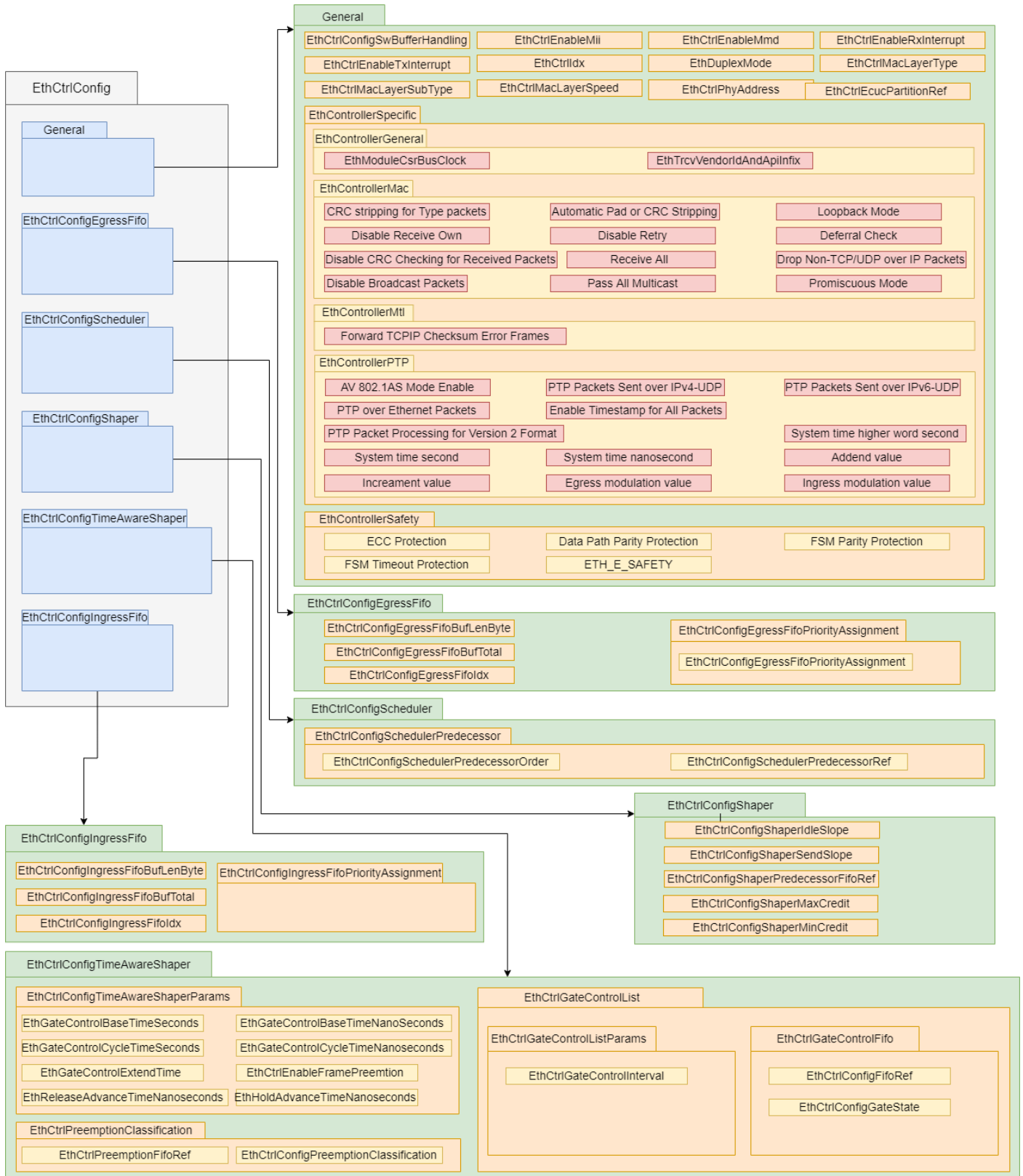


Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation

The container inclusion relation is shown as below:





CommonPublishedInformation

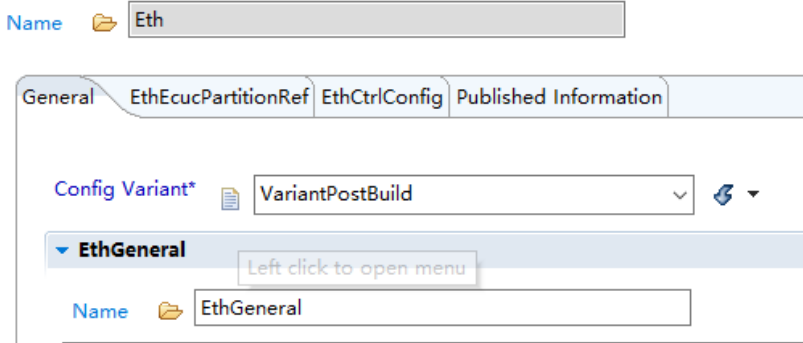
AUTOSAR Major Version	AUTOSAR Minor Version	AUTOSAR Patch Version	Module ID
Software Major Version	Software Minor Version	Software Patch Version	Vendor ID

3.2 Containers and Variables

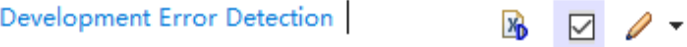
3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 EthGeneral

Container	EthGeneral	
Description	General configuration of Ethernet Driver module.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.1 EthDevErrorDetect

Container	EthDevErrorDetect	
Description	Development error detection	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SynbolicName	False
	Default	False

3.2.2.2 EthEnablePLCA

Container	EthEnablePLCA	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Boolean

	Origin	AUTOSAR_ECUC
	SynbolicName	False
	Default	False

3.2.2.3 EthGetCounterValuesApi

Container	EthGetCounterValuesApi	
Description	Enables / Disables Eth_GetCounterValues API.	
Screenshot	EthGetCounterValuesApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SynbolicName	False
	Default	True

3.2.2.4 EthGetRxStatsApi

Variable	EthGetRxStatsApi	
Description	Enables / Disables Eth_GetRxStats API.	
Screenshot	EthGetRxStatsApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

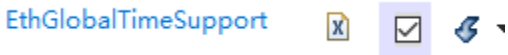
3.2.2.5 EthGetTxStatsApi

Variable	EthGetTxStatsApi	
Description	Enables/Disables Eth_GetTxStats API.	
Screenshot	EthGetTxStatsApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

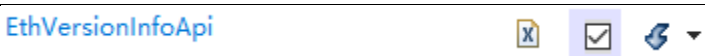
3.2.2.6 EthGetTxErrorCounterValuesApi

Variable	EthGetTxErrorCounterValuesApi	
Description	Enables/Disables Eth_GetTxErrorCounterValues API.	
Screenshot	EthGetTxErrorCounterValuesApi  ☒ 	
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.7 EthGlobalTimeSupport

Variable	EthGlobalTimeSupport	
Description	Enables/Disables the GlobalTime APIs used amongst others by Global Time Synchronization over Ethernet.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

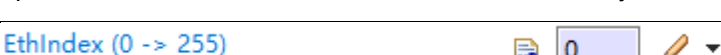
3.2.2.8 EthVersionInfoApi

Variable	EthVersionInfoApi	
Description	Enables / Disables version info API	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.9 EthUpdatePhysAddrFilter

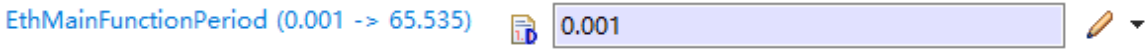
Variable	EthUpdatePhysAddrFilter	
Description	Enables / Disables Eth_UpdatePhysAddrFilter API	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.10 EthIndex

Variable	EthIndex	
Description	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.2.11 EthMainFunctionPeriod

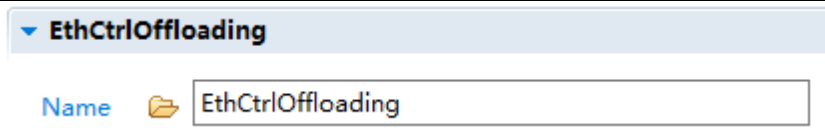
Variable	EthMainFunctionPeriod	
-----------------	-----------------------	--

Description	Specifies the period of main function Eth_MainFunction in seconds. Ethernet driver does not require this information but the BSW scheduler.	
Screenshot		
Properties	Property	Value
	Type	Variable: Float
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0.001

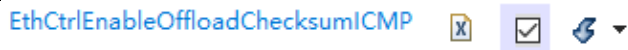
3.2.2.12 EthMaxCtrlsSupported

Variable	EthMaxCtrlsSupported	
Description	Limits the total number of supported controllers.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	1

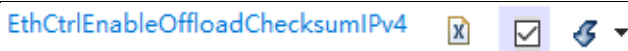
3.2.2.13 EthCtrlOffloading

Variable	EthCtrlOffloading	
Description	Configuration of hardware offloading features.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.13.1 EthCtrlEnableOffloadChecksumICMP

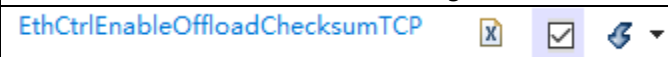
Variable	EthCtrlEnableOffloadChecksumICMP	
Description	Enables / Disables hardware offloading for ICMP checksums.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.13.2 EthCtrlEnableOffloadChecksumIPv4

Container	EthCtrlEnableOffloadChecksumIPv4	
Description	Enables / Disables hardware offloading for IPv4 checksums.	
Screenshot		
Properties	Property	Value
	Type	Boolean

	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

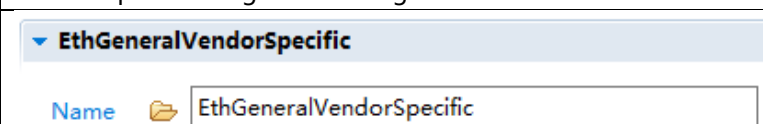
3.2.2.13.3 EthCtrlEnableOffloadChecksumTCP

Variable	EthCtrlEnableOffloadChecksumTCP	
Description	Enables / Disables hardware offloading for TCP checksums.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.13.4 EthCtrlEnableOffloadChecksumUDP

Variable	EthCtrlEnableOffloadChecksumUDP	
Description	Enables / Disables hardware offloading for UDP checksums.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.2.14 EthGeneralVendorSpecific

Container	EthGeneralVendorSpecific	
Description	Vendor specific Eth general configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.14.1 EthDisableDemEventDetect

Variable	EthDisableDemEventDetect	
Description	Disables / Enables Eth Diagnostic Event Module.	
Screenshot		
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.2.14.2 EthEnableUserModeSupport

Variable	EthEnableUserModeSupport	
Description	Enables / Disables user mode .	
Screenshot	N/A	
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

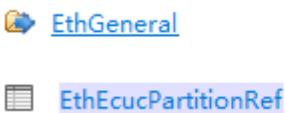
3.2.2.14.3 EthMulticoreSupport

Variable	EthMulticoreSupport	
Description	Switches multicore support on or off: False: For all variants, no EcucPartition shall be referenced in EthEcucPartitionRef. True: For all variants, at least one EcucPartition needs to be referenced in EthEcucPartitionRef.	
Screenshot	EthMulticoreSupport 	
Properties	Property	Value
	Type	Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.2.14.4 EthTimeoutCycles

Variable	EthTimeoutCycles	
Description	The number of attempts to wait for hardware response when the driver access to hardware.	
Screenshot	EthTimeoutCycles (1 -> 65535) 	
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	1500

3.2.2.15 EthEcucPartitionRef

Container	EthEcucPartitionRef	
Description	Maps the Ethernet driver to zero or multiple ECUC partitions to make the module's API available in this partition. The Ethernet driver will operate as an independent instance in each of the partitions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.2.15.1 EthEcucPartitionRef

Variable	EthEcucPartitionRef
-----------------	---------------------

Description	Reference to an ECUC partition.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

3.2.3 EthCtrlConfig

Variable	EthCtrlConfig	
Description	Configuration of the individual controller.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.1 EthCtrlConfigSwBufferHandling

Variable	EthCtrlConfigSwBufferHandling	
Description	Enables / Disables SW buffer management.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	False

3.2.3.2 EthCtrlEnableMii

Variable	EthCtrlEnableMii	
Description	Enables / Disables Media Independent Interface (MII) for transceiver access.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.3.3 EthCtrlEnableMmd

Variable	EthCtrlEnableMmd	
Description	Enables / Disables Media Independent Interface (MII) clause 45 protocol for transceiver access.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FILAGCHIP
	SymbolicNameValue	False

	Default	False
--	---------	-------

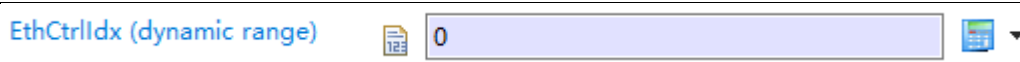
3.2.3.4 EthCtrlEnableRxInterrupt

Variable	EthCtrlEnableRxInterrupt	
Description	Enables / Disables receive interrupt.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

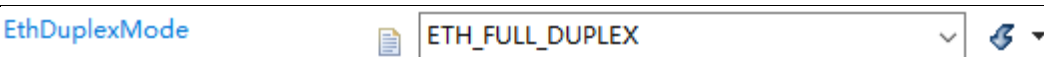
3.2.3.5 EthCtrlEnableTxInterrupt

Variable	EthCtrlEnableTxInterrupt	
Description	Enables / Disables transmit interrupt	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	True

3.2.3.6 EthCtrlIdx

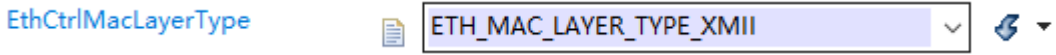
Container	EthCtrlIdx	
Description	Specifies the instance ID of the configured controller.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	0

3.2.3.7 EthDuplexMode

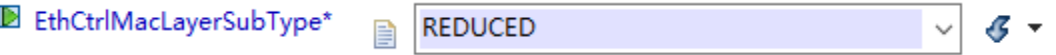
Variable	EthDuplexMode	
Description	Set Eth full duplex or half duplex mode connection.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	ETH_FULL_DUPLEX, ETH_HALF_DUPLEX
	Default	ETH_FULL_DUPLEX

3.2.3.8 EthCtrlMacLayerType

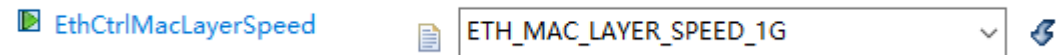
Variable	EthCtrlMacLayerType	
-----------------	---------------------	--

Description	Defines the MAC layer type of the ethernet controller.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETH_MAC_LAYER_TYPE_XMII, ETH_MAC_LAYER_TYPE_XGMII, ETH_MAC_LAYER_TYPE_XXGMII
	Default	ETH_MAC_LAYER_TYPE_XGMII

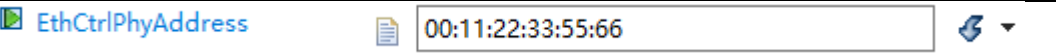
3.2.3.9 EthCtrlMacLayerSubType

Variable	EthCtrlMacLayerSubType	
Description	Defines the MAC layer subtype of a switch port.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	REDUCED, REVERSED, SERIAL, STANDARD, UNIVERSAL_SERIAL
	Default	REDUCED

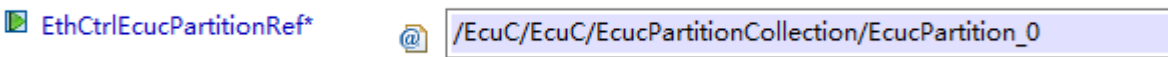
3.2.3.10 EthCtrlMacLayerSpeed

Variable	EthCtrlMacLayerSpeed	
Description	Defines the baud rate of the MAC layer.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	AUTOSAR_ECUC
	Range	ETH_MAC_LAYER_SPEED_10M ETH_MAC_LAYER_SPEED_100M ETH_MAC_LAYER_SPEED_1G ETH_MAC_LAYER_SPEED_10G
	Default	ETH_MAC_LAYER_SPEED_1G

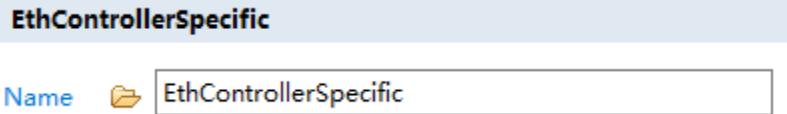
3.2.3.11 EthCtrlPhyAddress

Variable	EthCtrlPhyAddress	
Description	Specifies the unique 48-bit physical address (MAC address) of the controller in network byte order.	
Screenshot		
Properties	Property	Value
	Type	String
	Origin	AUTOSAR_ECUC
	Length	17
	Default	"00:11:22:33:55:66"

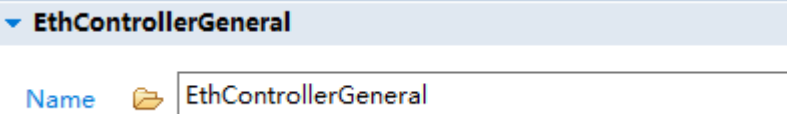
3.2.3.12 EthCtrlEcucPartitionRef

Variable	EthCtrlEcucPartitionRef	
Description	Maps the Ethernet controller to zero or one ECUC partitions. The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

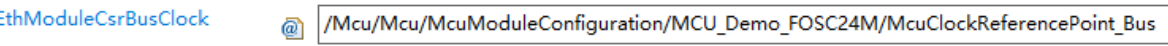
3.2.3.13 EthControllerSpecific

Variable	EthControllerSpecific	
Description	Controller specific configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

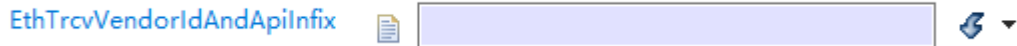
3.2.3.13.1 EthControllerGeneral

Variable	EthControllerGeneral	
Description	ETH Controller General Configuration.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.1.1 EthModuleCsrBusClock

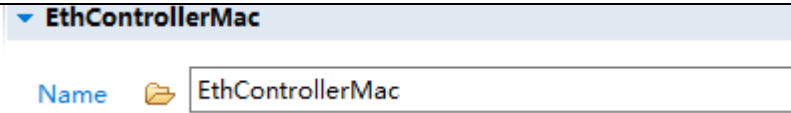
Variable	EthModuleCsrBusClock	
Description	Eth Module controller status registers Bus Clock selection.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	FLAGCHIP

3.2.3.13.1.2 EthTrcvVendorIdAndApiInfix

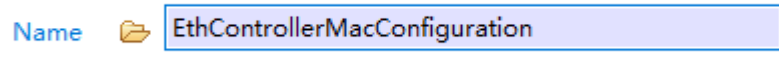
Variable	EthTrcvVendorIdAndApiInfix	
Description	Set the vendor id and vendor api infix for the EthTrcv driver. This is needed for Eth driver to include the transceiver header file and call mii indication functions.	
Screenshot		
Properties	Property	Value
	Type	Variable: String

	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	NULL

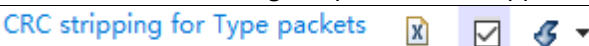
3.2.3.13.2 EthControllerMac

Variable	EthControllerMac	
Description	Controller specific MAC configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.2.1 EthControllerMacConfiguration

Variable	EthControllerMacConfiguration	
Description	MAC Controller Configuration.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.2.1.1 EthControllerMacConfiguration_CST

Variable	EthControllerMacConfiguration_CST	
Description	Enables / Disables stripping the last four bytes (FCS) of all packets of Ether type (type field greater than 1,536) before forwarding the packet to the application.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.13.2.1.2 EthControllerMacConfiguration_ACS

Variable	EthControllerMacConfiguration_ACS	
Description	Enables / Disables stripping the Pad or FCS field on the incoming packets only if the value of the length field is less than 1,536 bytes.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.13.2.1.3 EthControllerMacConfiguration_LM

Variable	EthControllerMacConfiguration_LM	
Description	Enables / Disables the MAC operating in the loopback mode.	
Screenshot	Loopback Mode 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.1.4 EthControllerMacConfiguration_DO

Variable	EthControllerMacConfiguration_DO	
Description	Disables / Enables receive own in the half-duplex mode.	
Screenshot	Disable Receive Own 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.13.2.1.5 EthControllerMacConfiguration_DR

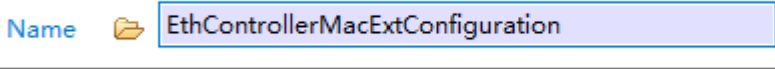
Variable	EthControllerMacConfiguration_DR	
Description	Disables / Enables retry when a collision occurs in the half-duplex mode.	
Screenshot	Disable Retry 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.1.6 EthControllerMacConfiguration_DC

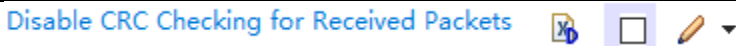
Variable	EthControllerMacConfiguration_DC	
Description	Enables / Disables the deferral check function.	
Screenshot	Deferral Check 	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.2 EthControllerMacExtConfiguration

Variable	EthControllerMacExtConfiguration
Description	MAC Controller Extend Configuration.

Screenshot		
Properties	Property	Value
	Type	Container
Requirements	FUNC_Eth_00033	

3.2.3.13.2.2.1 EthControllerMacExtConfiguration_DCRCC

Variable	EthControllerMacExtConfiguration_DCRCC	
Description	Disables / Enables CRC checking for received packets.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.3 EthControllerMacFilter

Variable	EthControllerMacFilter	
Description	MAC Controller Packet Filter Configuration	
Screenshot		
Properties	Property	Value
	Type	Container
Requirements	FUNC_Eth_00035	

3.2.3.13.2.3.1 EthControllerMacFilter_RA

Variable	EthControllerMacFilter_RA	
Description	Enables / Disables the MAC Receiver module to pass all received packets to the application, irrespective of whether they pass the address filter or not.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.3.2 EthControllerMacFilter_DNTU

Variable	EthControllerMacFilter_DNTU	
Description	Enables / Disables the MAC to drop the non-TCP or UDP over IP packets.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean

	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.4 EthControllerMacFilter_DBF

Variable	EthControllerMacFilter_DBF	
Description	Enables / Disables the MAC to block all the incoming broadcast packets. it overrides all other filter settings.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

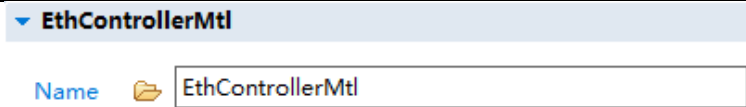
3.2.3.13.2.5 EthControllerMacFilter_PM

Variable	EthControllerMacFilter_PM	
Description	Enables / Disables the MAC to pass all the received packets with a multicast destination address to application.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.2.6 EthControllerMacFilter_PR

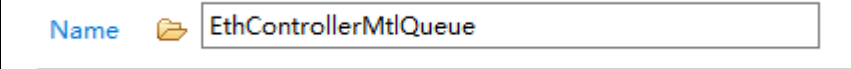
Variable	EthControllerMacFilter_PR	
Description	Enables / Disables the Address Filtering module passes all incoming packets irrespective of the destination or source address.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.3 EthControllerMtl

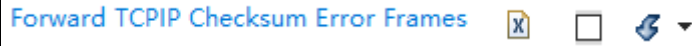
Variable	EthControllerMtl	
Description	Controller specific MAC transaction layer queue configuration.	
Screenshot		
Properties	Property	Value

	Type	Container
--	------	-----------

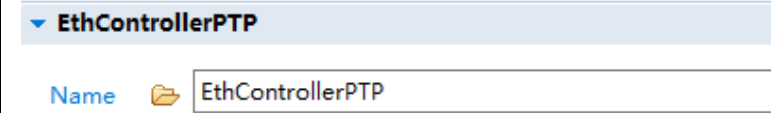
3.2.3.13.3.1 EthControllerMtlQueue

Variable	EthControllerMtlQueue	
Description	MTL Queue (0/1) Configuration.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.3.1.1 EthControllerMtlQueue_CHECKSUM_ERR

Variable	EthControllerMtlQueue_CHECKSUM_ERR	
Description	This value works with EthCtrlOffloading. If checksum engine is enabled, this value determines whether the tcpip frames that have checksum errors is forwarded to the application or discarded.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	Default	False

3.2.3.13.4 EthControllerPTP

Variable	EthControllerPTP	
Description	Controller specific IEEE1588 module configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.4.1 EthControllerPTP_AV8021ASMEN

Variable	EthControllerPTP_AV8021ASMEN	
Description	Enables / Disables the MAC to process only untagged PTP over Ethernet packets for providing PTP status and capturing timestamp snapshots, that is, IEEE 802.1AS mode of operation.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.4.2 EthControllerPTP_TSIPV4ENA

Variable	EthControllerPTP_TSIPV4ENA	
Description	Enables / Disables the MAC to process the PTP packets encapsulated in IPv4-UDP packets.	

Screenshot	PTP Packets Sent over IPv4-UDP    ▼	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.13.4.3 EthControllerPTP_TSIPV6ENA

Variable	EthControllerPTP_TSIPV6ENA	
Description	Enables / Disables the MAC to process the PTP packets encapsulated in IPv6-UDP packets.	
Screenshot	PTP Packets Sent over IPv6-UDP  	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.4.4 EthControllerPTP_TSIPENA

Variable	EthControllerPTP_TSIPENA	
Description	Enables / Disables the MAC to process the PTP packets encapsulated directly in the Ethernet packets.	
Screenshot	PTP over Ethernet Packets    ▼	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.4.5 EthControllerPTP_TSVER2ENA

Variable	EthControllerPTP_TSVER2ENA	
Description	Enables / Disables the MAC to process the PTP version 2 packets.	
Screenshot	PTP Packet Processing for Version 2 Format  	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	True

3.2.3.13.4.6 EthControllerPTP_TSENALL

Variable	EthControllerPTP_TSENALL	
Description	Enables / Disables the timestamp snapshotting for all packets received by the MAC.	
Screenshot	Enable Timestamp for All Packets    ▼	
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP

	SymbolicNameValue	False
	Default	False

3.2.3.13.4.7 EthControllerPTP_HiInitialVal

Variable	EthControllerPTP_HiInitialVal	
Description	Set the system time seconds high 16bit word.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 65535
	Default	0

3.2.3.13.4.8 EthControllerPTP_InitialVal

Variable	EthControllerPTP_InitialVal	
Description	Set the system time seconds low 32bit word.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 4294967296
	Default	0

3.2.3.13.4.9 EthControllerPTP_NanoInitialVal

Variable	EthControllerPTP_NanoInitialVal	
Description	Set the system time nanoseconds.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 999999999
	Default	0

3.2.3.13.4.10 EthControllerPTP_Addend

Variable	EthControllerPTP_Addend	
Description	This value added to a 32-bit accumulator in every PTP clock cycle and the system time is updated when the accumulator overflows.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	1 - 4294967296
	Default	1431655765

3.2.3.13.4.11 EthControllerPTP_IncrementVal

Variable	EthControllerPTP_IncrementVal	
Description	This value to be added to the system time when 32-bit accumulator overflows.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	1 - 255
	Default	20

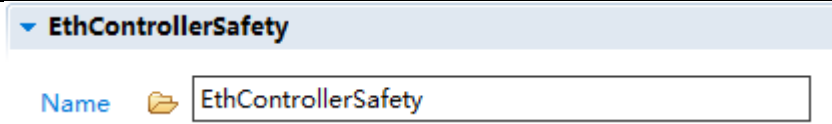
3.2.3.13.4.12 EthControllerPTP_EgressModulationVal

Variable	EthControllerPTP_EgressModulationVal	
Description	The correction value in nanoseconds to be used with the captured timestamp value in the egress path.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.3.13.4.13 EthControllerPTP_IngressModulationVal

Variable	EthControllerPTP_IngressModulationVal	
Description	The correction value in nanoseconds to be used with the captured timestamp value in the ingress path.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.3.13.5 EthControllerSafety

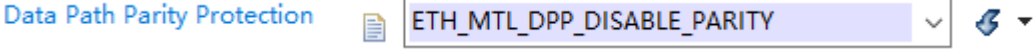
Variable	EthControllerSafety	
Description	Controller specific safety feature configurations.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.13.5.1 EthControllerSafety_ECCEN

Variable	EthControllerSafety_ECCEN	
Description	Enable or disable ECC Protection for MTL TX FIFO, RX FIFO and EST memory.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean

	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.5.2 EthControllerSafety_DataPathParity

Variable	EthControllerSafety_DataPathParity	
Description	Enable or disable parity protection for Transmit and Receive data paths.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	ETH_MTL_DPP_DISABLE_PARITY ETH_MTL_DPP_EVEN_PARITY ETH_MTL_DPP_ODD_PARITY
	Default	ETH_MTL_DPP_DISABLE_PARITY

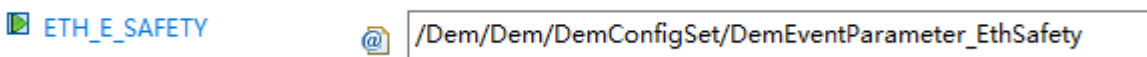
3.2.3.13.5.3 EthControllerSafety_FSMParityEN

Variable	EthControllerSafety_FSMParityEN	
Description	Enable or disable FSM State Parity Protection.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.5.4 EthControllerSafety_FSMTimeoutEN

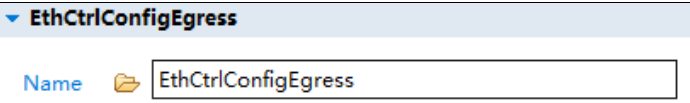
Variable	EthControllerSafety_FSMTimeoutEN	
Description	Enable or disable FSM Timeout Protection.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.3.13.5.5 ETH_E_SAFETY

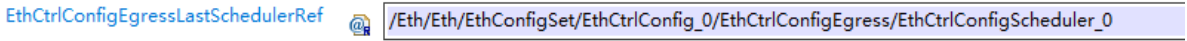
Variable	ETH_E_SAFETY	
Description	Reference to the DemEventParameter which shall be issued when the 'Safety errors' has occurred.	
Screenshot		
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]

	Origin	FLAGCHIP
--	--------	----------

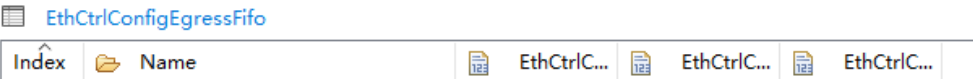
3.2.3.14 EthCtrlConfigEgress

Variable	EthCtrlConfigEgress	
Description	Configuration of one Ethernet controller egress behavior.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.14.1 EthCtrlConfigEgressLastSchedulerRef

Variable	EthCtrlConfigEgressLastSchedulerRef	
Description	Reference to the scheduler which is the last in the egress structure.	
Screenshot		
Properties	Property	Value
	Type	Container
Requirements	ECUC_Eth_00052	

3.2.3.14.2 EthCtrlConfigEgressFifo

Variable	EthCtrlConfigEgressFifo	
Description	Represents a Fifo at the egress side.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.14.2.1 EthCtrlConfigEgressFifoBufLenByte

Variable	EthCtrlConfigEgressFifoBufLenByte	
Description	Length of Fifo elements in bytes.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	64 - 65535
	Default	1536

3.2.3.14.2.2 EthCtrlConfigEgressFifoBufTotal

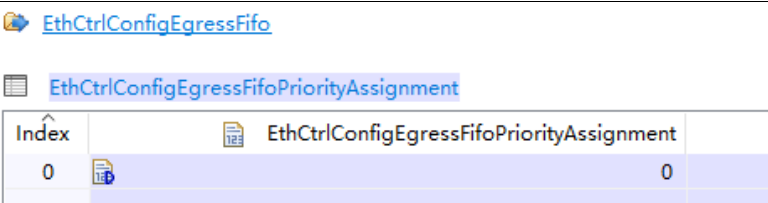
Variable	EthCtrlConfigEgressFifoBufTotal	
Description	Fifo buffer count.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer

	Origin	AUTOSAR_ECUC
	Range	4 - 255
	Default	6

3.2.3.14.2.3 EthCtrlConfigEgressFifoldx

Variable	EthCtrlConfigEgressFifoldx	
Description	Egress Fifo index.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 255
	Default	0

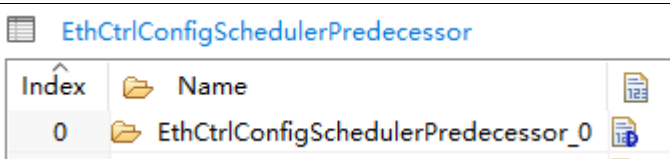
3.2.3.14.2.4 EthCtrlConfigEgressFifoPriorityAssignment

Variable	EthCtrlConfigEgressFifoPriorityAssignment	
Description	Message egress priority assignment.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 7
	Default	0

3.2.3.14.3 EthCtrlConfigScheduler

Variable	EthCtrlConfigScheduler	
Description	Represents a Scheduler on the egress side.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.14.3.1 EthCtrlConfigSchedulerPredecessor

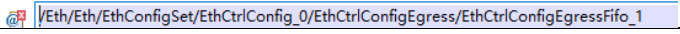
Variable	EthCtrlConfigSchedulerPredecessor	
Description	Defines an ordered list of predecessors for this scheduler.	
Screenshot		
Properties	Property	Value
	Type	Container

	Origin	AUTOSAR_ECUC
--	--------	--------------

3.2.3.14.3.1.1 EthCtrlConfigSchedulerPredecessorOrder

Variable	EthCtrlConfigSchedulerPredecessorOrder	
Description	Defines the order of the scheduler predecessors.	
Screenshot	EthCtrlConfigSchedulerPredecessorOrder (0 -> 255) 	
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 18446744073709551615
	Default	0

3.2.3.14.3.1.2 EthCtrlConfigSchedulerPredecessorRef

Variable	EthCtrlConfigSchedulerPredecessorRef	
Description	Reference to the scheduler predecessor.	
Screenshot	EthCtrlConfigSchedulerPredecessorRef* 	
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

3.2.3.14.4 EthCtrlConfigShaper

Variable	EthCtrlConfigShaper	
Description	Represents a Shaper on the egress side.	
Screenshot	 EthCtrlConfigShaper	
Properties	Property	Value
	Type	Container

3.2.3.14.4.1 EthCtrlConfigShaperIdleSlope

Variable	EthCtrlConfigShaperIdleSlope	
Description	Defines the increase of credit in bits per second for the AVB shaper.	
Screenshot	EthCtrlConfigShaperIdleSlope (dynamic range)*  	
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - dynamic range
	Default	0

3.2.3.14.4.2 EthCtrlConfigShaperSendSlope

Variable	EthCtrlConfigShaperSendSlope	
Description	Rate of credits consumed in kilobits per second during transmission.	
Screenshot	EthCtrlConfigShaperSendSlope*  	
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - dynamic range

	Default	0
--	---------	---

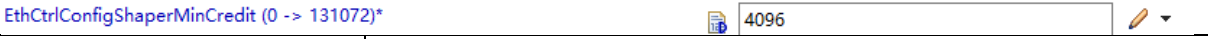
3.2.3.14.4.3 EthCtrlConfigShaperPredecessorFifoRef

Variable	EthCtrlConfigShaperPredecessorFifoRef	
Description	Reference to the fifo which is the predecessor for this shaper.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	AUTOSAR_ECUC

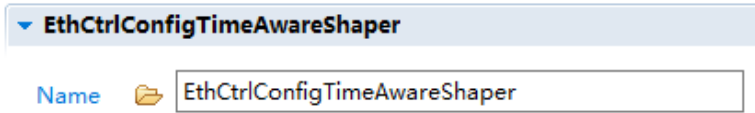
3.2.3.14.4.4 EthCtrlConfigShaperMaxCredit

Variable	EthCtrlConfigShaperMaxCredit	
Description	Maximum amount of credits that can be accumulated for a queue.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 131072
	Default	0

3.2.3.14.4.5 EthCtrlConfigShaperMinCredit

Variable	EthCtrlConfigShaperMinCredit	
Description	Minimum amount of credits in bytes that can be accumulated for a queue.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 131072
	Default	0

3.2.3.15 EthCtrlConfigTimeAwareShaper

Variable	EthCtrlConfigTimeAwareShaper	
Description	Time Aware Shaper configurations on the Egress side. This is the implementation of time sensitive network.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.15.1 EthGateControlBaseTimeSeconds

Variable	EthGateControlBaseTimeSeconds	
Description	The upper 32-bits (in seconds) of the base time to hold the start time of the Gate control list.	
Screenshot		
Properties	Property	Value

	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 4294967295
	Default	0

3.2.3.15.2 EthGateControlBaseTimeNanoSeconds

Variable	EthGateControlBaseTimeNanoSeconds	
Description	The lower 32-bits (in nanoseconds) of the base time to hold the start time of the Gate control list.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 999999999
	Default	0

3.2.3.15.3 EthGateControlCycleTimeSeconds

Variable	EthGateControlCycleTimeSeconds	
Description	The upper 8-bits (in seconds) of the cycle time to hold the time at which the execution of the GCL should be repeated.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 255
	Default	0

3.2.3.15.4 EthGateControlCycleTimeNanoseconds

Variable	EthGateControlCycleTimeNanoseconds	
Description	The lower 32-bits (in nanoseconds) of the cycle time to hold the time at which the execution of the GCL should be repeated.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 999999999
	Default	0

3.2.3.15.5 EthGateControlExtendTime

Variable	EthGateControlExtendTime	
Description	This parameter holds the amount of time (in nano seconds) the current Gate Control List can be extended before switching to the newGate Control List. This is useful to avoid small fragments of the current list before switching to a new list.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer

	Origin	FLAGCHIP
	Range	0 - 2147483647
	Default	0

3.2.3.15.6 EthCtrlEnableFramePreemption

Variable	EthCtrlEnableFramePreemption	
Description	Enabled - Frame preemption is enabled.	
	Disabled - Frame preemption is disabled.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

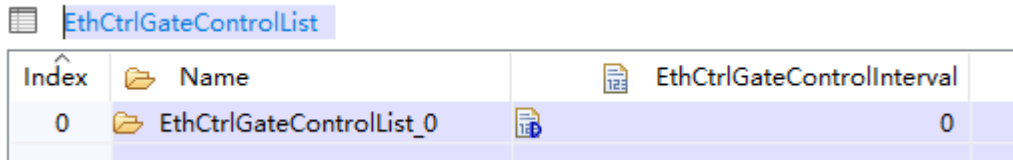
3.2.3.15.7 EthReleaseAdvanceTimeNanoseconds

Variable	EthReleaseAdvanceTimeNanoseconds	
Description	The maximum time in nanoseconds that can elapse between issuing a RELEASE to the MAC and the MAC being ready to resume transmission of preemptable frames.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	1 - 65535
	Default	1

3.2.3.15.8 EthHoldAdvanceTimeNanoseconds

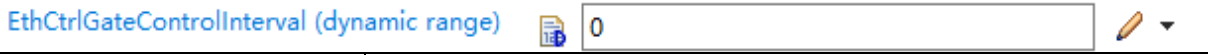
Variable	EthHoldAdvanceTimeNanoseconds	
Description	The maximum time in nanoseconds that can elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit any preemptable frame that is in the process of transmission or any preemptable frames that are queued for transmission.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	1 - 65535
	Default	1

3.2.3.15.9 EthCtrlGateControlList

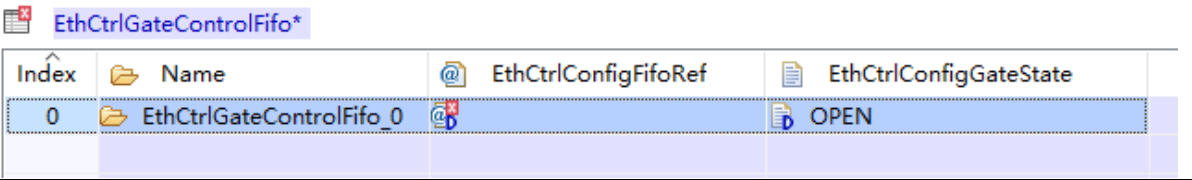
Variable	EthCtrlGateControlList	
Description	The list in the hardware memory that holds the gate controls and the associated time intervals.	
Screenshot		

Properties	Property	Value
	Type	Container
	Origin	FLAGCHIP

3.2.3.15.9.1 EthCtrlGateControlInterval

Variable	EthCtrlGateControlInterval	
Description	The time in nanoseconds for which the gate controls are valid and should be applied before reading the next gate controls from the list.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	Range	0 - 16000000
	Default	0

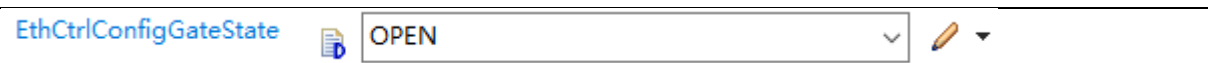
3.2.3.15.9.2 EthCtrlGateControlFifo

Variable	EthCtrlGateControlFifo	
Description	Configure the open or close state of the Fifo in duration interval.	
Screenshot		
Properties	Property	Value
	Type	Container
	Origin	FLAGCHIP

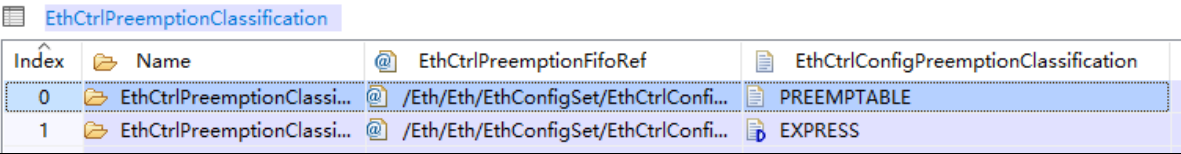
3.2.3.15.9.2.1 EthCtrlConfigFifoRef

Variable	EthCtrlConfigFifoRef	
Description	Reference to the FIFO which is the predecessor for this time aware shaper.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	FLAGCHIP

3.2.3.15.9.2.2 EthCtrlConfigGateState

Variable	EthCtrlConfigGateState	
Description	This parameter select the Open (O) represented by logic 1 or Close (C) represented by logic 0 state for the gate of each Fifo.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	CLOSE OPEN

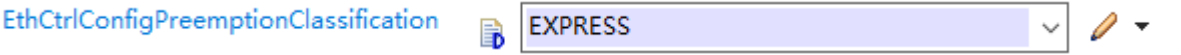
3.2.3.15.10 EthCtrlPreemptionClassification

Variable	EthCtrlPreemptionClassification	
Description	This parameter indicates that the corresponding Fifo is classified as preemptable or express.	
Screenshot		
Properties	Property	Value
	Type	Container
	Origin	FLAGCHIP

3.2.3.15.10.1 EthCtrlPreemptionFifoRef

Variable	EthCtrlPreemptionFifoRef	
Description	Reference to the FIFO which is the predecessor for preemption.	
Screenshot		
Properties	Property	Value
	Type	Reference
	Origin	FLAGCHIP

3.2.3.15.10.2 EthCtrlConfigPreemptionClassification

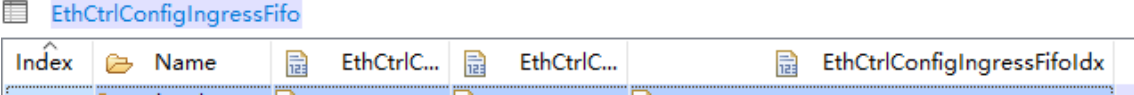
Variable	EthCtrlConfigPreemptionClassification	
Description	This parameters to select the queue is preemptable or express.	
Screenshot		
Properties	Property	Value
	Type	Enumeration
	Origin	FLAGCHIP
	Range	PREEMPTABLE, EXPRESS
	Default	EXPRESS

3.2.3.16 EthCtrlConfigIngress

Variable	EthCtrlConfigIngress	
Description	Configuration of one Ethernet controller ingress behavior.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.16.1 EthCtrlConfigIngressFifo

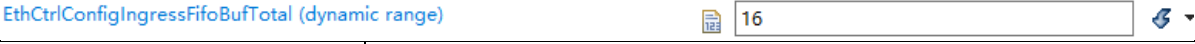
Variable	EthCtrlConfigIngressFifo	
Description	Represents a Fifo at the ingress side.	

Screenshot		
	Properties	Value
	Type	Container

3.2.3.16.1.1 EthCtrlConfigIngressFifoBufLenByte

Variable	EthCtrlConfigIngressFifoBufLenByte	
Description	Length of Fifo elements in bytes.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	64 - 65535
	Default	1536

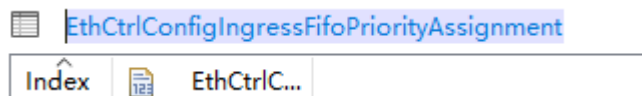
3.2.3.16.1.2 EthCtrlConfigIngressFifoBufTotal

Variable	EthCtrlConfigIngressFifoBufTotal	
Description	Fifo buffer count.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	4 - 255
	Default	16

3.2.3.16.1.3 EthCtrlConfigIngressFifoldx

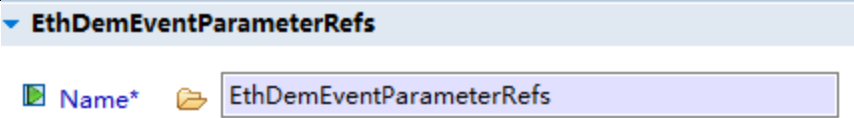
Variable	EthCtrlConfigIngressFifoldx	
Description	Ingress Fifo index.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 255
	Default	0

3.2.3.16.1.4 EthCtrlConfigIngressFifoPriorityAssignment

Variable	EthCtrlConfigIngressFifoPriorityAssignment	
Description	Message ingress priority assignment.	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	AUTOSAR_ECUC
	Range	0 - 7

	Default	0
--	---------	---

3.2.3.17 EthDemEventParameterRefs

Variable	EthDemEventParameterRefs	
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId symbolic value. The standardized errors are provided in this container and can be extended by vendor- specific error references.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.17.1 ETH_E_ACCESS

Variable	ETH_E_ACCESS	
Description	Reference to the DemEventParameter which shall be issued when the error "Controller access failed" has occurred.	
Screenshot		
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.2 ETH_E_ALIGNMENT

Variable	ETH_E_ALIGNMENT	
Description	Reference to the DemEventParameter which shall be issued when the error " Alignment Error" has occurred.	
Screenshot		
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.3 ETH_E_CRC

Variable	ETH_E_CRC	
Description	Reference to the DemEventParameter which shall be issued when the error "CRC Failure" has occurred.	
Screenshot		
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.4 ETH_E_LATECOLLISION

Variable	ETH_E_LATECOLLISION	
Description	Reference to the DemEventParameter which shall be issued when the error "Late Collisions" has	

	occured.	
Screenshot	 	
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.5 ETH_E_MULTIPLECOLLISION

Variable	ETH_E_MULTIPLECOLLISION	
Description	Reference to the DemEventParameter which shall be issued when the error " Multiple Collisions" has occured.	
Screenshot	 	
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.6 ETH_E_OVERSIZEFRAME

Variable	ETH_E_OVERSIZEFRAME	
Description	Reference to the DemEventParameter which shall be issued when the error " Oversized Frame" has occured.	
Screenshot	 	
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.7 ETH_E_RX_FRAMES_LOST

Variable	ETH_E_RX_FRAMES_LOST	
Description	Reference to the DemEventParameter which shall be issued when the error "receive frames lost" has occured.	
Screenshot	 	
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

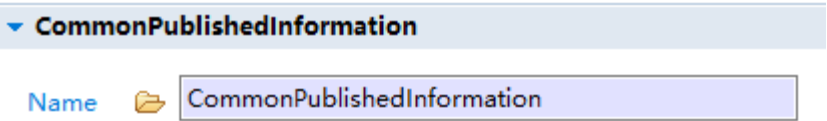
3.2.3.17.8 ETH_E_SINGLECOLLISION

Variable	ETH_E_SINGLECOLLISION	
Description	Reference to the DemEventParameter which shall be issued when the error " Single Collisions" has occured.	
Screenshot	 	
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.3.17.9 ETH_E_UNDERSIZEFRAME

Variable	ETH_E_UNDERSIZEFRAME	
Description	Reference to the DemEventParameter which shall be issued when the error " Undersized Frame" has occurred.	
Screenshot		
Properties	Property	Value
	Type	Symbolic name reference to [DemEventParameter]
	Origin	AUTOSAR_ECUC

3.2.4 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	IDENTIFIABLE

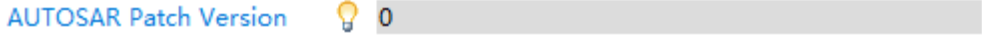
3.2.4.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	4
	Origin	FLAGCHIP

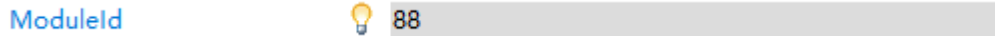
3.2.4.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	6
	Origin	FLAGCHIP

3.2.4.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	0
	Origin	FLAGCHIP

3.2.4.4 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	88
	Origin	FLAGCHIP

3.2.4.5 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	1
	Origin	FLAGCHIP

3.2.4.6 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	1
	Origin	FLAGCHIP

3.2.4.7 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	0
	Origin	FLAGCHIP

3.2.4.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list. Note: Implementation Specific Parameter	
Screenshot		
Properties	Property	Value
	Type	INTEGER_LABEL
	SynmbolicName	False
	Default	174
	Origin	FLAGCHIP

Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- 1) At least one egress and ingress FIFO should be configured for transmit and receive.
- 2) Disable Broadcast Packets should not be chosen, if TCP/IP stack is used, the ARP protocol will not be completed.
- 3) Loopback Mode is only used for test, and should not be chosen for normal usage.
- 4) If any of Receive All, Pass All Multicast and Promiscuous Mode is chosen, the EthUpdatePhysAddrFilter has no effect.
- 5) If EthCtrlConfigShaper is selected in the EthCtrlConfigScheduler, the AVB mode is automatically set for the egress FIFO associated with shaper.
- 6) For FC7300F8MDT and FC7240 series, TX checksum offload is only supported for queue0 by hardware controller, but the FC7300F8MDQ and FC7300F4MDD support TX checksum offload for queue0 and queue1.

Eth Usage Common Steps

- 1) Before configure Eth, you shall configure and enable ENET peripheral clocks and SCM clock in the MCU module.

General	FOSC	SOSC	SIRC	SIRC32K	FIRC	PLL0	PLL1	System Clock	PCC	Clock Reference Point	CSC0	SCM
McuPeripheralClockConfig												
Index	Name	Mcu Peri...	Enable p...	Peripher...	Peripher...	Peripheral Clock Frequency (0 ~ 300000000)(Hz)						
15	McuPeripheralClockConfig_46	FLEXCAN5	☒	FOSCDIVH	1	2.4E7						
16	McuPeripheralClockConfig_10	PTIMER0	☒	OFF	1	0.0						
17	McuPeripheralClockConfig_16	FCIIC0	☒	FOSCDIVM	1	1.2E7						
18	McuPeripheralClockConfig_17	FCUART0	☒	FOSCDIVM	1	1.2E7						
19	McuPeripheralClockConfig_18	PTIMER1	☒	OFF	1	0.0						
20	McuPeripheralClockConfig_20	FCUART5	☒	OFF	1	0.0						
21	McuPeripheralClockConfig_21	FTU4	☒	FOSCDIVH	1	2.4E7						
22	McuPeripheralClockConfig_0	FCSPI1	☒	FOSCDIVM	1	1.2E7						
23	McuPeripheralClockConfig_4	FTU3	☒	OFF	1	0.0						
24	McuPeripheralClockConfig_9	FCUART2	☒	FOSCDIVM	1	1.2E7						
25	McuPeripheralClockConfig_22	FCUART1	☒	FOSCDIVM	1	1.2E7						
26	McuPeripheralClockConfig_24	FCSPI0	☒	FOSCDIVM	1	1.2E7						
27	ETH_PTPCLKSEL	ENET	☒	PLL0DIVH	1	3.0E8						
28	McuPeripheralClockConfig_6	CMU1	☒	OFF	1	0.0						
29	McuPeripheralClockConfig_7	CMU2	☒	OFF	1	0.0						
30	McuPeripheralClockConfig_8	CMU3	☒	OFF	1	0.0						
31	McuPeripheralClockConfig_15	CMU4	☒	OFF	1	0.0						
32	McuPeripheralClockConfig_5	TRGSEL0	☒	OFF	1	0.0						
33	McuPeripheralClockConfig_19	TRGSEL1	☒	OFF	1	0.0						

McuEnetClockConfigure

Name

- Enable Enet ☒ ☒ ☒ Enable Enet MII Tx Clock output buffer ☒ ☒ ☒
- Enable Enet PLL0 Divider ☒ ☐ ☒
- Enet PLL0 Divider(1 ~ 16)
- Enet RGMII Clock select
- Enet TX Clock Select
- Enet PTP Clock Select

- 2) Users shall also configure the PortPin mode for Eth module.

20	 ETH_MDC		21		5	 PTB5	 ☒	 MII_RMII_MDC_TRGSEL_IN0
21	 ETH_MDIO		22		4	 PTB4	 ☒	 MII_RMII_MDIO_TRGSEL_IN1
0	 ETH_RXD1		64		0	 PTC0	 ☒	 ENET_RXD1
1	 ETH_RXD0		65		1	 PTC1	 ☒	 ENET_RXD0
2	 ETH_TXD0		66		2	 PTC2	 ☒	 ENET_TXD0
15	 ETH_RXD3		79		15	 PTC15	 ☒	 ENET_RXD3
5	 ETH_TXD3		101		5	 PTD5	 ☒	 ENET_TXD3
6	 ETH_TXD2		102		6	 PTD6	 ☒	 ENET_TXD2
7	 ETH_TXD1		103		7	 PTD7	 ☒	 ENET_TXD1
8	 ETH_RXDV		104		8	 PTD8	 ☒	 ENET_RX_DV_CTL
9	 ETH_RXD2		105		9	 PTD9	 ☒	 ENET_RXD2
10	 ETH_RXCLK		106		10	 PTD10	 ☒	 ENET_RX_CLK
11	 ETH_TXCLK		107		11	 PTD11	 ☒	 ENET_TX_CLK
12	 ETH_TXEN		108		12	 PTD12	 ☒	 ENET_TX_EN_CTL

- 3) Users shall configure the general configurations for Eth. This includes the Eth functions enable, the Eth offloading, and vendor specific configurations, etc.

▼ EthGeneral

Name

ETHGeneral

Development Error Detection

✓

EthGetCounterValuesApi

✓

EthGetRxStatsApi

✓

EthGetTxStatsApi

✓

EthGetTxErrorCounterValuesApi

✓

EthGlobalTimeSupport

✓

EthVersionInfoApi

✓

EthIndex (0 -> 255)

0

EthMainFunctionPeriod (0.001 -> 65.535)

0.001

EthMaxCtrlsSupported (dynamic range)

1

▼ EthCtrlOffloading

Name

EthCtrlOffloading

EthCtrlEnableOffloadChecksumICMP

✓

EthCtrlEnableOffloadChecksumIPv4

✓

EthCtrlEnableOffloadChecksumTCP

✓

EthCtrlEnableOffloadChecksumUDP

✓

▼ EthGeneralVendorSpecific

Name

EthGeneralVendorSpecific

Eth Disable Diagnostic Event Module

EthEnableUserModeSupport

EthMulticoreSupport

EthUpdatePhysAddrFilterApi

✓

EthTimeoutCycles (1 -> 65535)

1500

EthModuleMcuCoreClock

1.2E8

- 4) Users shall create at least one item in EthCtrlConfig. This container contains the configuration parameters of the Eth controller(s). The Eth controller includes the general functions, controller specific configurations, egress FIFOs, ingress FIFOs, and traffic shaper, Time aware shaper etc.

General EthCtrlConfigEgressFifo EthCtrlConfigScheduler EthCtrlConfigShaper EthCtrlConfigTimeAwareShaper EthCtrlConfigIngressFifo

EthCtrlEnableMii ☒ ☐ ☐ EthCtrlEnableRxInterrupt ☒ ☐ ☐

EthCtrlEnableTxInterrupt ☒ ☐ ☐

EthCtrlIdx (dynamic range)

EthDuplexMode

EthCtrlMacLayerType

EthCtrlMacLayerSubType

EthCtrlMacLayerSpeed*

EthCtrlPhyAddress

EthCtrlEcucPartitionRef

EthControllerSpecific

Name

- 5) Users shall create at least one item in EthCtrlConfigEgressFifo.

General EthCtrlConfigEgressFifo EthCtrlConfigScheduler EthCtrlConfigShaper EthCtrlConfigTimeAwareShaper EthCtrlConfigIngressFifo

EthCtrlConfigEgress

EthCtrlConfigEgressLastSchedulerRef

EthCtrlConfigEgressFifo

Index	Name	EthCtrlC...	EthCtrlC...	EthCtrlC...
0	EthCtrlConfigEgressFifo_0	1536	4	0
1	EthCtrlConfigEgressFifo_1	1536	4	1

- 6) Users shall create at least one item in EthCtrlConfigScheduler.

General EthCtrlConfigEgressFifo EthCtrlConfigScheduler EthCtrlConfigShaper EthCtrlConfigTimeAwareShaper EthCtrlConfigIngressFifo

EthCtrlConfigEgress

EthCtrlConfigScheduler

Index	Name
0	EthCtrlConfigScheduler_0

- 7) Users shall create at least one item in EthCtrlConfigIngressFifo.

General EthCtrlConfigEgressFifo EthCtrlConfigScheduler EthCtrlConfigShaper EthCtrlConfigTimeAwareShaper EthCtrlConfigIngressFifo						
EthCtrlConfigIngress						
EthCtrlConfigIngressFifo						
Index	Name	EthCtrlC...	EthCtrlC...	EthCtrlConfigIngressFifoIdx		
0	EthCtrlCo...	1536	16	0		
1	EthCtrlCo...	1536	16	1		

- 8) If you want to use AVB mode, you should create an item in EthCtrlConfigShaper, and select the shaper in EthCtrlConfigScheduler /EthCtrlConfigSchedulerPredecessor/ EthCtrlConfigSchedulerPredecessorRef.

EthCtrlConfigShaper

Name

EthCtrlConfigShaper

EthCtrlConfigShaperIdleSlope (dynamic range)

EthCtrlConfigShaperSendSlope

EthCtrlConfigShaperPredecessorFifoRef

EthCtrlConfigShaperMaxCredit (0 -> 131072)

EthCtrlConfigShaperMinCredit (0 -> 131072)

- 9) For EthControllerMac and EthControllerMtl, the default Settings are recommended.

EthControllerMac

Name

Name

CRC stripping for Type packets ☒	Automatic Pad or CRC Stripping ☒
Jumbo Packet Enable ☐	Loopback Mode ☐
Disable Receive Own ☒	Disable Retry ☐
Deferral Check ☐	

Name

Disable CRC Checking for Received Packets ☐

Name

Receive All ☐	Drop Non-TCP/UDP over IP Packets ☐
Disable Broadcast Packets ☐	Pass All Multicast ☐
Promiscuous Mode ☐	

▼

EthControllerMtl

Name

📁

EthControllerMtl

Name

📁

EthControllerMtlQueue

Forward TCPIP Checksum Error Frames

📄

☐

✎

▼

- 10) If EthGlobalTimeSupport is enabled, ENET PTP clock selection, Addend value and Increment value should be configured properly.

▼

EthControllerPTP

Name

📁

EthControllerPTP

AV 802.1AS Mode Enable

📄

☐

✎

▼

PTP Packets Sent over IPv4-UDP

📄

☑

✎

▼

PTP Packets Sent over IPv6-UDP

📄

☐

✎

▼

PTP over Ethernet Packets

📄

☐

✎

▼

PTP Packet Processing for Version 2 Format

📄

☑

✎

▼

Enable Timestamp for All Packets

📄

☑

✎

▼

System time higher word second (0 -> 65535)

📄

0

✎

▼

System time second (0 -> 4294967295)

📄

0

✎

▼

System time nanosecond (0 -> 999999999)

📄

0

✎

▼

Addend value (0 -> 4294967295)

📄

1431655765

🌐

▼

Increment value(nanosecond) (1 -> 255)

📄

20

✎

▼

Egress modulation value (0 -> 255)

📄

0

✎

▼

Ingress modulation value (0 -> 255)

📄

0

✎

▼

Properties for EthControllerPTP_IncrementVal

Description	Increment step value (nanoseconds) for the PTP, Addend value * Increment value = 1000000000 * 2 ³² / ptpclk
Information	
Problems	
PostBuildVariantConditions	

- 11) When configuration completed, click the Generate Project to generate relevant code.

